Western and Central Pacific Fisheries Commission

Workflow demonstration

Assessment workflow demo: ALB

WORKFLOW DEMO

2026-09-05

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1 Executive summary

This is a workflow demonstration, not a stock assessment. It uses ALB outputs to demonstrate the model-to-report tools. The diagnostic-model figures use the archived MFCL-compatible KIMSA fit from job 24781. Stepwise, starting-value, self-test, retrospective, sensitivity, ensemble and additional-output figures use illustrative display values to demonstrate report construction. They are not new assessment estimates.

Replace the illustrative study bundles with fitted results, write the assessment interpretation in the section files, and update the report cover before submission.

2 Introduction

3 Data compilation

The diagnostic model retains the ALB observation timing and two-region structure. Catch, abundance-index, length-composition and conditional age-at-length observations are available in the interactive diagnostic supplement.

Table 2 describes the fishery roles and observation periods. Figure 3 shows data availability. Index observations and predictions follow the model likelihood's geometric-mean centering convention.

4 Model description

The archived ALB fit uses the MFCL-compatible age-region state, age-based selectivity and five-pass hybrid F, without tagging. Its original F upper bound of 3 is retained for reproduction; new model specifications default to 6.

Length compositions use square_fita with fishery divisors and a raw-record sample-size cap of 1000. The composition settings table records the fitted likelihood configuration.

Figure 5, Figure 6 and Figure 7 describe the biological inputs and estimates; Figure 33 shows fishery selectivity. Additional-format panels illustrate output types that are not estimated by this tag-free ALB reference model.

5 Model development

Figure 8 illustrates how model-development results are assembled. Each study retains its model definitions and change descriptions. Replace the example model collections with the actual development sequence before interpreting the differences.

6 Results

Figure 32 shows annual reference-model trajectories. Stocks use time-weighted annual means; recruitment and catches use annual totals. Depletion is the ratio of annual spawning-potential summaries. Index points and predictions use annual geometric means. The interactive viewer retains the original time resolution.

Length-residual panels use raw-sample-weighted annual averages within each fitted bin; terminal bins retain the model's compressed-tail support. These displays do not define a new annual likelihood.

Index fits and residuals (Figure 69, Figure 9), length-composition residuals (Figure 10) and conditional age-at-length residuals (Figure 19) are generated from the same result bundle used by the interactive viewer.

The following studies demonstrate the intended diagnostic presentation: starting-value trials (Figure 21), simulation-estimation recovery (Figure 24), retrospective analysis (Figure 26), objective profiles (Figure 27) and sensitivity comparisons. Their displayed values are illustrative.

7 Reporting examples

The ensemble, management-quantity and projection panels demonstrate the reporting formats. They do not establish ALB stock status. Supply actual reference points, model retention decisions, uncertainty draws and projection results before writing management conclusions.

8 Discussion

9 Tables

Table 1: Resources accompanying the workflow demonstration.

Resource	Contents
Interactive results	Complete fitted records and study galleries
Editable manuscript	Quarto sections, captions and replaceable artifacts
Model provenance	Archived ALB job 24781; illustrative auxiliary studies

Table 2: Fishery roles, regions, observation periods and units.

fishery	region	role	first_year	last_year	unit
1	1	Catch	1954	2022	fish
2	1	Catch	1954	2022	fish
3	1	Catch	1954	2022	fish
4	1	Catch	1954	2022	fish
5	1	Catch	1992	2022	fish
6	1	Catch	1982	2022	fish
7	1	Catch	1987	2022	fish
8	1	Catch	1954	2022	fish
9	1	Catch	1984	2022	fish
10	1	Catch	1985	2022	fish
11	1	Catch	1982	2022	t
12	1	Catch	1987	2022	t
13	1	Catch	1983	1990	t
14	2	Catch	1957	2022	fish
15	2	Catch	1961	2021	fish
16	2	Catch	1963	2021	fish
17	2	Catch	1986	2006	t
18	1	Index	1954	2022	relative index
19	1	Index	1992	2022	relative index
20	2	Index	1954	2022	relative index

Table 3: Illustrative tag-release programmes.

Programme	Years	Releases	Mixing_period
A	2000–2004	30000	4 quarters
B	2005–2008	26000	4 quarters

Table 4: Illustrative management quantities and supplied ranges.

Quantity	Median	Lower	Upper
Dynamic spawning depletion	4.4e-01	3.5e-01	5.3e-01
Spawning potential relative to MSY	2.1e+00	1.5e+00	2.8e+00
Fishing mortality relative to MSY	6.5e-01	4.2e-01	8.8e-01
MSY (t)	9.5e+04	8.0e+04	1.1e+05

Table 5: Reference quantities used in the illustrative status panels.

Quantity	Definition
Unfished spawning potential	Dynamic unfished spawning potential
Spawning potential at MSY	Equilibrium spawning potential at maximum sustainable yield
Fishing mortality at MSY	Fishing mortality at maximum sustainable yield

Table 6: Illustrative equilibrium reference points and supplied ranges.

Quantity	Median	Lower	Upper
MSY (t)	95000	80000	110000

Table 7: Illustrative between-model ranges for management quantities.

Quantity	Median	Lower	Upper
Dynamic spawning depletion	0.44	0.35	0.53
Spawning potential relative to MSY	2.10	1.50	2.80
Fishing mortality relative to MSY	0.65	0.42	0.88

Table 8: Illustrative projected spawning depletion by catch scenario.

Year	Scenario	Depletion
2025	Current catch	0.43
2030	Current catch	0.41
2035	Current catch	0.40
2025	Lower catch	0.45
2030	Lower catch	0.49
2035	Lower catch	0.52

Table 9: Model configurations.

Model	Parent	Change
alb	NA	No change description supplied

Table 10: Recorded numerical checks by model.

Model	Label	Status	Passed	Failed	Unknown
alb	ALB reference	passed	3	0	0

Table 11: Length-composition likelihood and weighting settings by fishery.

fishery	likelihood	divisor	record_cap	epsilon_scale	flag161
3	square_fita	51	1000	1	0
4	square_fita	164	1000	1	0
6	square_fita	89	1000	1	0
7	square_fita	29	1000	1	0
8	square_fita	372	1000	1	0
9	square_fita	212	1000	1	0
10	square_fita	125	1000	1	0
11	square_fita	27	1000	1	0
12	square_fita	37	1000	1	0
13	square_fita	358	1000	1	0
14	square_fita	123	1000	1	0
15	square_fita	241	1000	1	0
16	square_fita	450	1000	1	0
17	square_fita	1	1000	1	0
18	square_fita	746	1000	1	0
19	square_fita	55	1000	1	0
20	square_fita	61	1000	1	0

Table 12: Numbers of fitted observations and composition samples by series.

Kind	Series	First	Last	N
catch	F1_R1	1954	2022	274
catch	F10_R1	1985	2022	147
conditional_age	F10_R1	2010	2010	17
length	F10_R1	1989	2016	74
catch	F11_R1	1982	2022	344
conditional_age	F11_R1	2006	2010	63

Kind	Series	First	Last	N
length	F11_R1	1996	2022	143
catch	F12_R1	1987	2022	161
length	F12_R1	1987	1998	27
catch	F13_R1	1983	1990	17
length	F13_R1	1988	1989	2
catch	F14_R2	1957	2022	262
length	F14_R2	1966	2022	119
catch	F15_R2	1961	2021	229
length	F15_R2	1968	2021	65
catch	F16_R2	1963	2021	195
length	F16_R2	1968	2021	17
catch	F17_R2	1986	2006	22
length	F17_R2	1992	1992	1
length	F18_R1	1965	2022	57
index	F18_R1_G18	1954	2022	69
length	F19_R1	1997	2022	25
index	F19_R1_G19	1992	2022	31
catch	F2_R1	1954	2022	274
length	F20_R2	1966	2018	39
index	F20_R2_G20	1954	2022	69
catch	F3_R1	1954	2022	275
conditional_age	F3_R1	2009	2009	4
length	F3_R1	1965	2022	178
catch	F4_R1	1954	2022	274
length	F4_R1	1965	2022	145
catch	F5_R1	1992	2022	121
catch	F6_R1	1982	2022	161
conditional_age	F6_R1	2009	2010	81
length	F6_R1	1992	2022	116
catch	F7_R1	1987	2022	135
conditional_age	F7_R1	2009	2010	30
length	F7_R1	1993	2013	25
catch	F8_R1	1954	2022	271
conditional_age	F8_R1	2010	2010	15
length	F8_R1	1965	2022	89
catch	F9_R1	1984	2022	126
conditional_age	F9_R1	2007	2013	45
length	F9_R1	1999	2020	17

Table 13: Index fit on the log scale; the first decade starts at each index's first observation.

Index	N	Log_RMSE	First_decade_RMSE
18	69	0.180	0.269
19	31	0.349	0.502
20	69	0.209	0.127

Table 14: Study inventory; auxiliary studies use illustrative display values.

study	type	models	passed	failed	unknown
diagnostic	comparison	1	1	0	0
stepwise	stepwise	4	4	0	0
jitter	jitter	9	8	1	0
selftest	selftest	6	6	0	0
retrospective	retrospective	6	6	0	0
sensitivity	sensitivity	3	3	0	0
ensemble	ensemble	7	7	0	0
formats	comparison	1	1	0	0

Table 15: Model configurations.

Model	Parent	Change
initial		Initial configuration
data-update	initial	Update observations
biology-update	data-update	Update biological assumptions
reference	biology-update	Select diagnostic configuration

Table 16: Recorded numerical checks by model.

Model	Label	Status	Passed	Failed	Unknown
initial	Initial	passed	3	0	0
data-update	Data update	passed	3	0	0
biology-update	Biology update	passed	3	0	0
reference	Reference	passed	3	0	0

Table 17: Numbers of fitted observations and composition samples by series.

Kind	Series	First	Last	N
index	F18_R1_G18	1954	2022	69

Kind	Series	First	Last	N
index	F19_R1_G19	1992	2022	31
index	F20_R2_G20	1954	2022	69

Table 18: Illustrative model-development sequence.

id	parent	change
initial		Initial configuration
data-update	initial	Update observations
biology-update	data-update	Update biological assumptions
reference	biology-update	Select diagnostic configuration

Table 19: Model configurations.

Model	Parent	Change
reference	NA	No change description supplied
trial-1	NA	No change description supplied
trial-2	NA	No change description supplied
trial-3	NA	No change description supplied
trial-4	NA	No change description supplied
trial-5	NA	No change description supplied
trial-6	NA	No change description supplied
trial-7	NA	No change description supplied
trial-8	NA	No change description supplied

Table 20: Recorded numerical checks by model.

Model	Label	Status	Passed	Failed	Unknown
reference	Reference	passed	3	0	0
trial-1	Trial 1	passed	3	0	0
trial-2	Trial 2	passed	3	0	0
trial-3	Trial 3	passed	3	0	0
trial-4	Trial 4	passed	3	0	0
trial-5	Trial 5	passed	3	0	0
trial-6	Trial 6	passed	3	0	0
trial-7	Trial 7	passed	3	0	0
trial-8	Trial 8	failed	2	1	0

Table 21: Numbers of fitted observations and composition samples by series.

Kind	Series	First	Last	N
index	F18_R1_G18	1954	2022	69
index	F19_R1_G19	1992	2022	31
index	F20_R2_G20	1954	2022	69

Table 22: Model configurations.

Model	Parent	Change
truth	NA	No change description supplied
replicate-1	NA	No change description supplied
replicate-2	NA	No change description supplied
replicate-3	NA	No change description supplied
replicate-4	NA	No change description supplied
replicate-5	NA	No change description supplied

Table 23: Recorded numerical checks by model.

Model	Label	Status	Passed	Failed	Unknown
truth	Generating truth	passed	3	0	0
replicate-1	Refit 1	passed	3	0	0
replicate-2	Refit 2	passed	3	0	0
replicate-3	Refit 3	passed	3	0	0
replicate-4	Refit 4	passed	3	0	0
replicate-5	Refit 5	passed	3	0	0

Table 24: Numbers of fitted observations and composition samples by series.

Kind	Series	First	Last	N
index	F18_R1_G18	1954	2022	69
index	F19_R1_G19	1992	2022	31
index	F20_R2_G20	1954	2022	69

Table 25: Model configurations.

Model	Parent	Change
reference	NA	No change description supplied
peel-1	NA	No change description supplied

Model	Parent	Change
peel-2	NA	No change description supplied
peel-3	NA	No change description supplied
peel-4	NA	No change description supplied
peel-5	NA	No change description supplied

Table 26: Recorded numerical checks by model.

Model	Label	Status	Passed	Failed	Unknown
reference	Reference	passed	3	0	0
peel-1	Peel 1	passed	3	0	0
peel-2	Peel 2	passed	3	0	0
peel-3	Peel 3	passed	3	0	0
peel-4	Peel 4	passed	3	0	0
peel-5	Peel 5	passed	3	0	0

Table 27: Numbers of fitted observations and composition samples by series.

Kind	Series	First	Last	N
index	F18_R1_G18	1954	2022	69
index	F19_R1_G19	1992	2022	31
index	F20_R2_G20	1954	2022	69

Table 28: Model configurations.

Model	Parent	Change
reference	NA	No change description supplied
lower	NA	No change description supplied
upper	NA	No change description supplied

Table 29: Recorded numerical checks by model.

Model	Label	Status	Passed	Failed	Unknown
reference	Reference	passed	3	0	0
lower	Lower	passed	3	0	0
upper	Upper	passed	3	0	0

Table 30: Numbers of fitted observations and composition samples by series.

Kind	Series	First	Last	N
index	F18_R1_G18	1954	2022	69
index	F19_R1_G19	1992	2022	31
index	F20_R2_G20	1954	2022	69

Table 31: Model configurations.

Model	Parent	Change
reference	NA	No change description supplied
member-1	NA	No change description supplied
member-2	NA	No change description supplied
member-3	NA	No change description supplied
member-4	NA	No change description supplied
member-5	NA	No change description supplied
member-6	NA	No change description supplied

Table 32: Recorded numerical checks by model.

Model	Label	Status	Passed	Failed	Unknown
reference	Reference	passed	3	0	0
member-1	Member 1	passed	3	0	0
member-2	Member 2	passed	3	0	0
member-3	Member 3	passed	3	0	0
member-4	Member 4	passed	3	0	0
member-5	Member 5	passed	3	0	0
member-6	Member 6	passed	3	0	0

Table 33: Numbers of fitted observations and composition samples by series.

Kind	Series	First	Last	N
index	F18_R1_G18	1954	2022	69
index	F19_R1_G19	1992	2022	31
index	F20_R2_G20	1954	2022	69

Table 34: Model configurations.

Model	Parent	Change
formats	NA	No change description supplied

Table 35: Recorded numerical checks by model.

Model	Label	Status	Passed	Failed	Unknown
formats	Illustrative ALB outputs	passed	3	0	0

Table 36: Numbers of fitted observations and composition samples by series.

Kind	Series	First	Last	N
index	F18_R1_G18	1954	2022	69
index	F19_R1_G19	1992	2022	31
index	F20_R2_G20	1954	2022	69

10 Figures

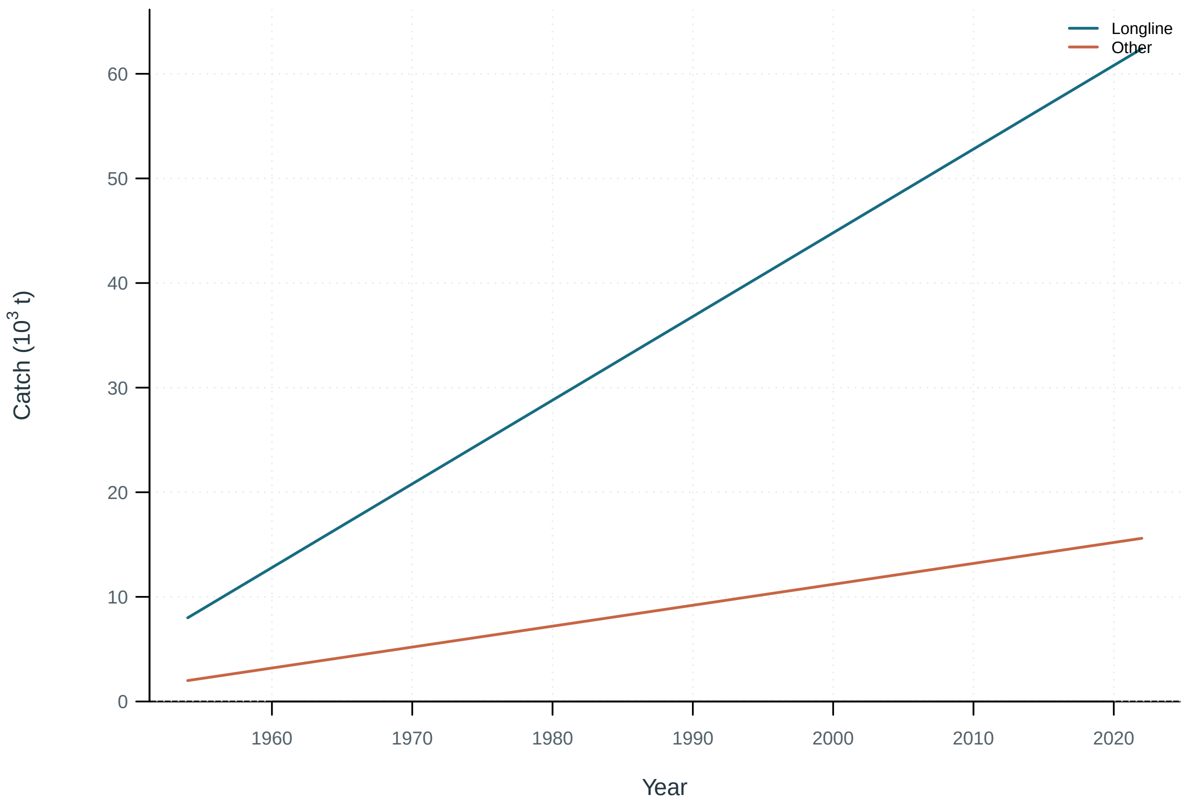

Figure 1: Annual catch by gear.

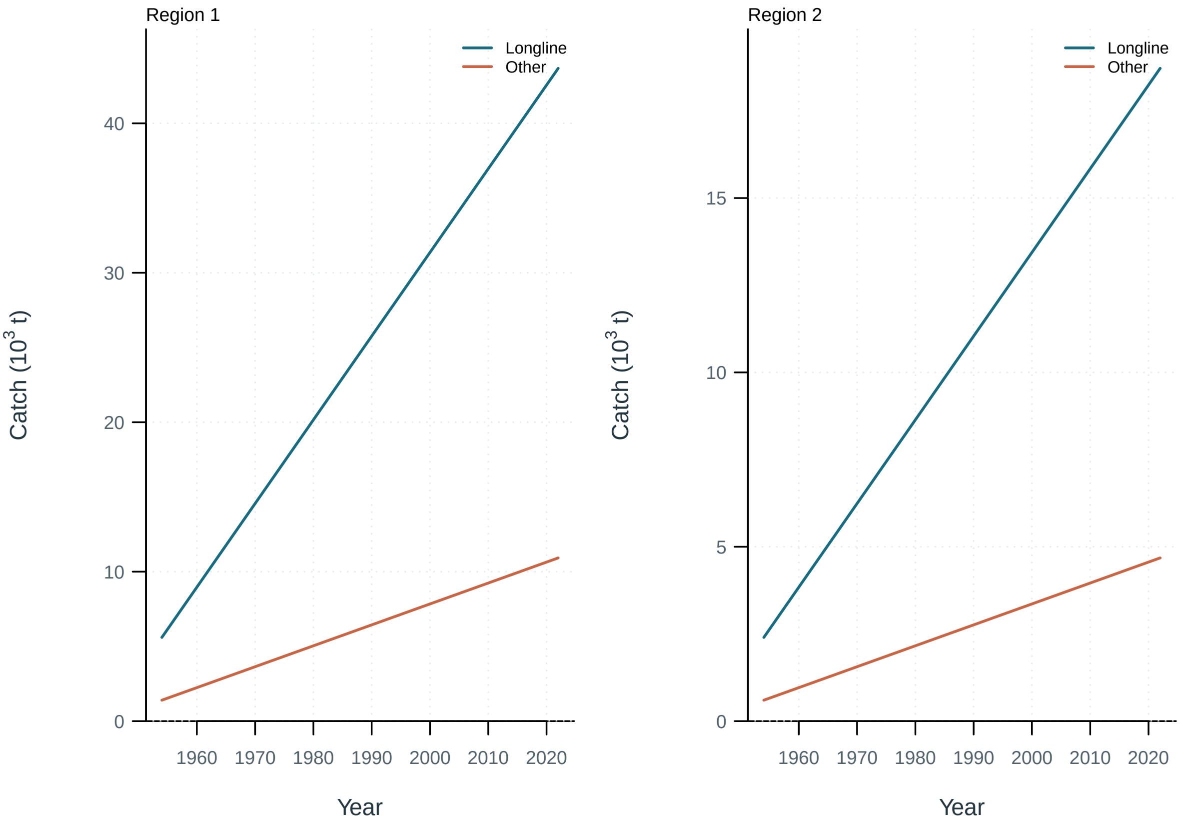

Figure 2: Annual catch by model region and gear.

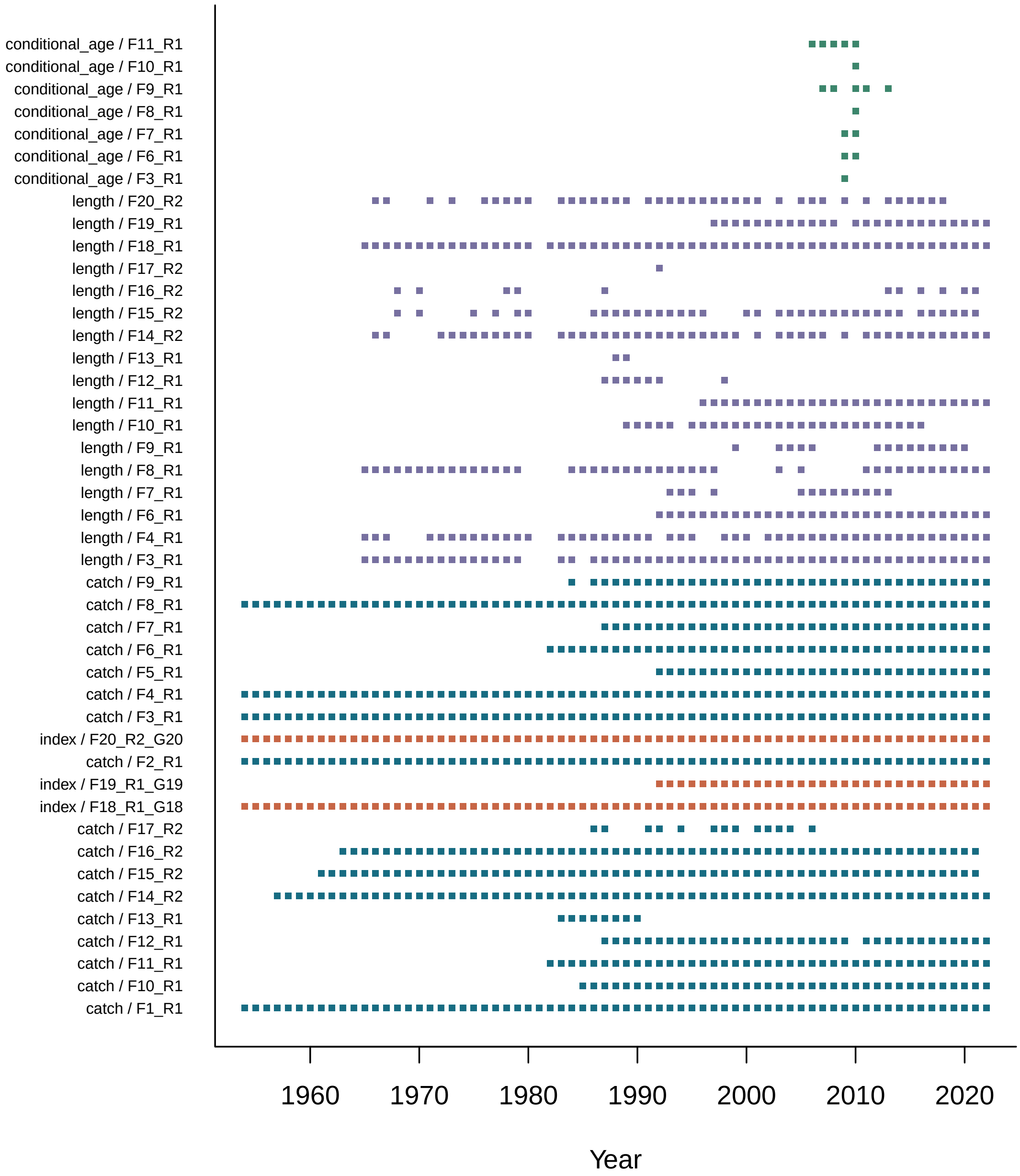

Figure 3: Availability of catch, abundance-index and composition observations by series and year.

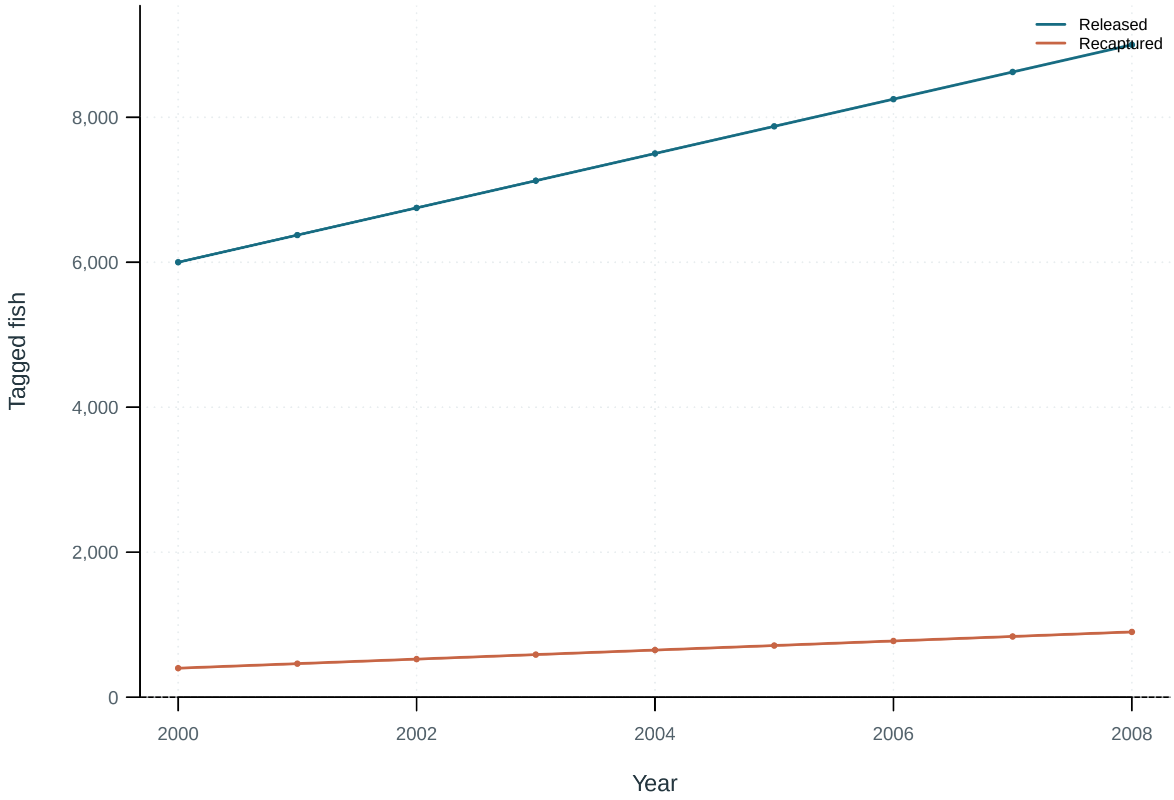

Figure 4: Tag releases and recaptures by year.

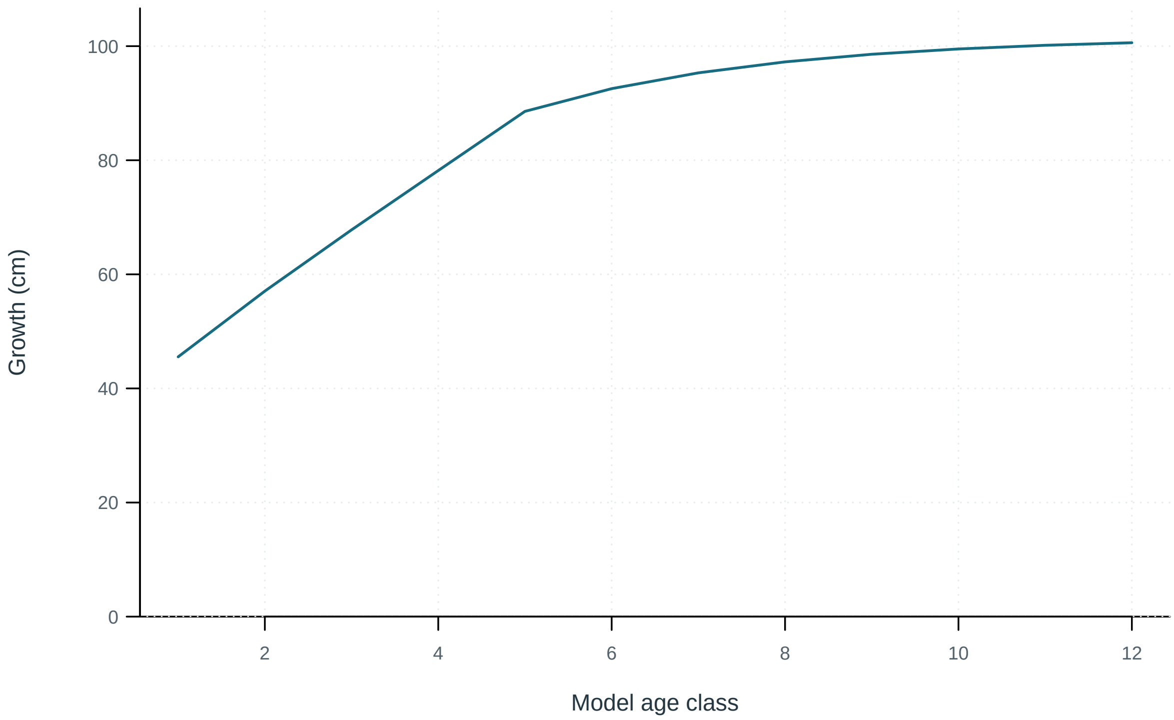

Figure 5: Mean length at the start of each model age class.

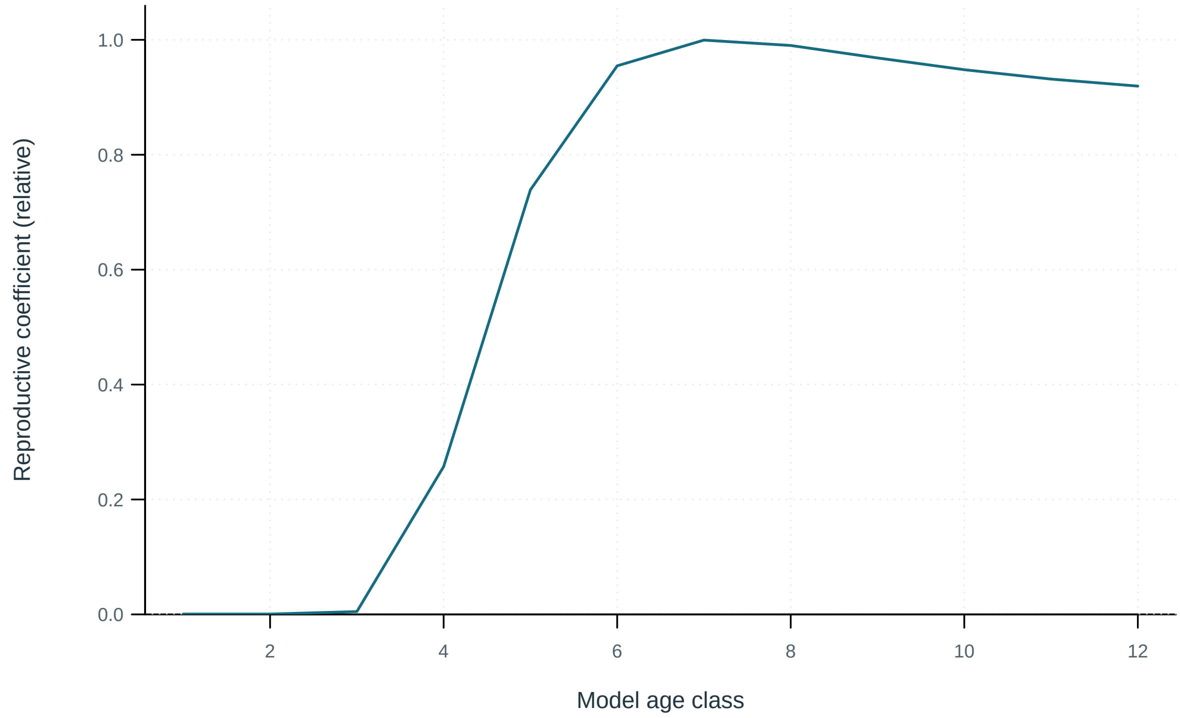

Figure 6: Model reproductive coefficient at age.

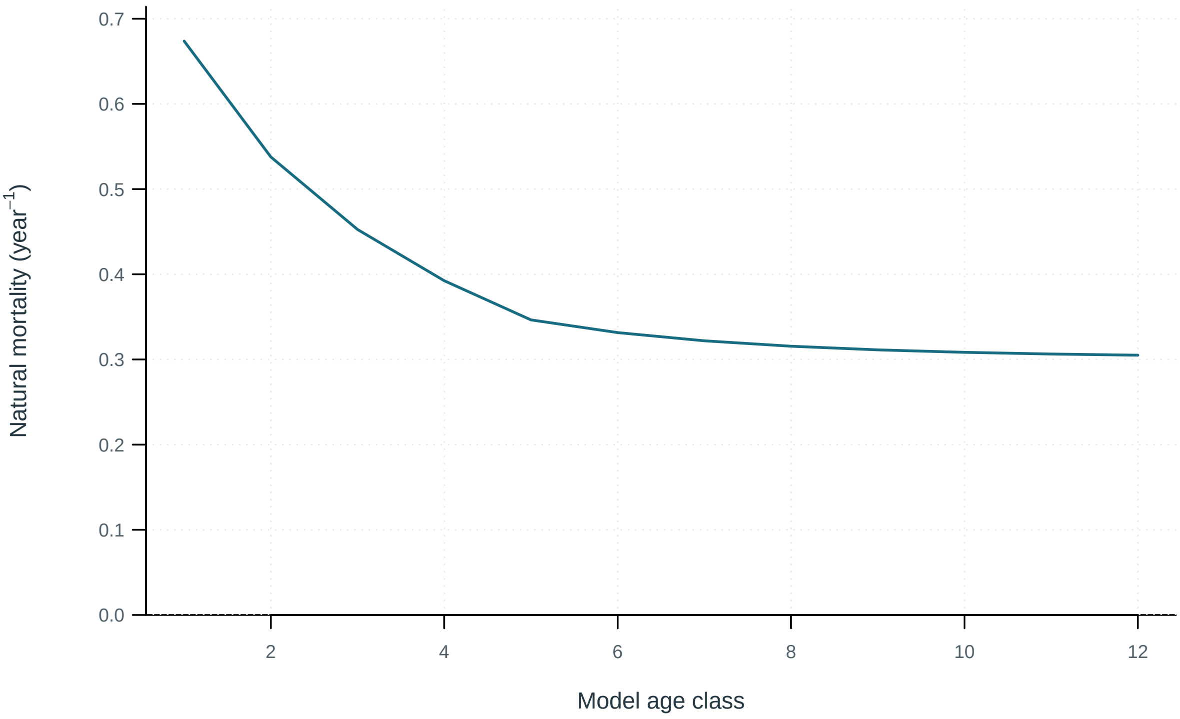

Figure 7: Annual instantaneous natural mortality at age.

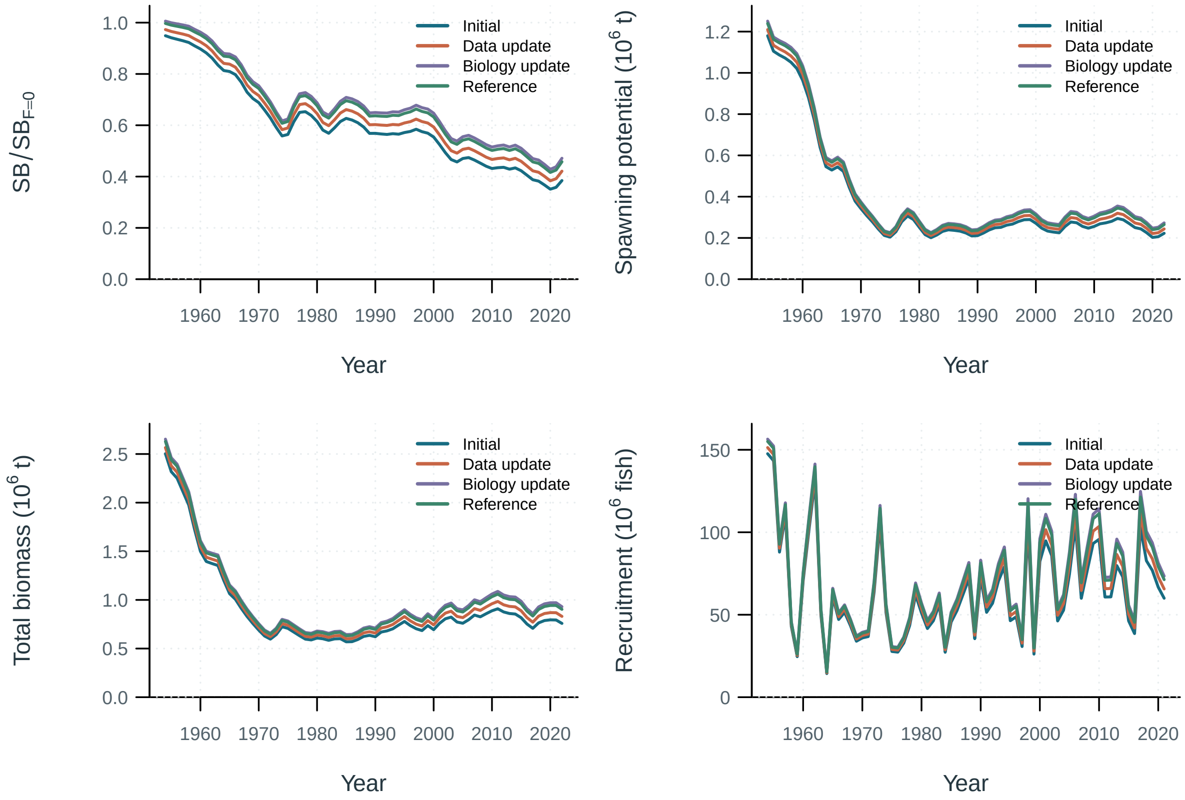

Figure 8: Stepwise model trajectories.

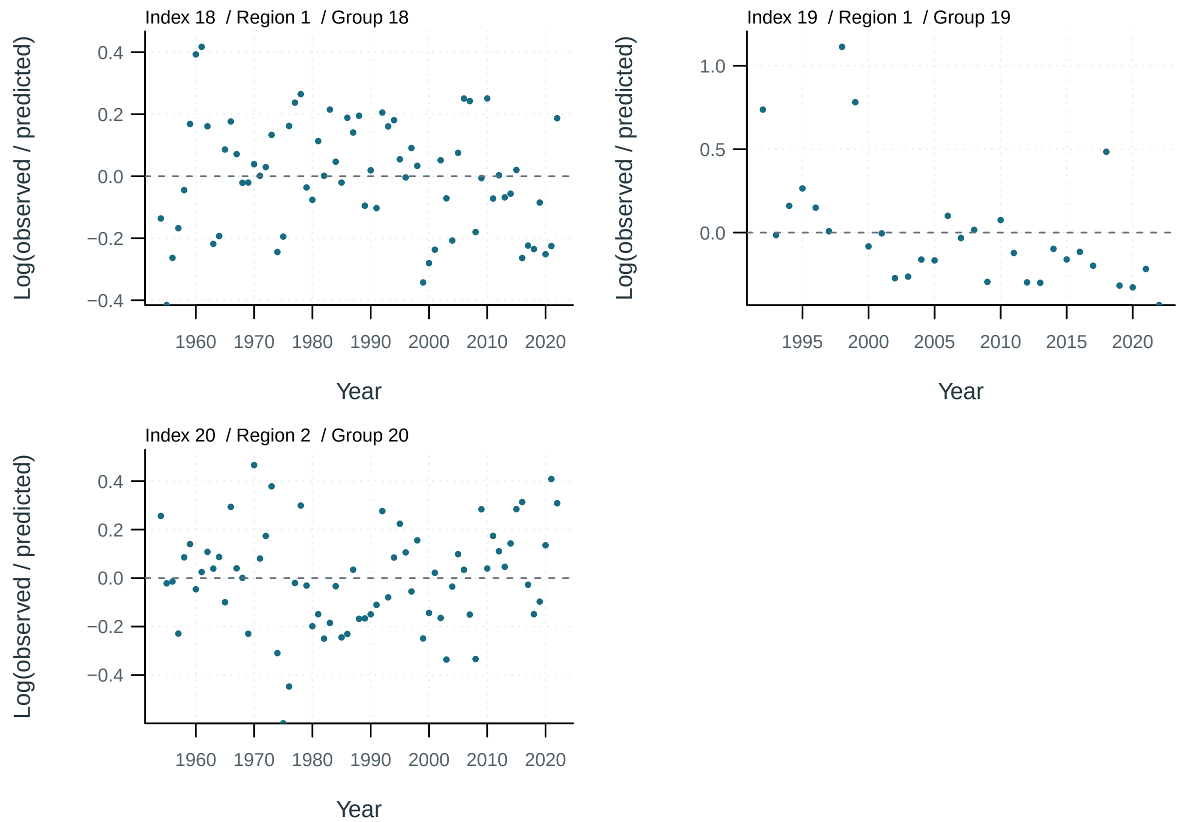

Figure 9: Abundance-index log residuals, log(observed/predicted), by series.

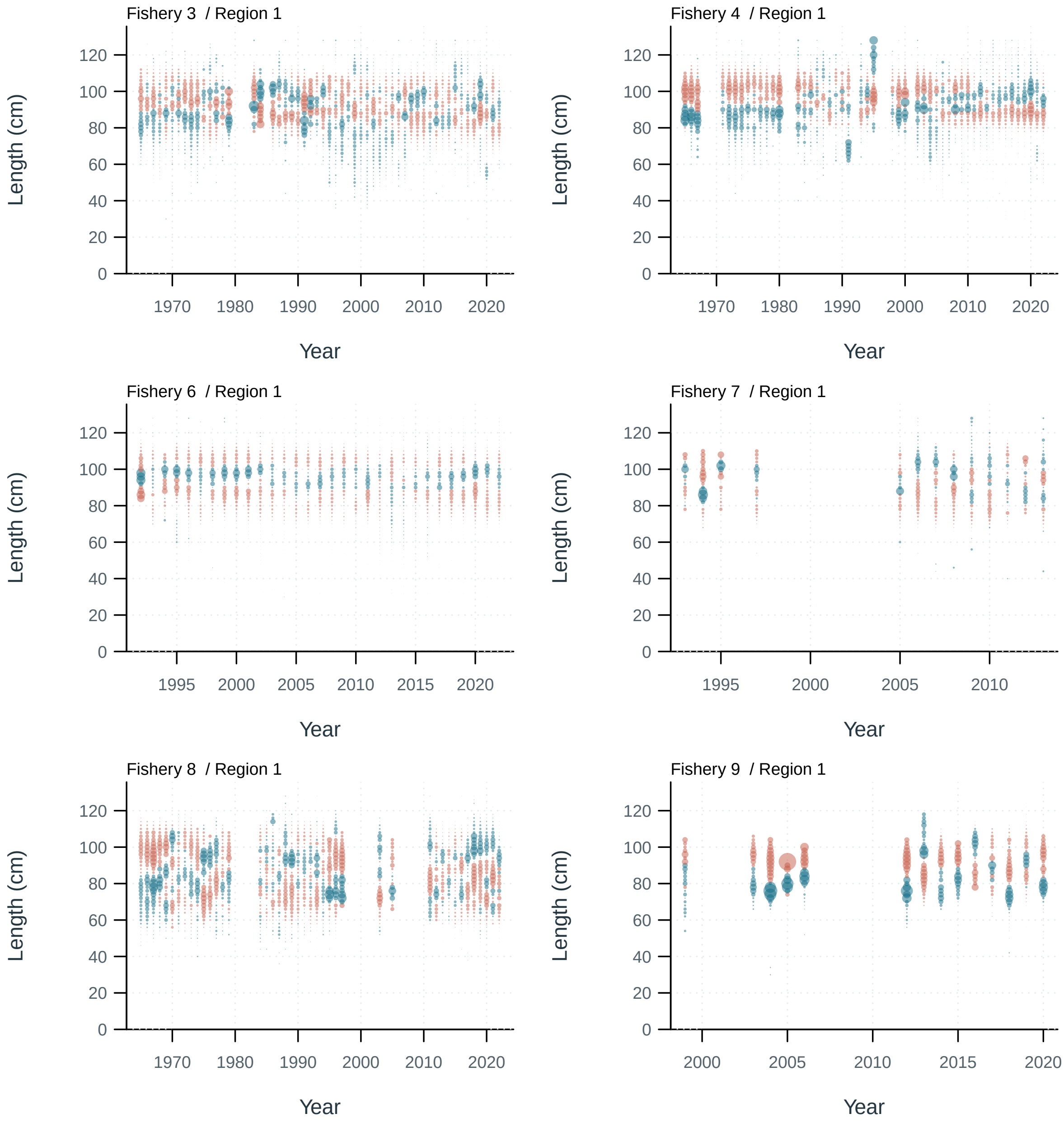

Figure 10: Length composition residuals by fishery. Blue: observed exceeds predicted; red: predicted exceeds observed. Circle area is proportional to absolute residual. Terminal bins include compressed tails.

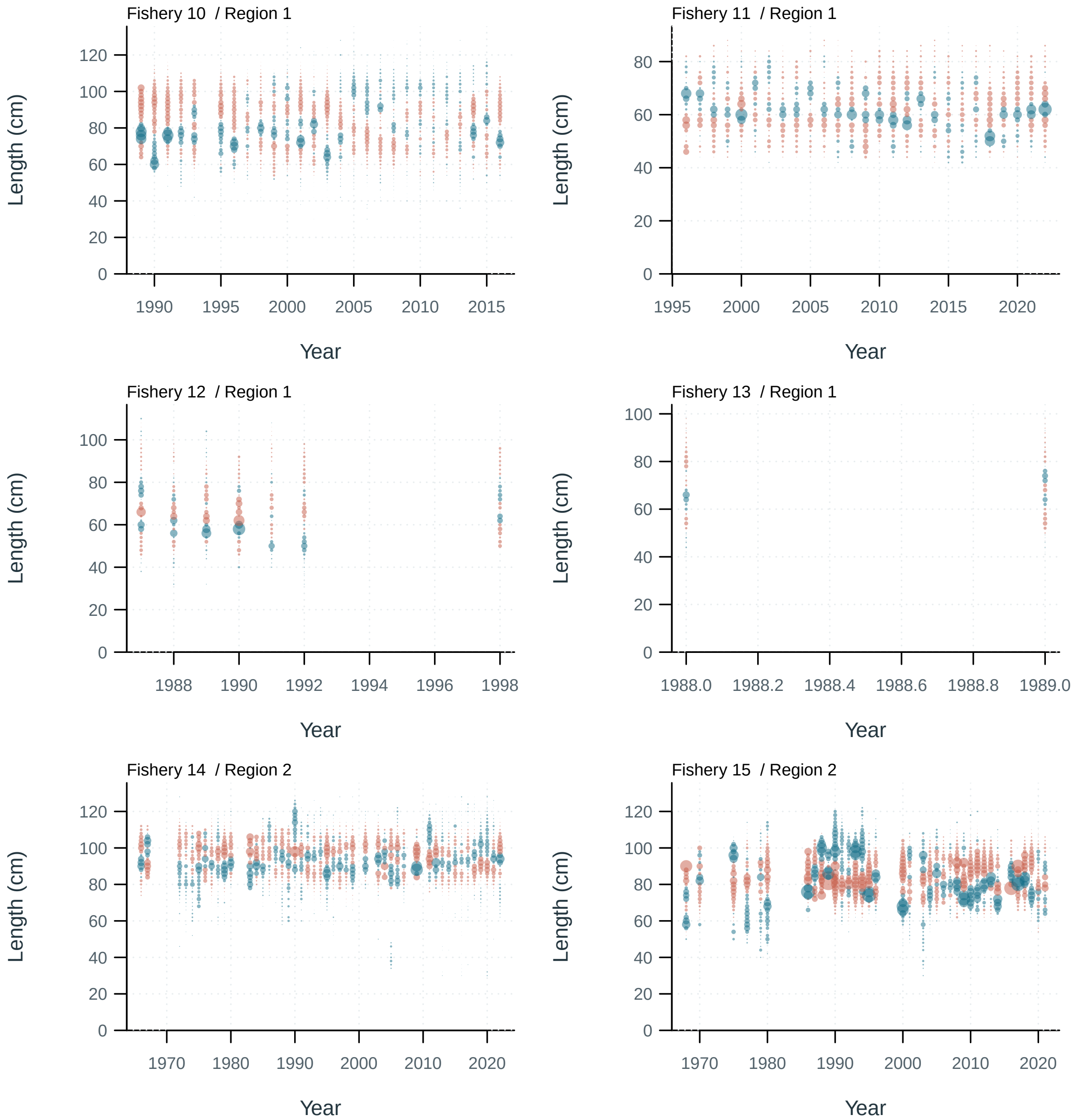

Figure 11: Length composition residuals by fishery. Blue: observed exceeds predicted; red: predicted exceeds observed. Circle area is proportional to absolute residual. Terminal bins include compressed tails.

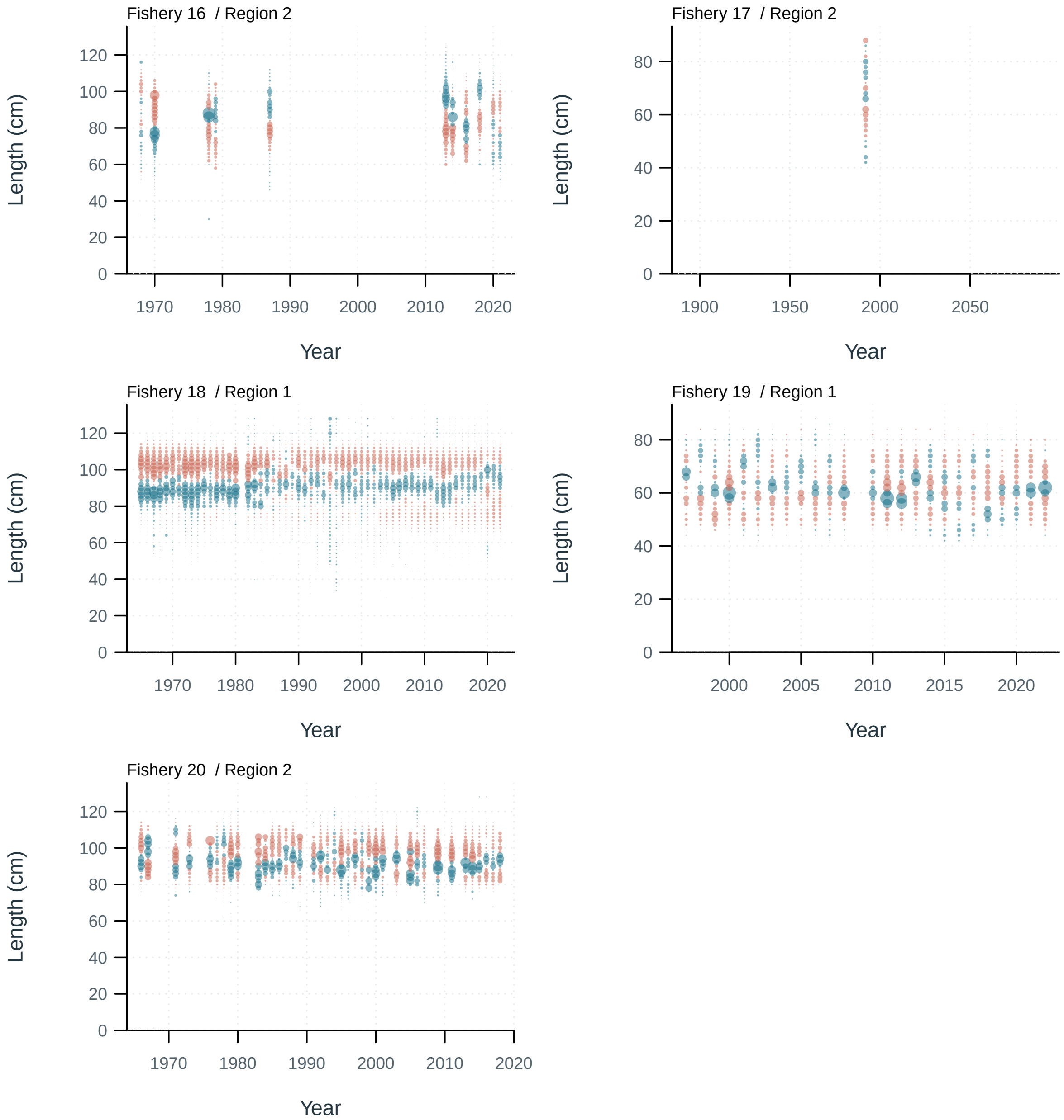

Figure 12: Length composition residuals by fishery. Blue: observed exceeds predicted; red: predicted exceeds observed. Circle area is proportional to absolute residual. Terminal bins include compressed tails.

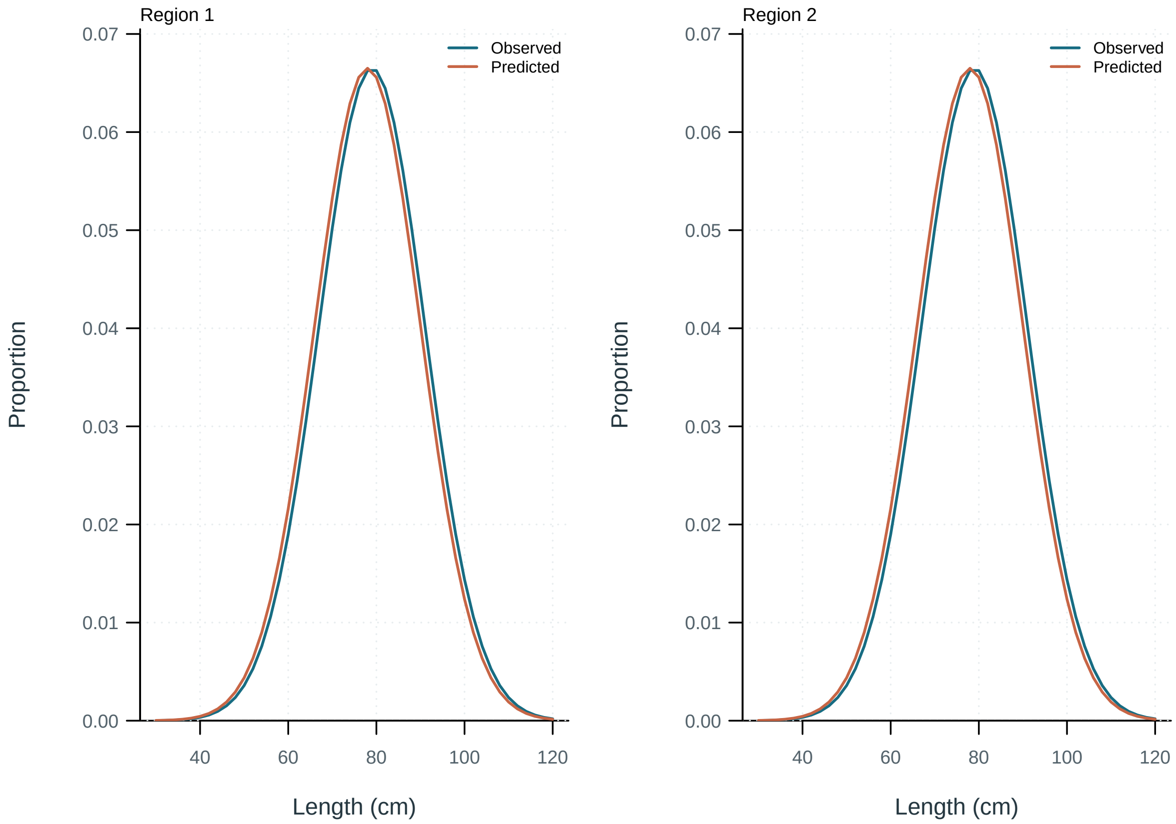

Figure 13: Observed and predicted length proportions by region.

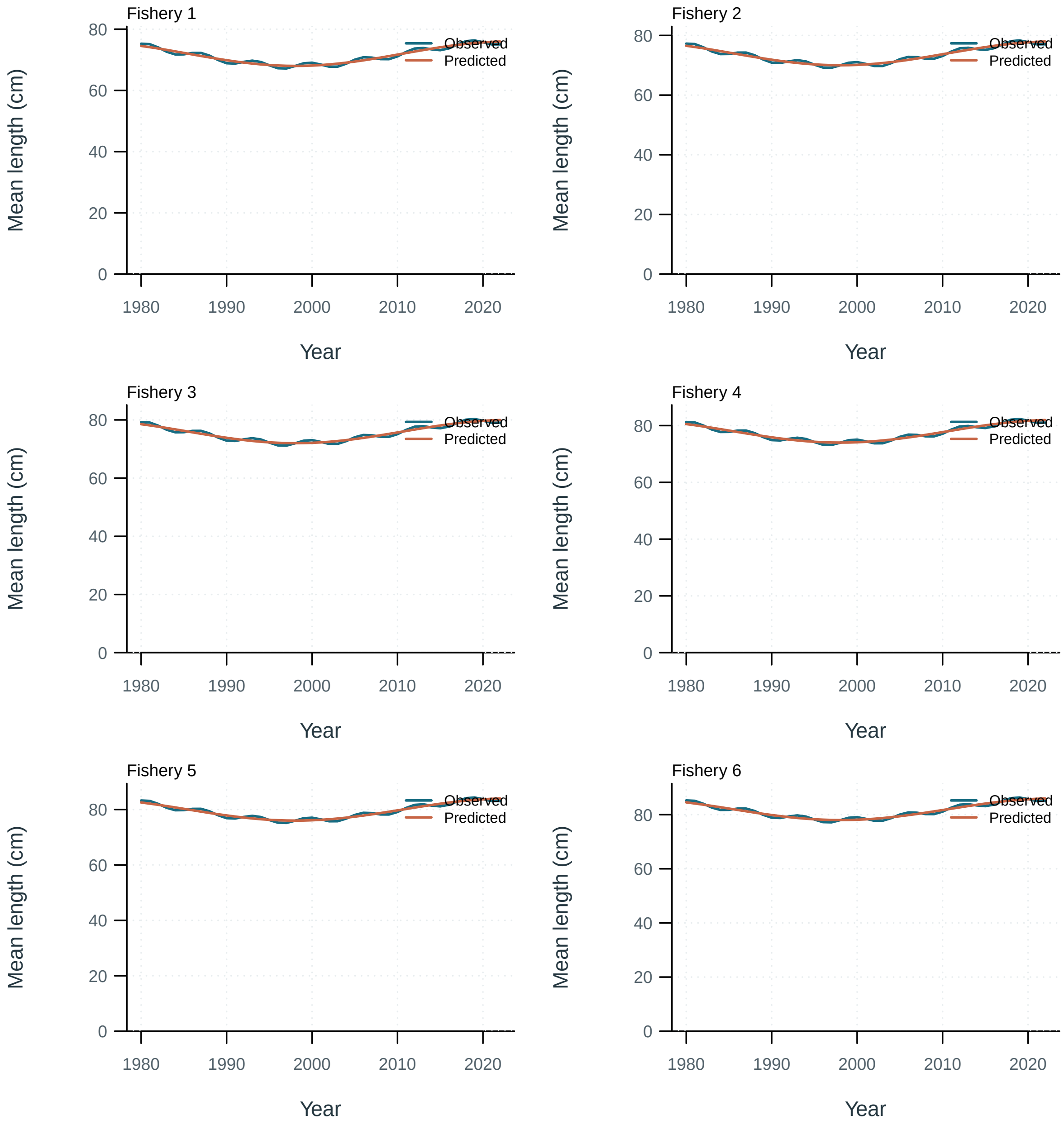

Figure 14: Observed and predicted mean lengths by longline fishery.

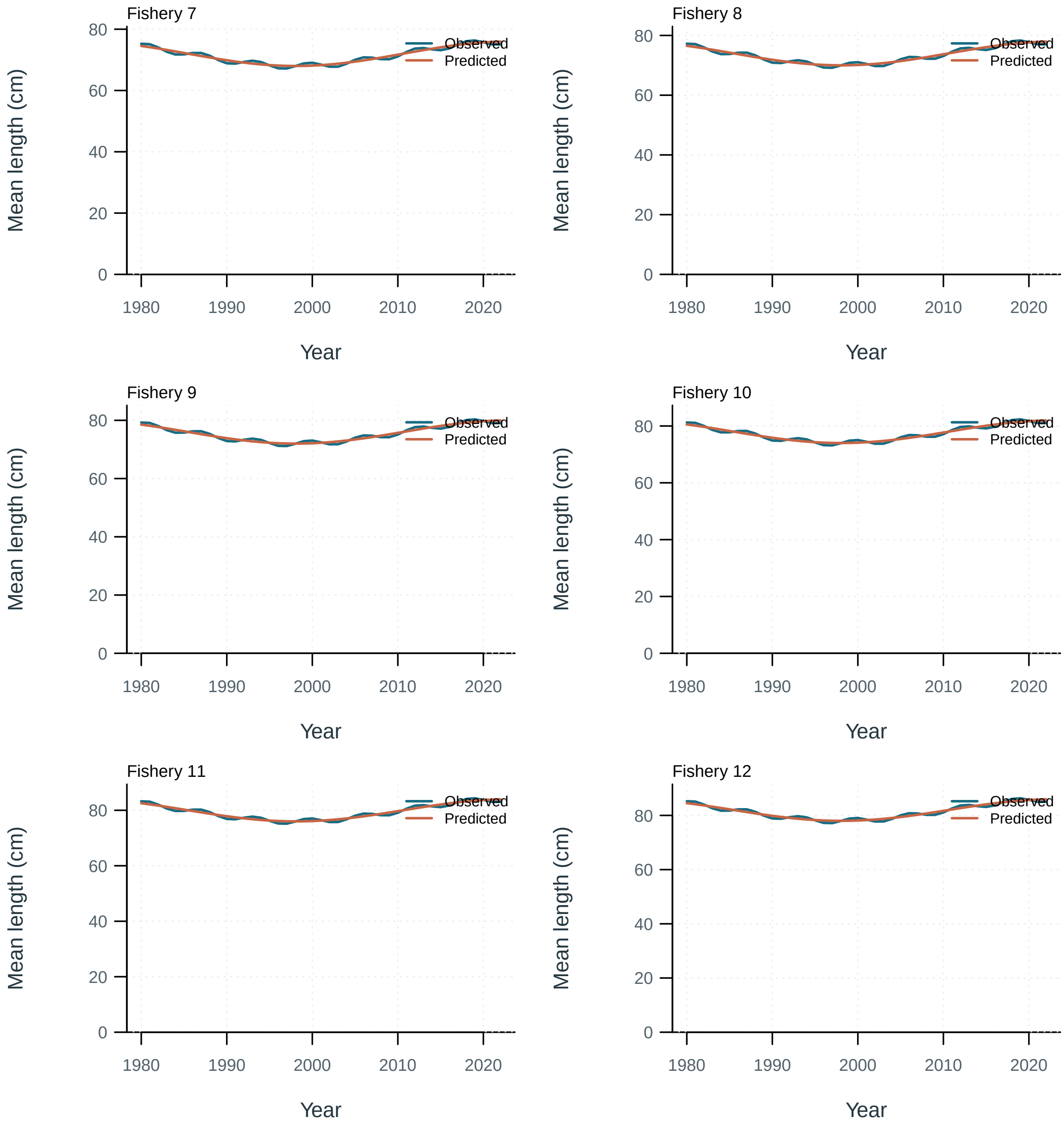

Figure 15: Observed and predicted mean lengths for other fisheries.

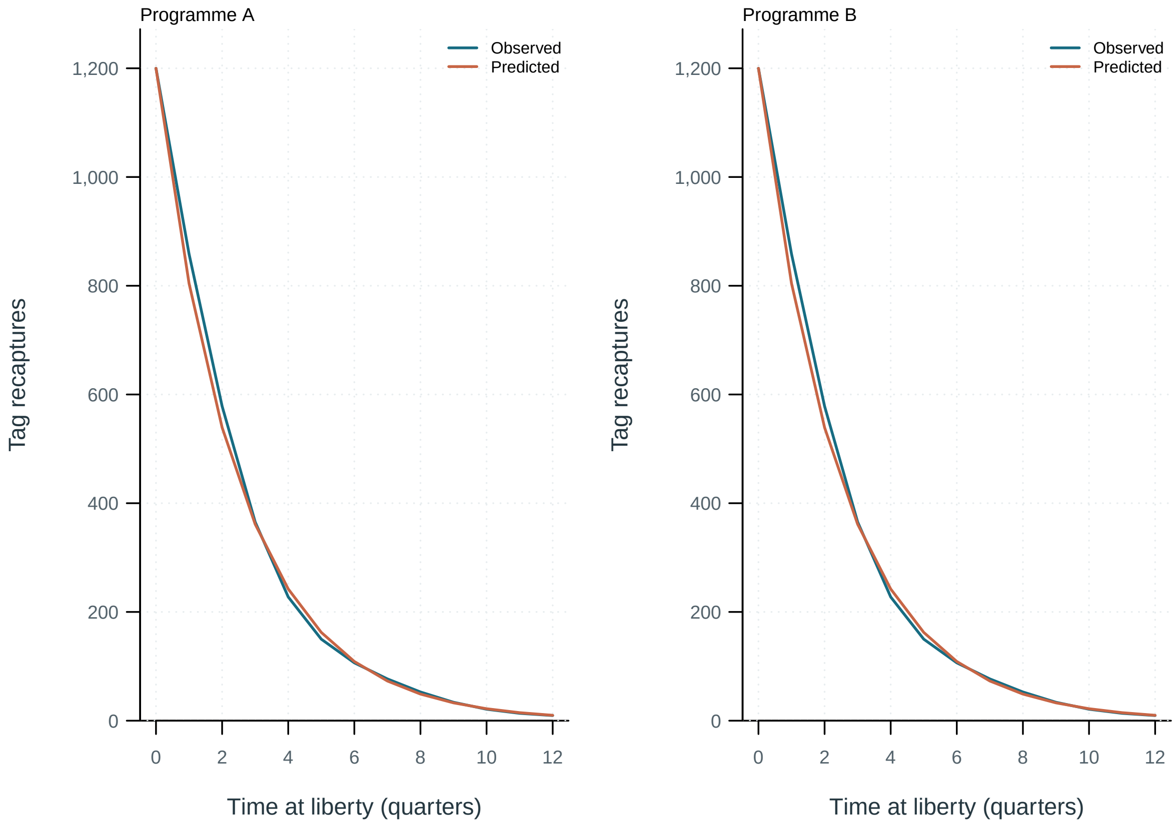

Figure 16: Observed and predicted tag recaptures by time at liberty and release programme.

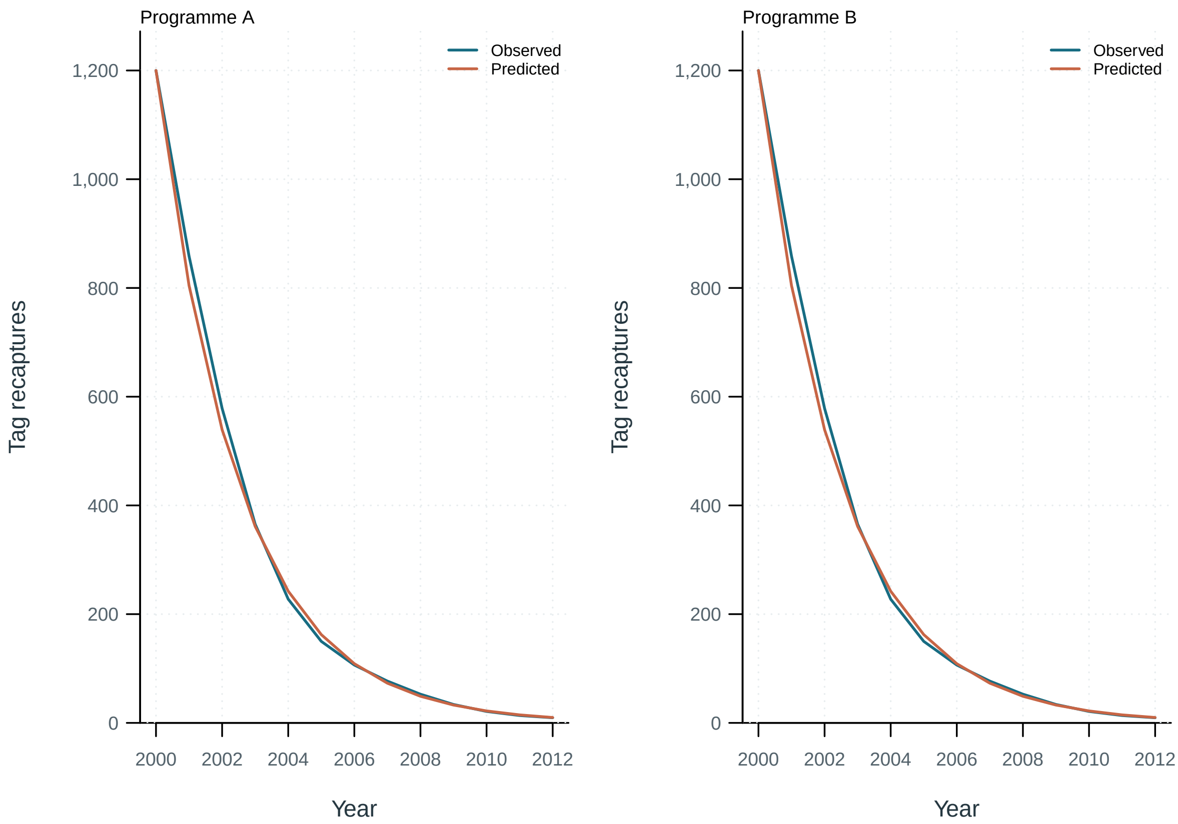

Figure 17: Observed and predicted tag recaptures by year and programme.

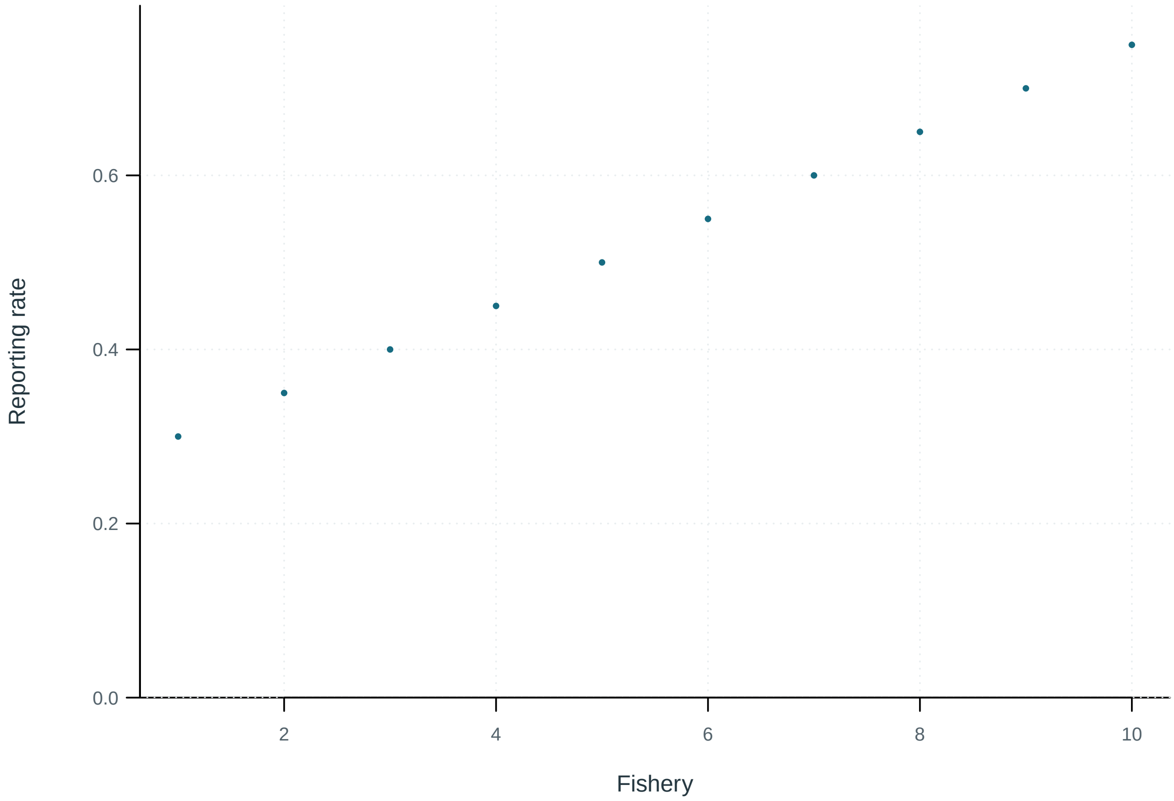

Figure 18: Tag-reporting rates by fishery.

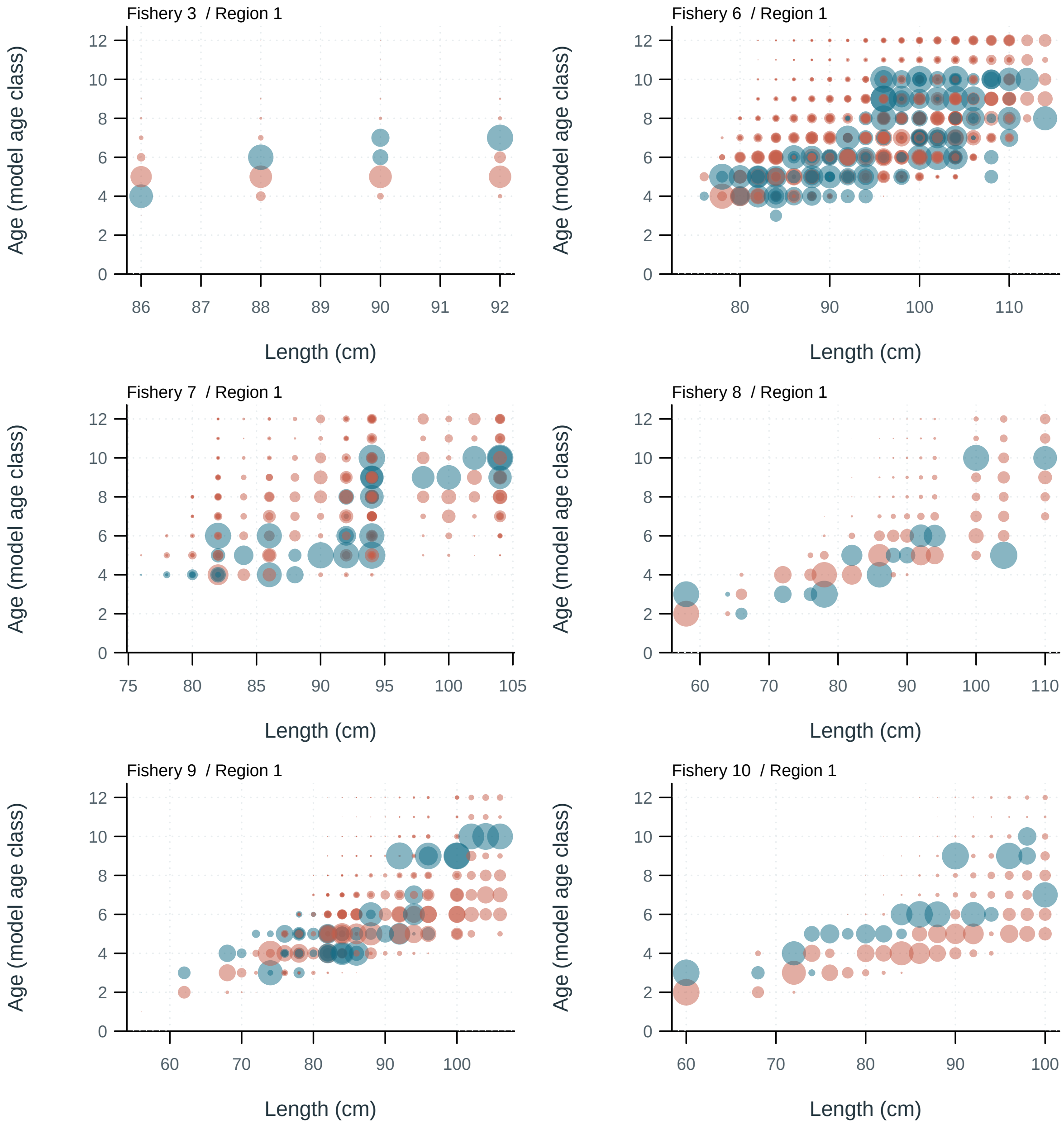

Figure 19: Conditional age-at-length composition residuals by fishery. Blue: observed exceeds predicted; red: predicted exceeds observed. Circle area is proportional to absolute residual.

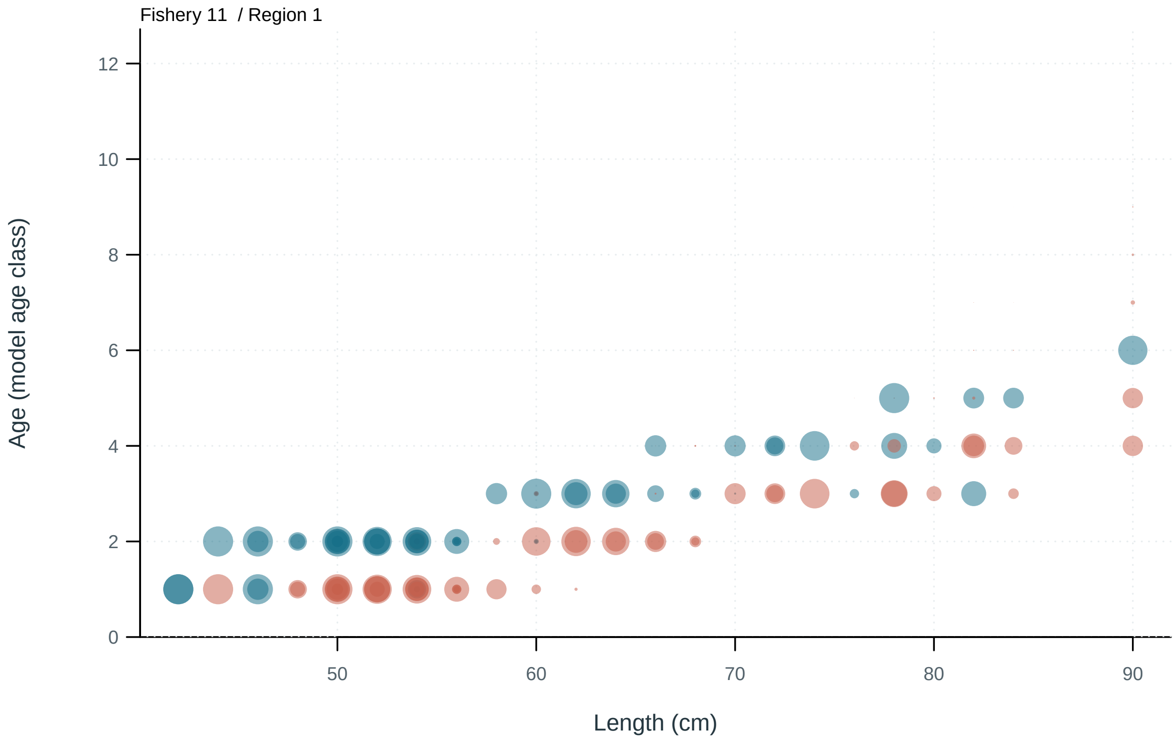

Figure 20: Conditional age-at-length composition residuals by fishery. Blue: observed exceeds predicted; red: predicted exceeds observed. Circle area is proportional to absolute residual.

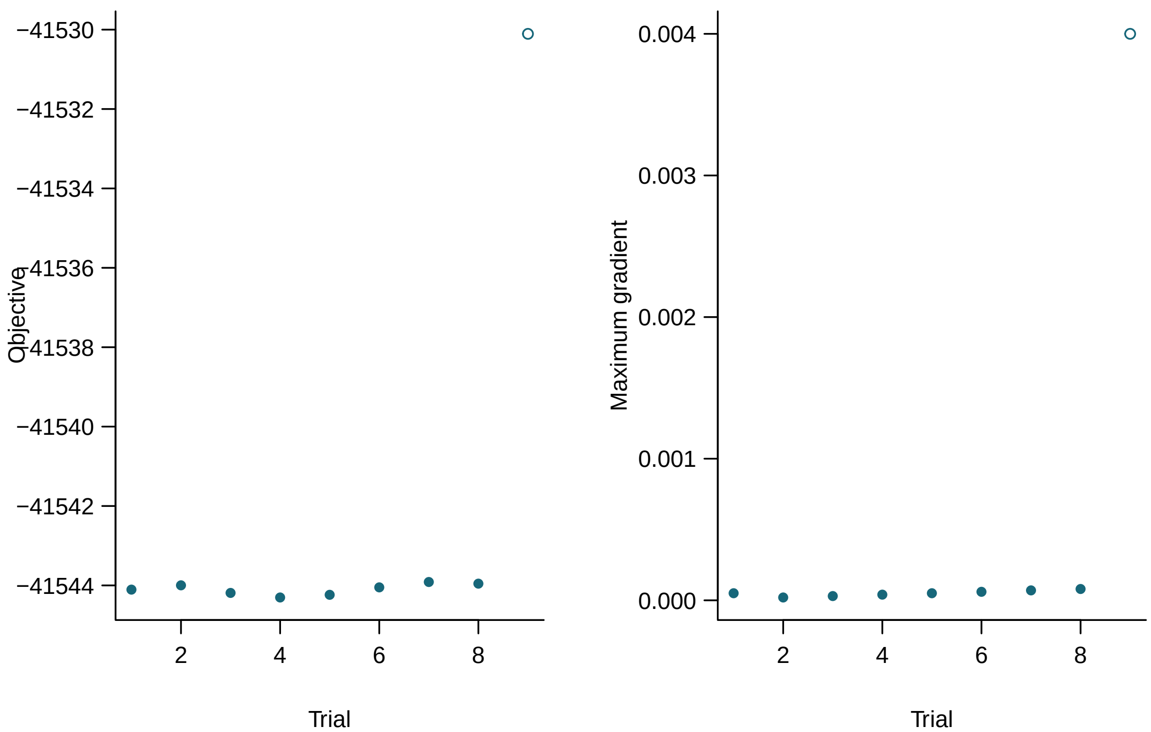

Figure 21: Final objectives and maximum gradients for independent starting values. Filled points meet all recorded numerical checks.

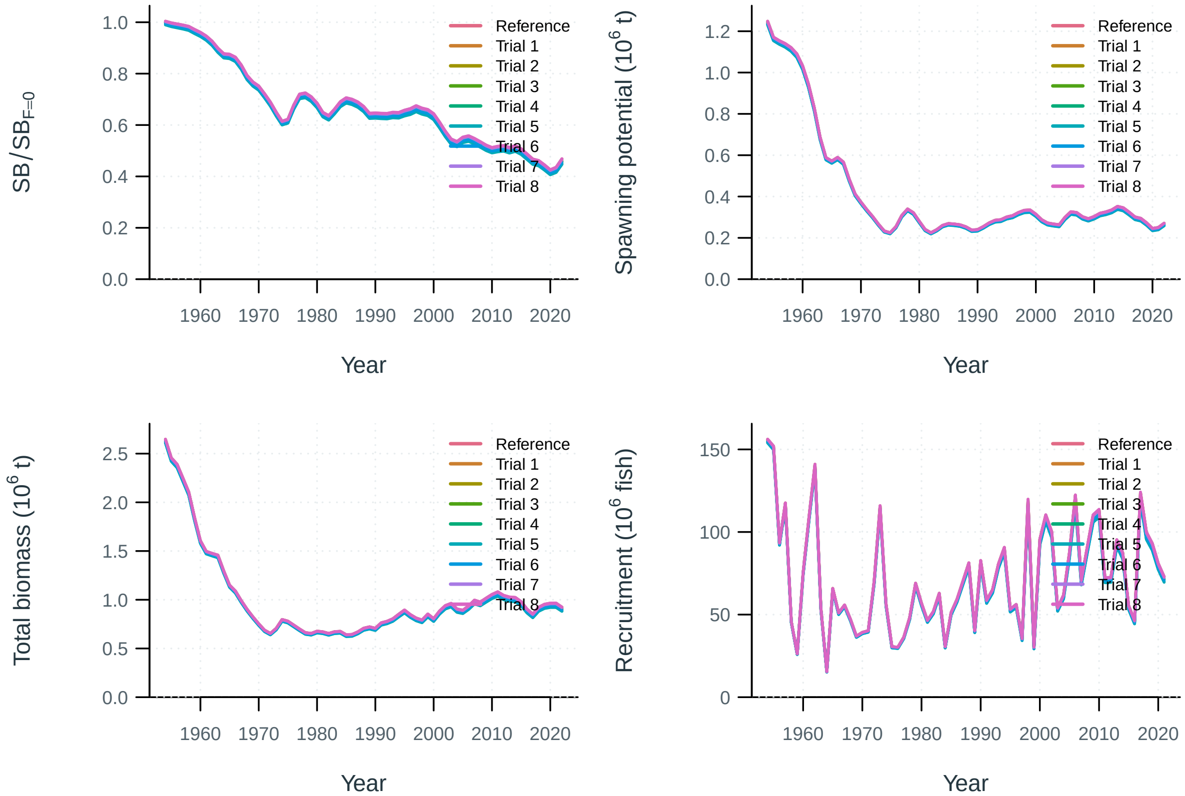

Figure 22: Jitter model trajectories.

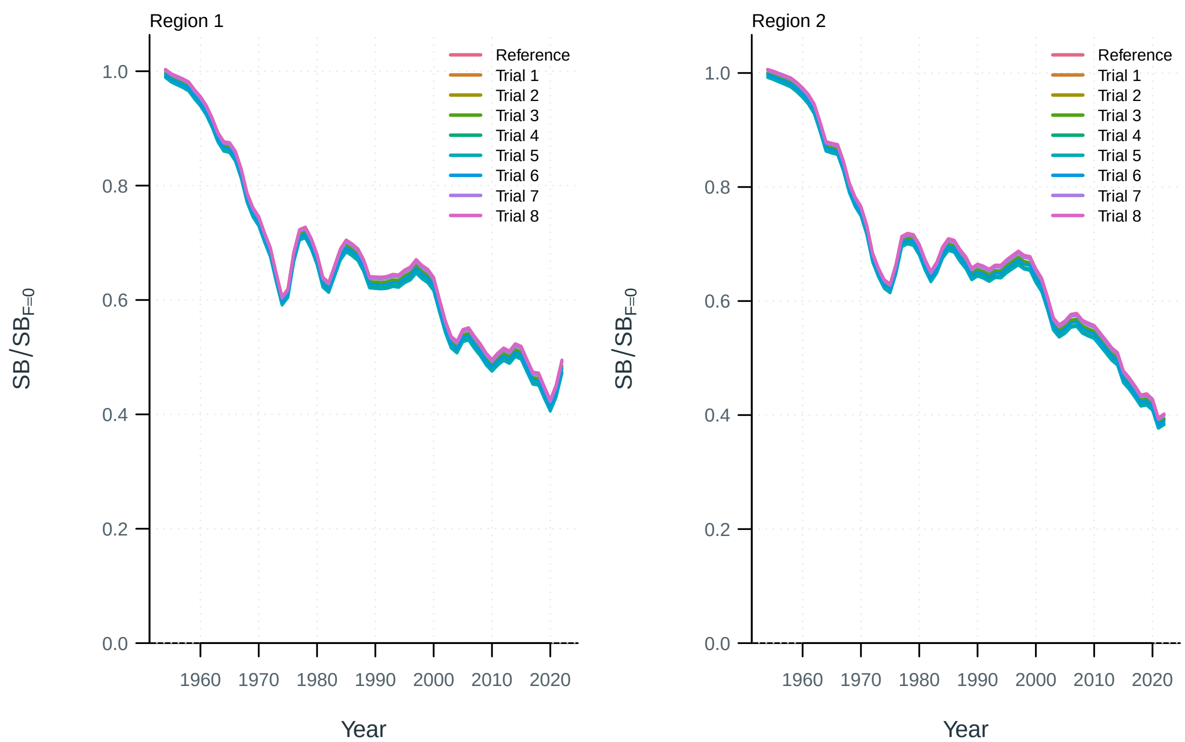

Figure 23: Spawning depletion by model and region, relative to each model's dynamic unfished reference.

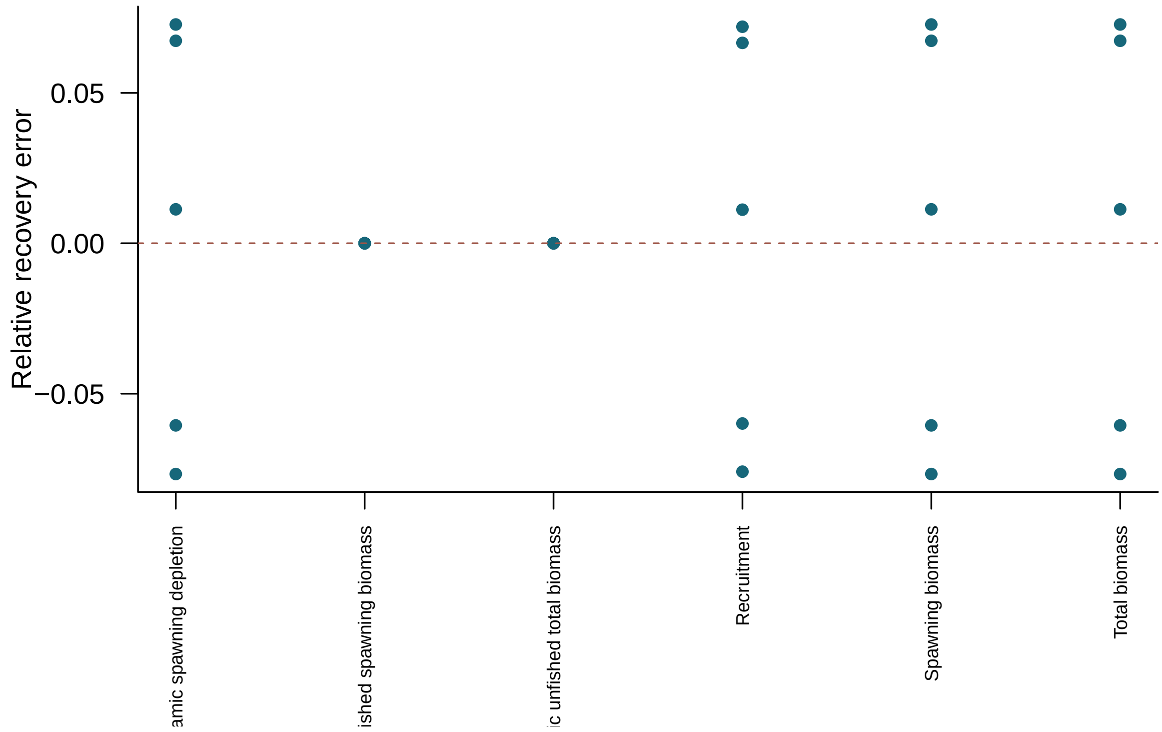

Figure 24: Relative recovery error in terminal quantities for converged simulated-data refits.

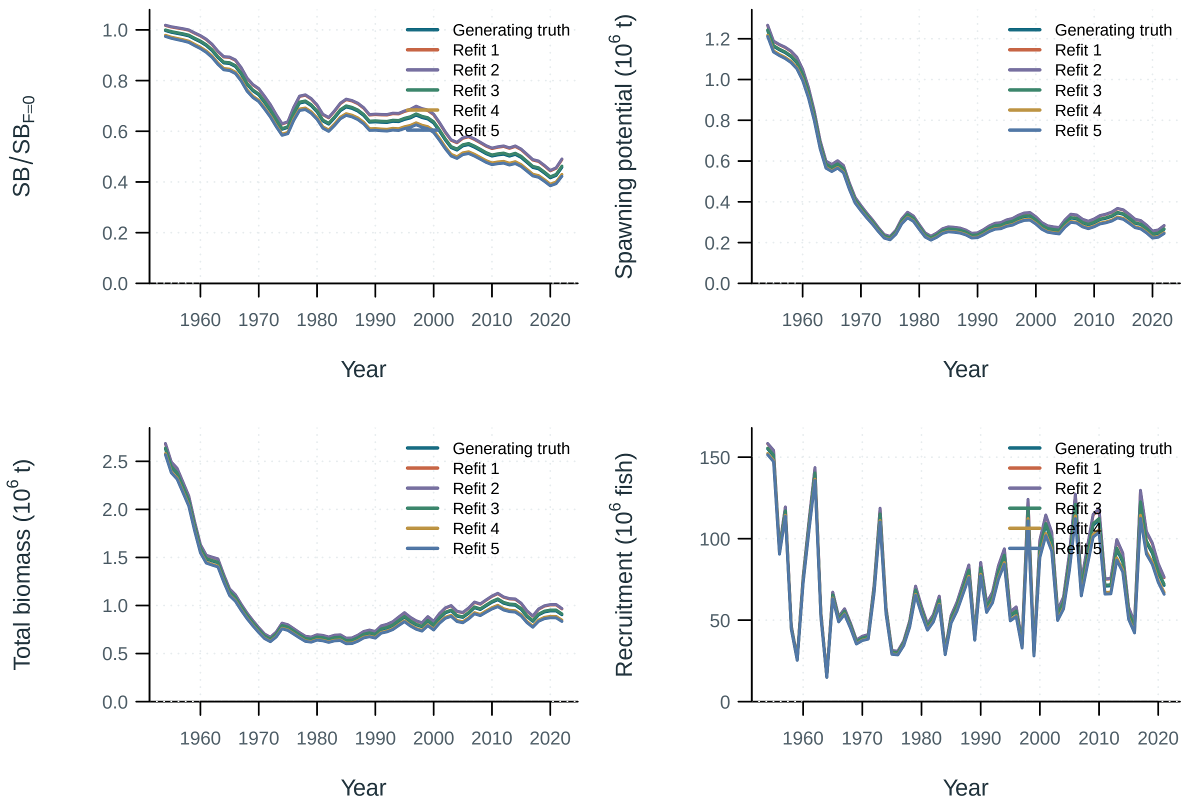

Figure 25: Selftest model trajectories.

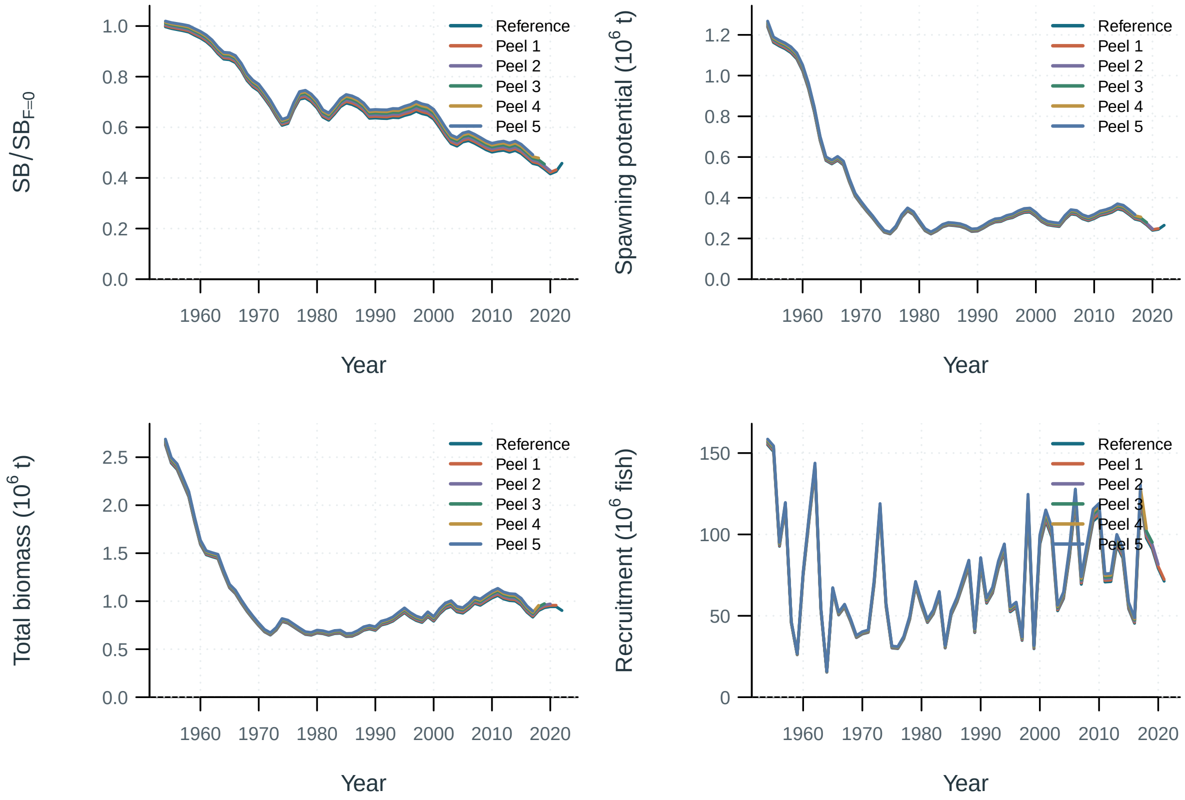

Figure 26: Retrospective model trajectories.

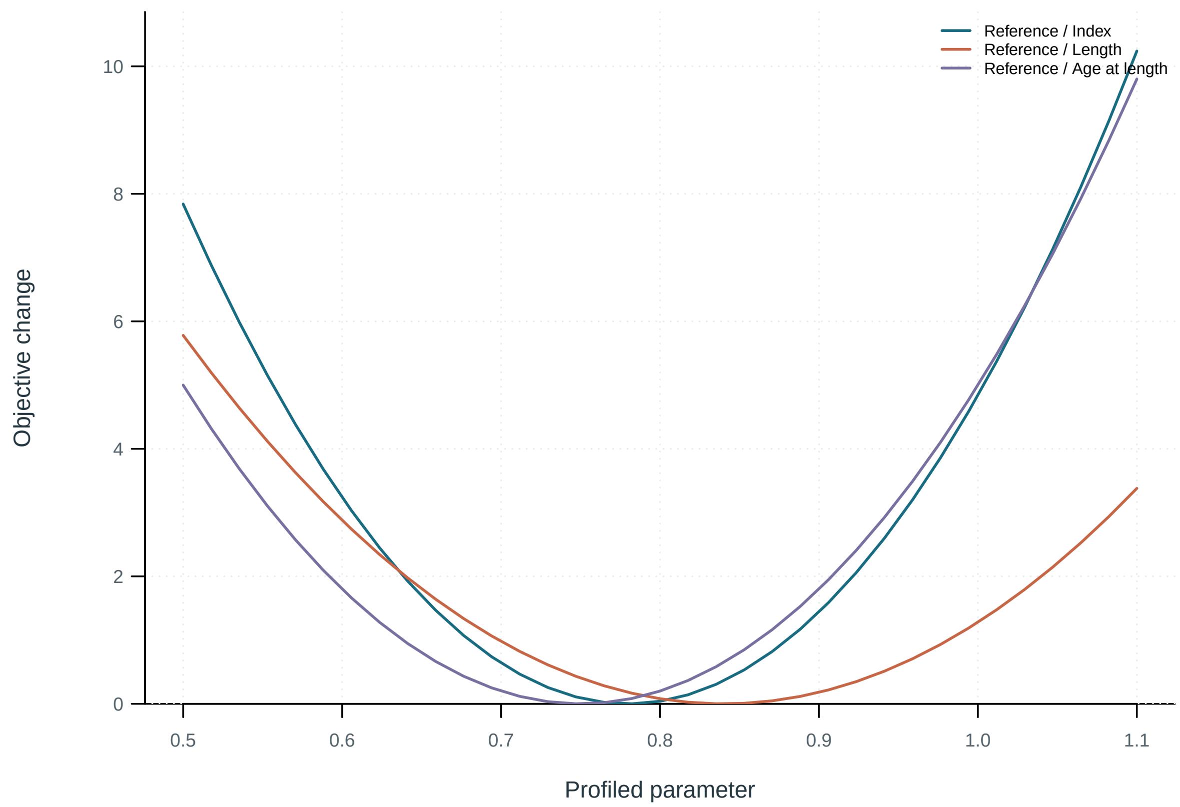

Figure 27: Objective profiles by data component.

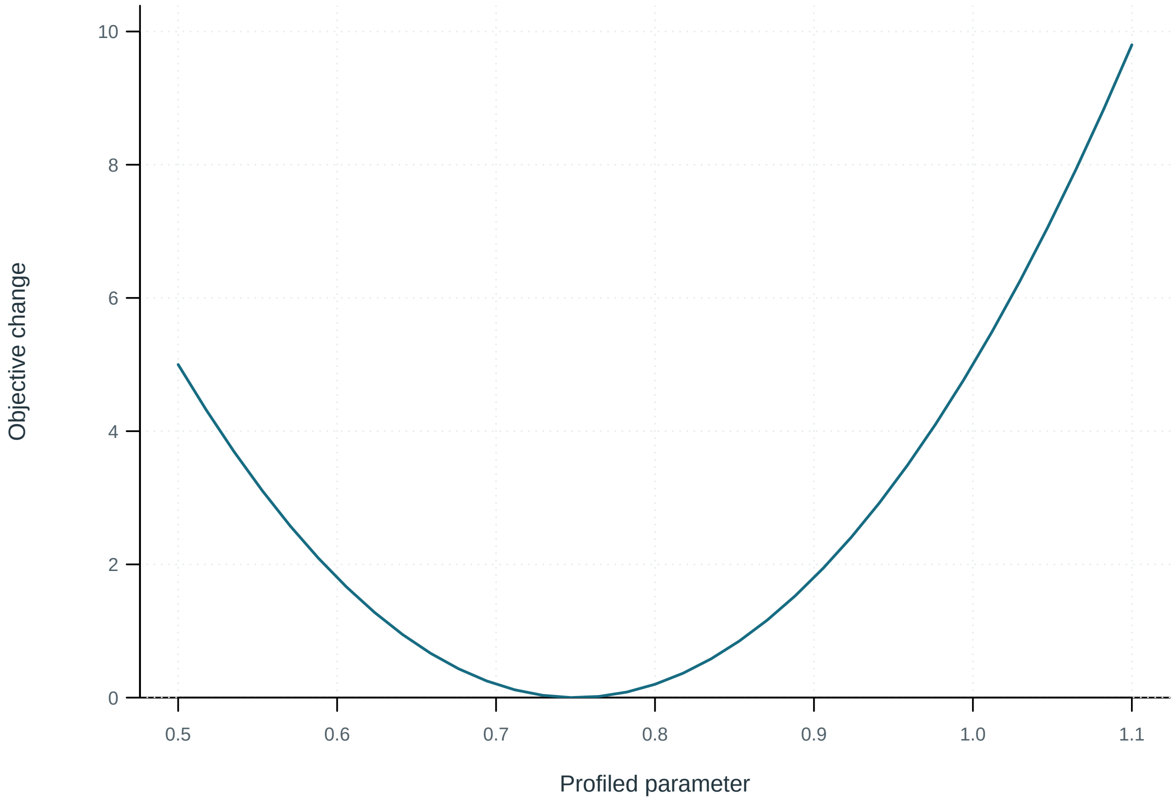

Figure 28: Index component of the objective profile.

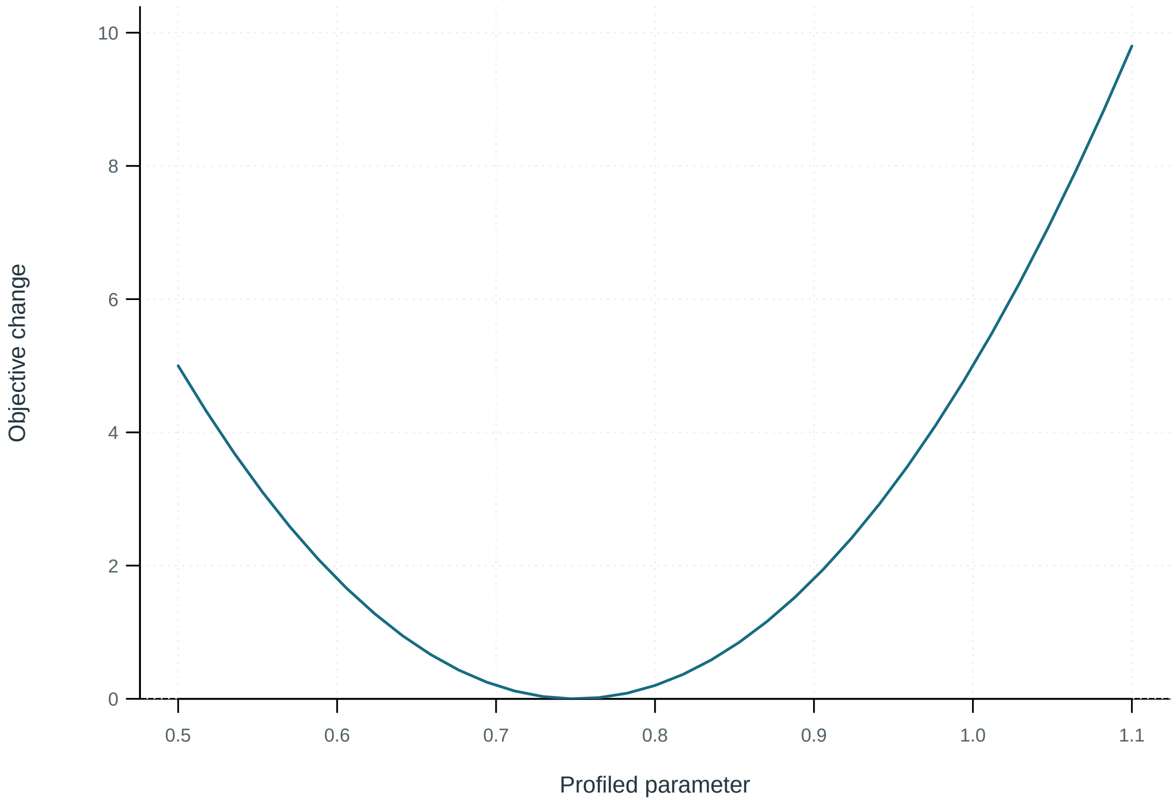

Figure 29: Length component of the objective profile.

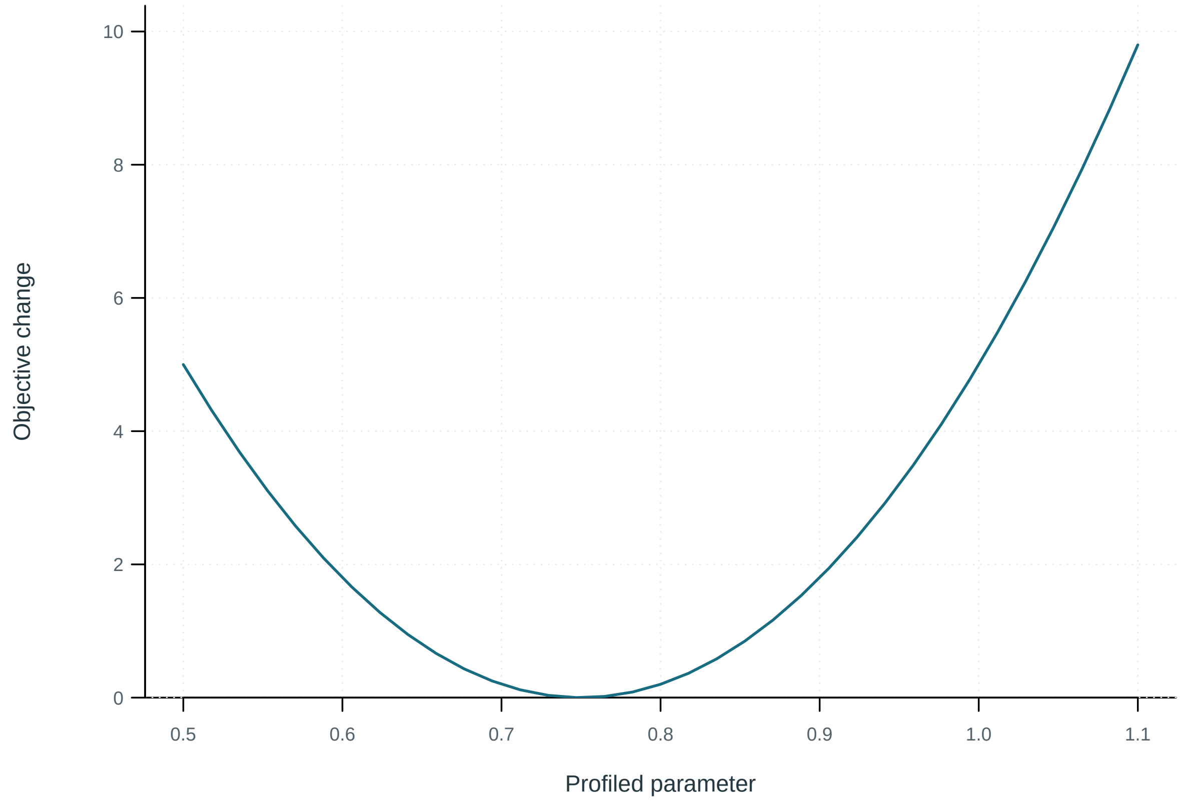

Figure 30: Age at length component of the objective profile.

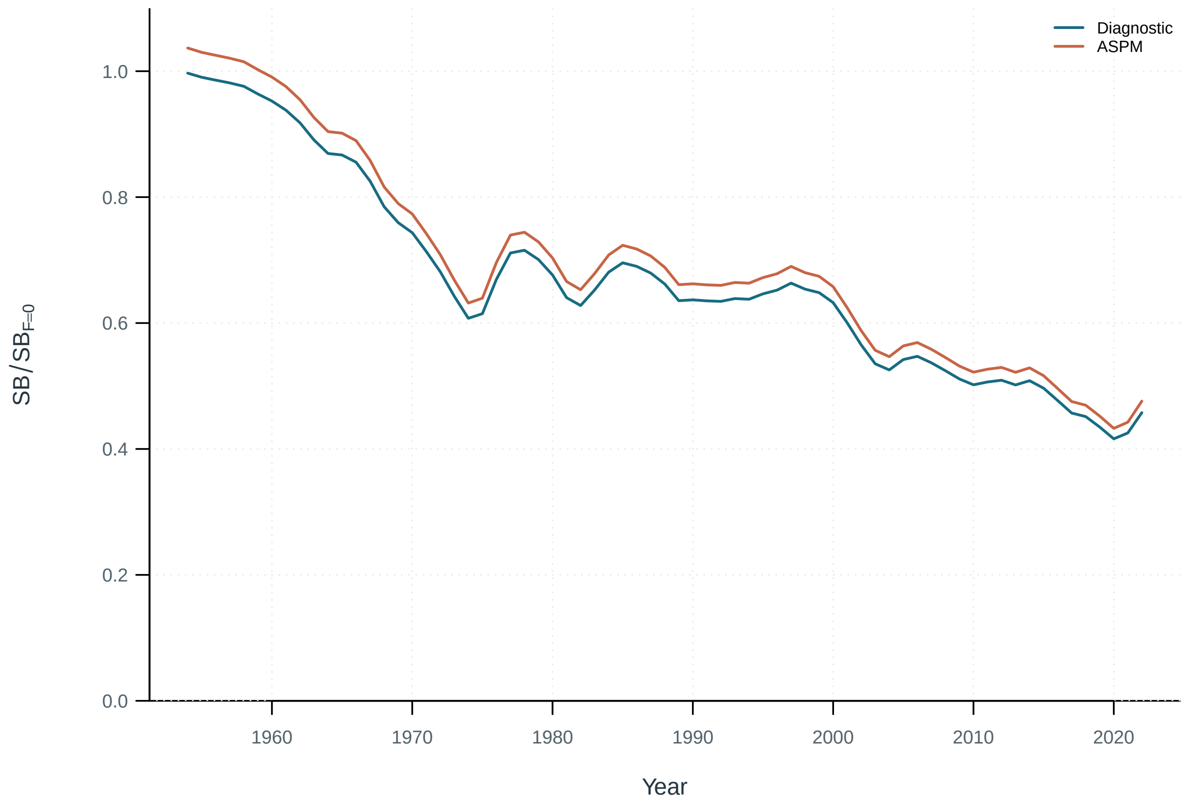

Figure 31: Spawning depletion from the diagnostic and age-structured production models.

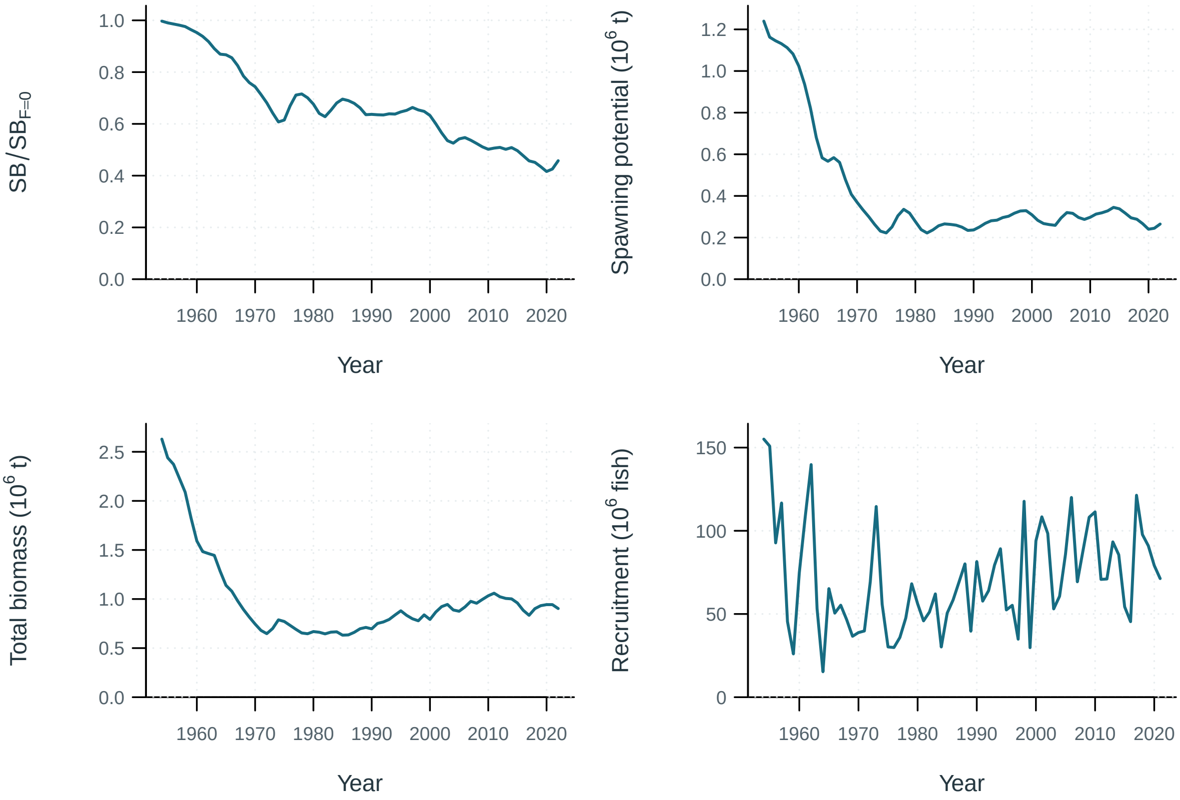

Figure 32: Estimated population trajectories.

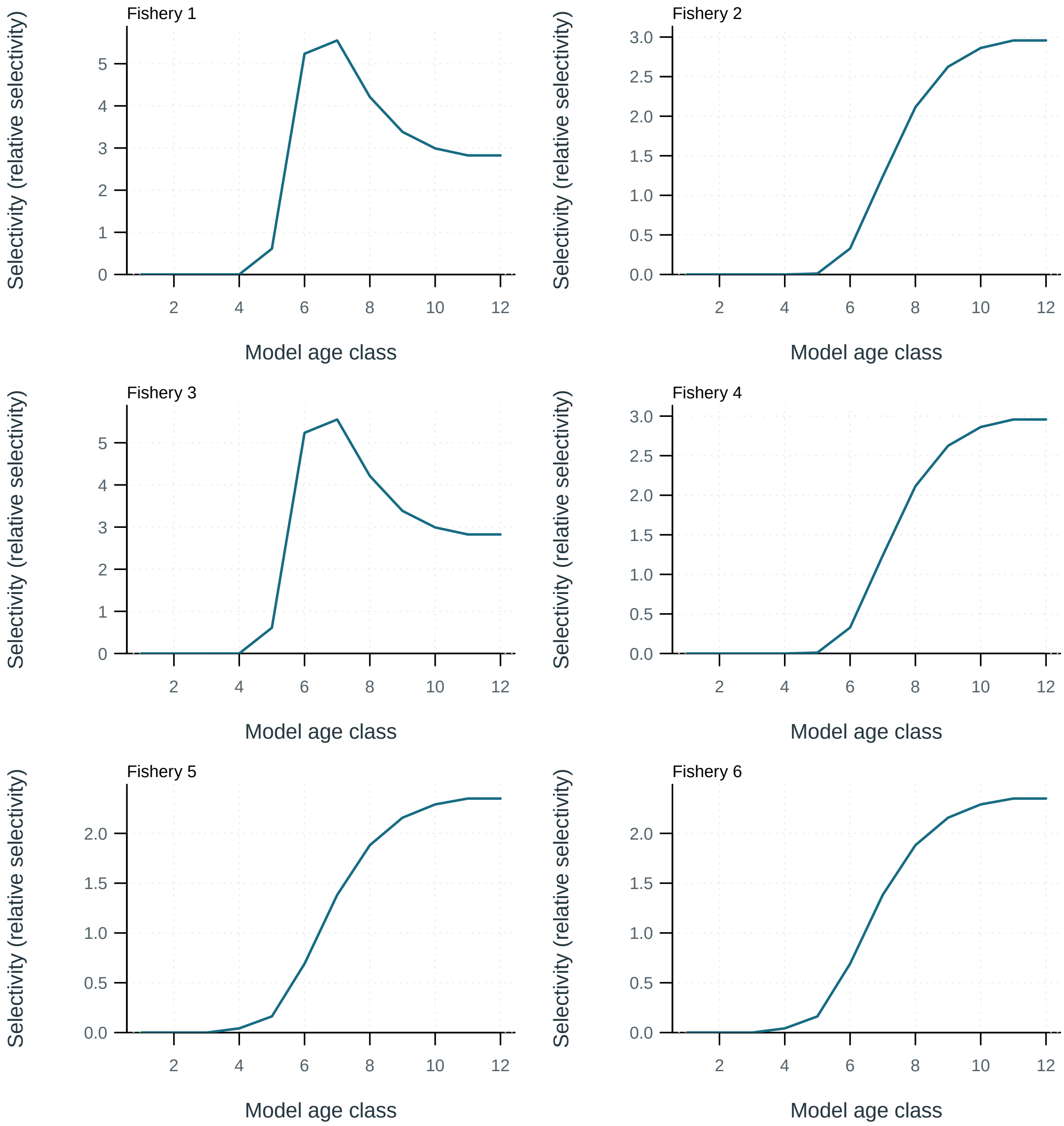

Figure 33: Fishery selectivity at age.

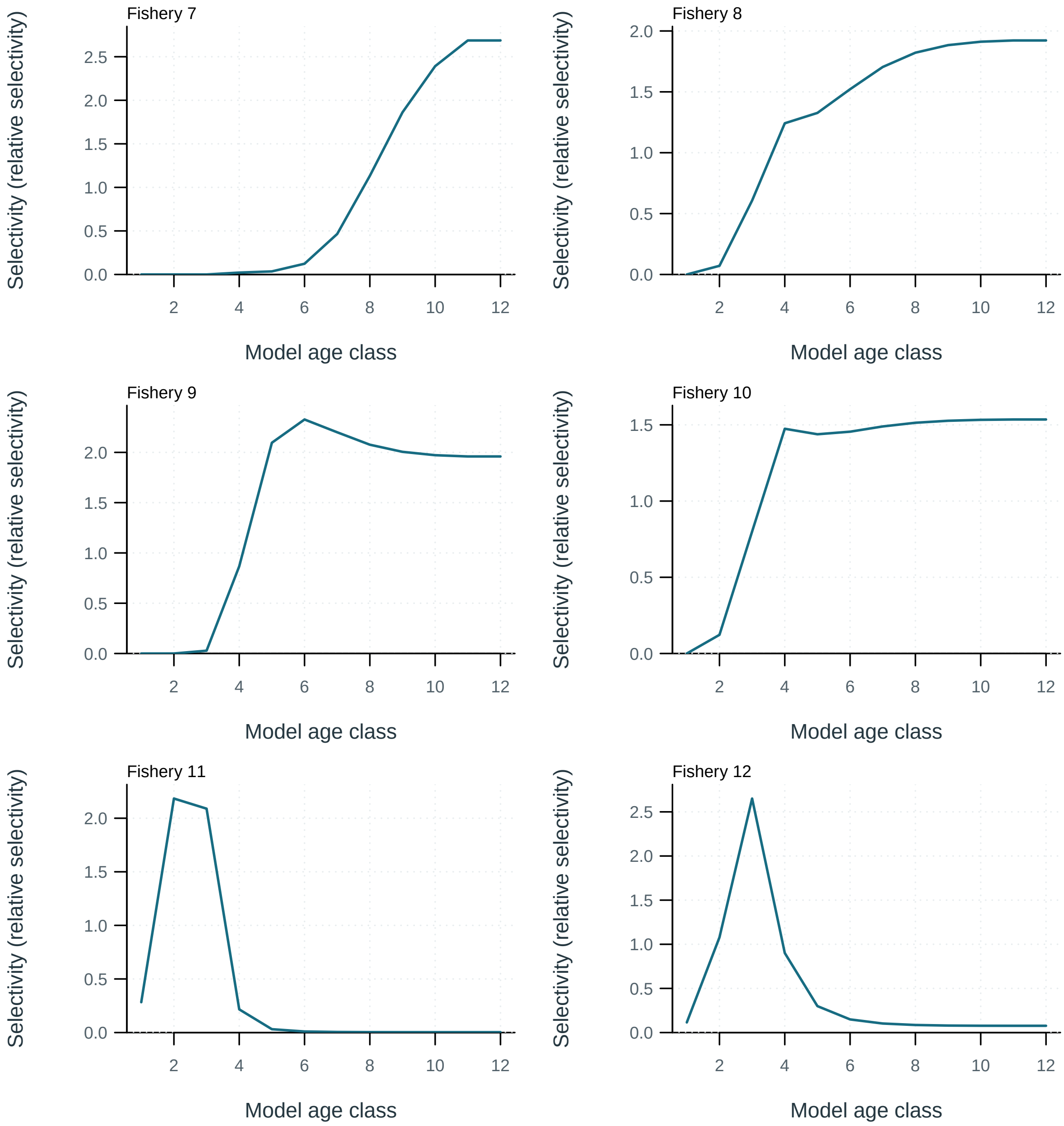

Figure 34: Fishery selectivity at age.

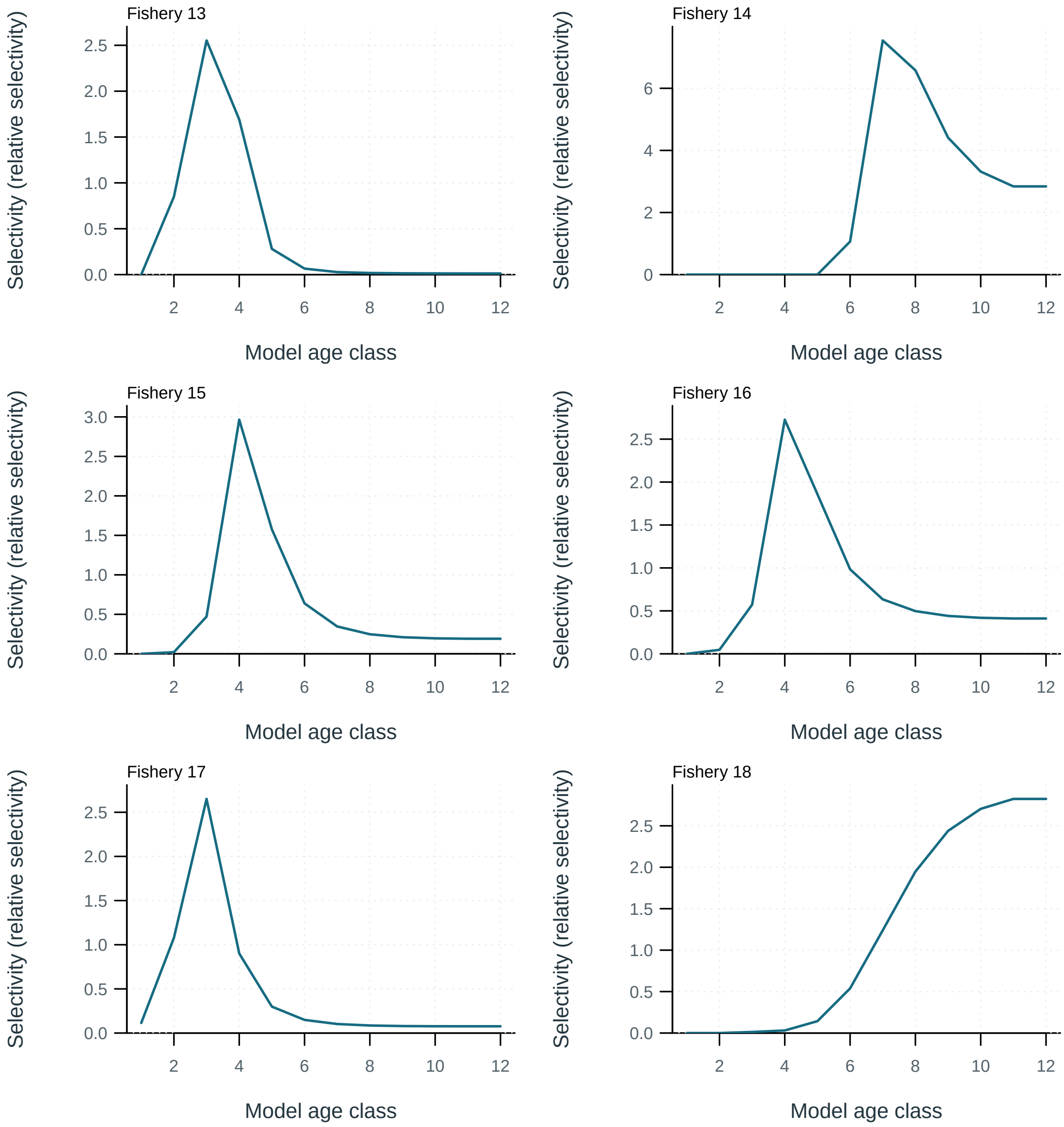

Figure 35: Fishery selectivity at age.

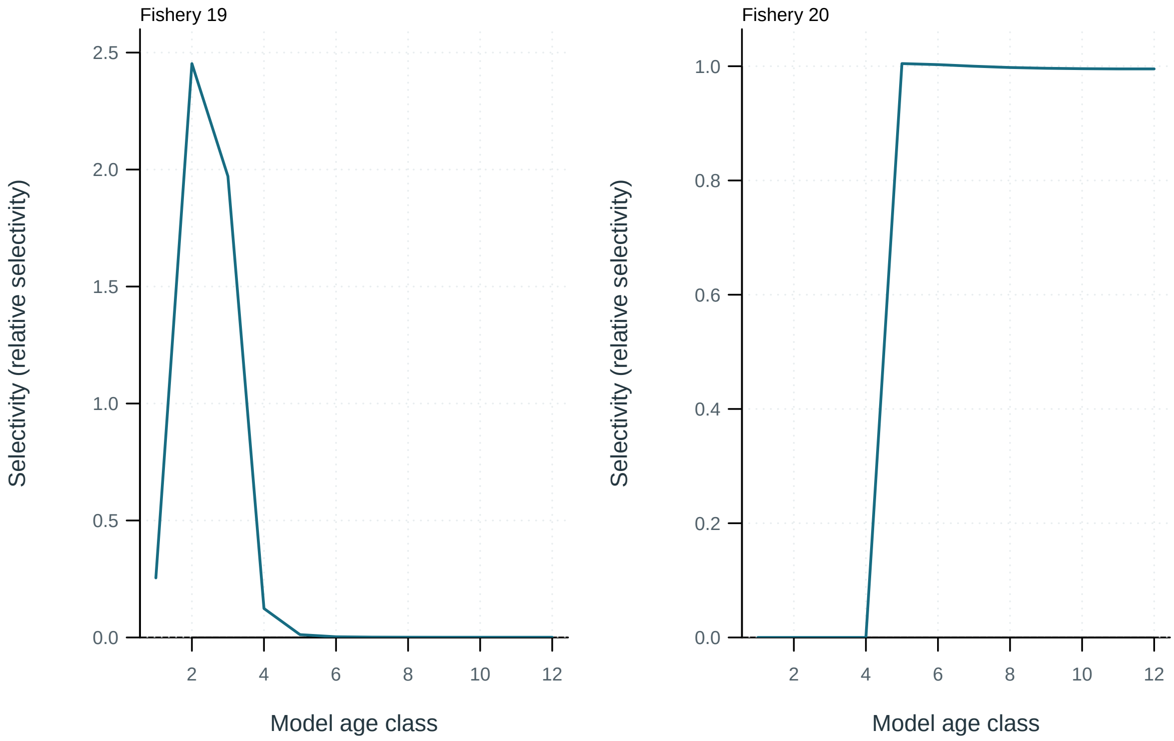

Figure 36: Fishery selectivity at age.

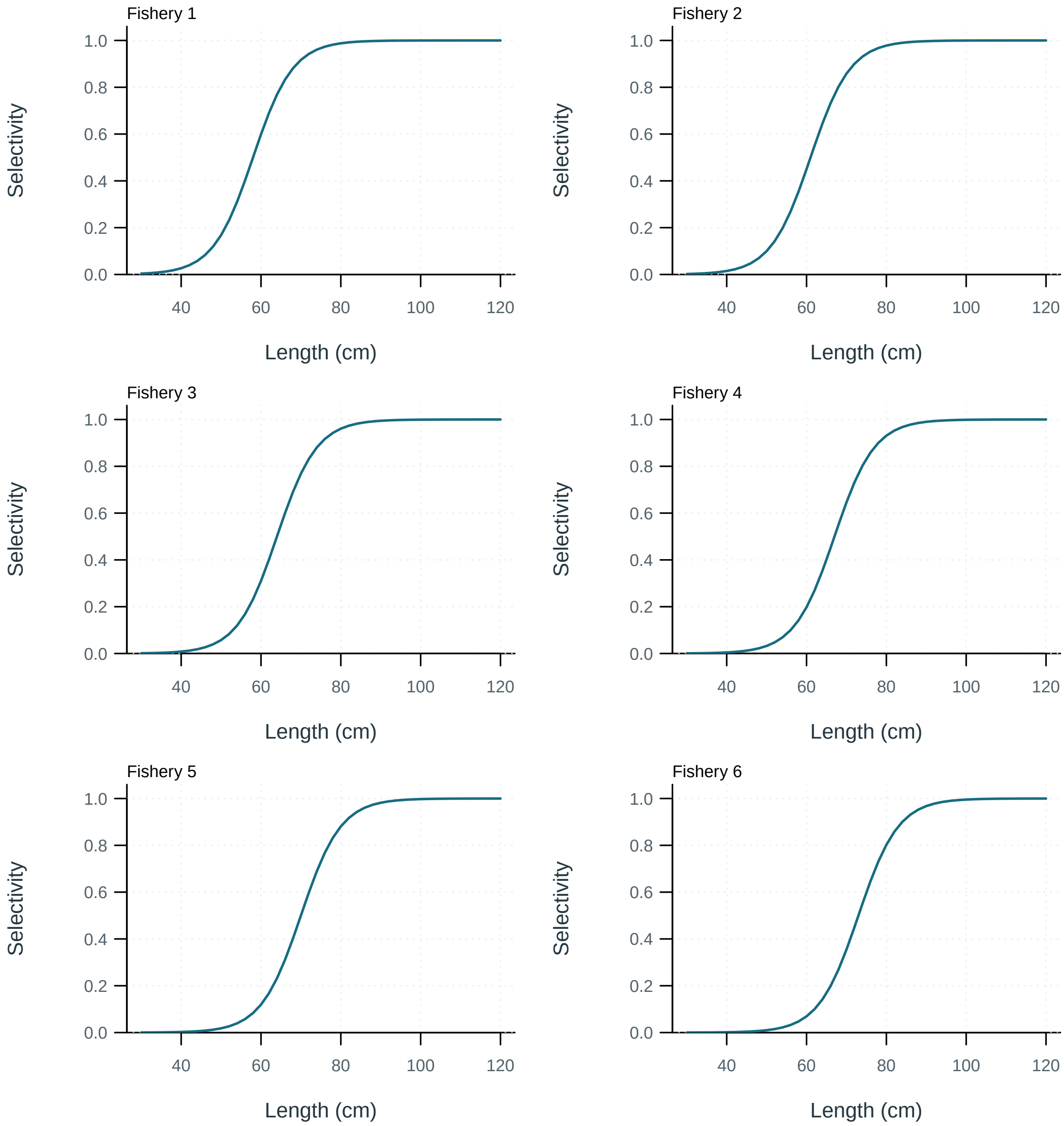

Figure 37: Selectivity at length by fishery.

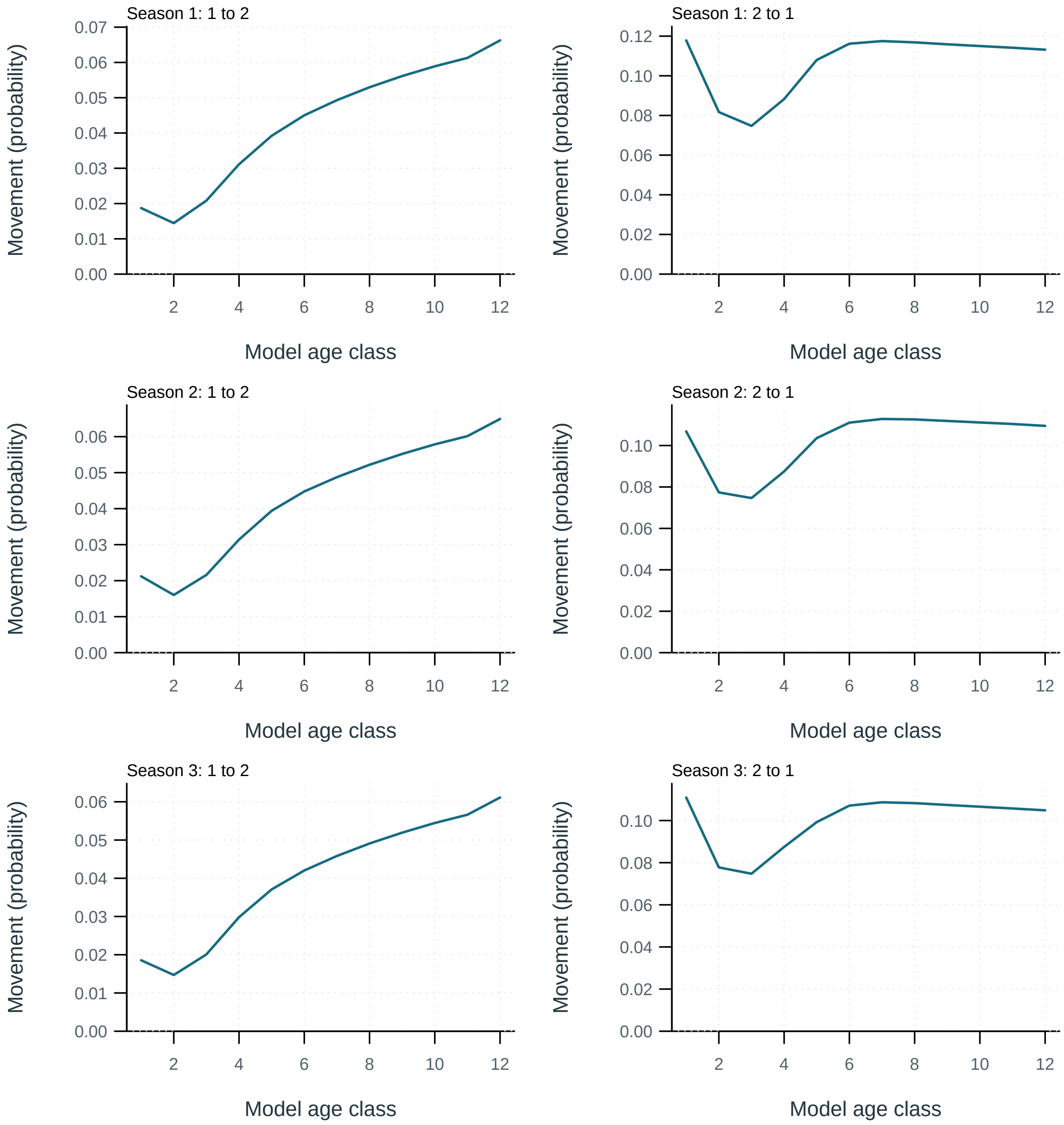

Figure 38: Probability of movement from source to destination region per movement event.

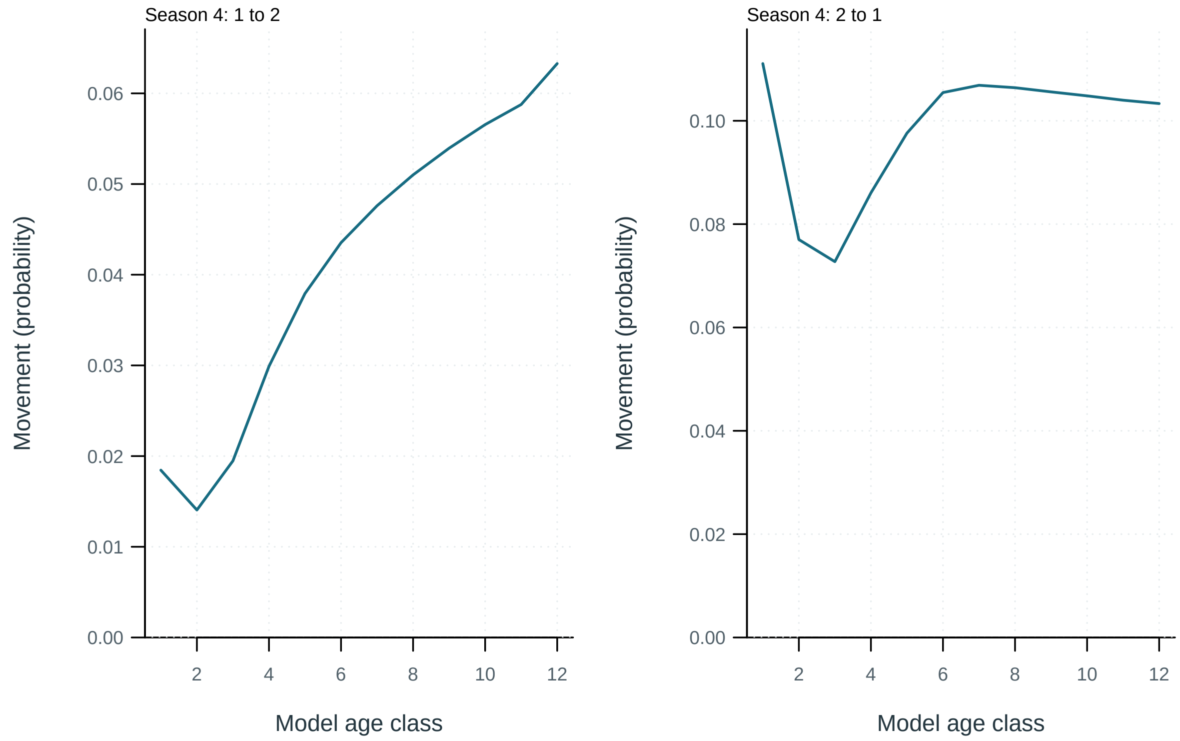

Figure 39: Probability of movement from source to destination region per movement event.

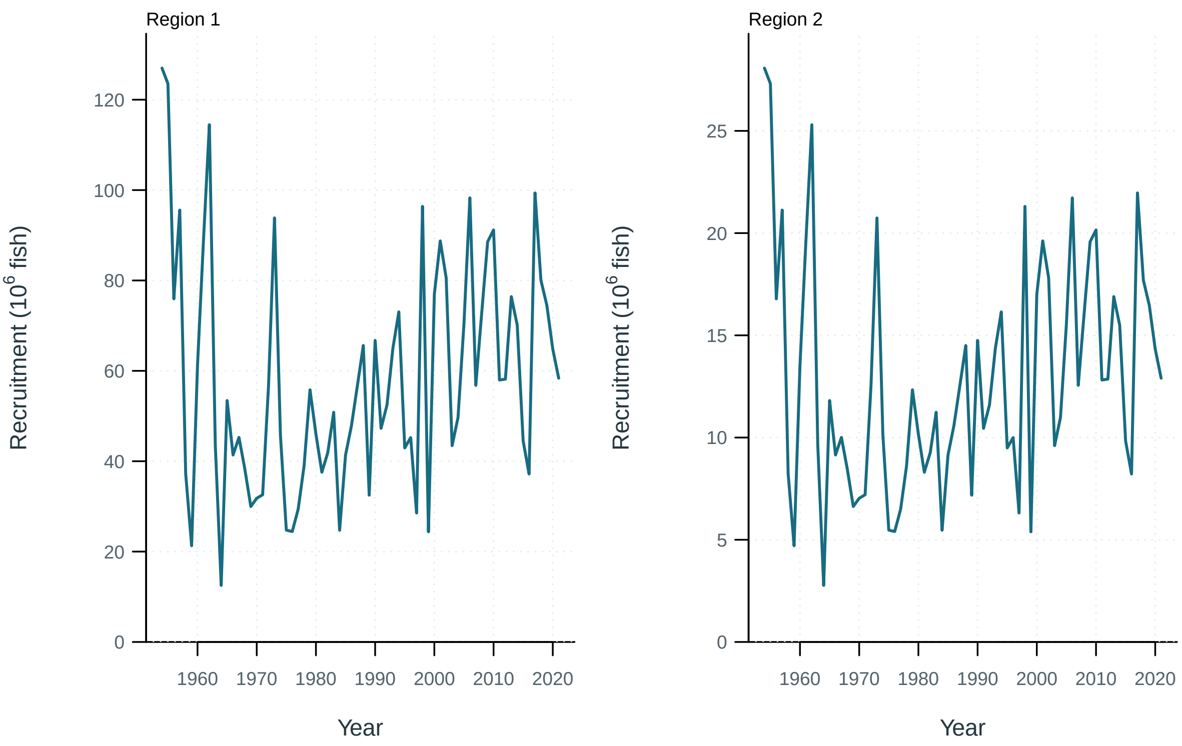

Figure 40: Recruitment by model region.

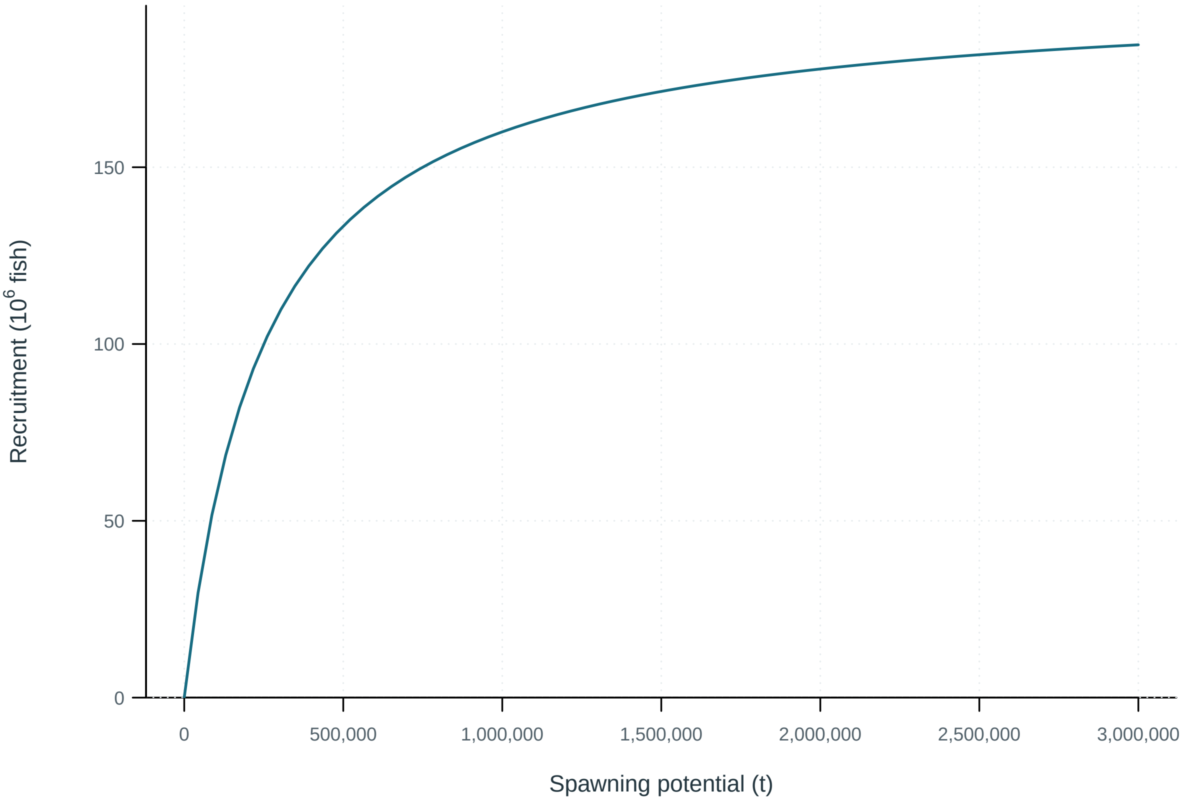

Figure 41: Stock–recruitment relationship.

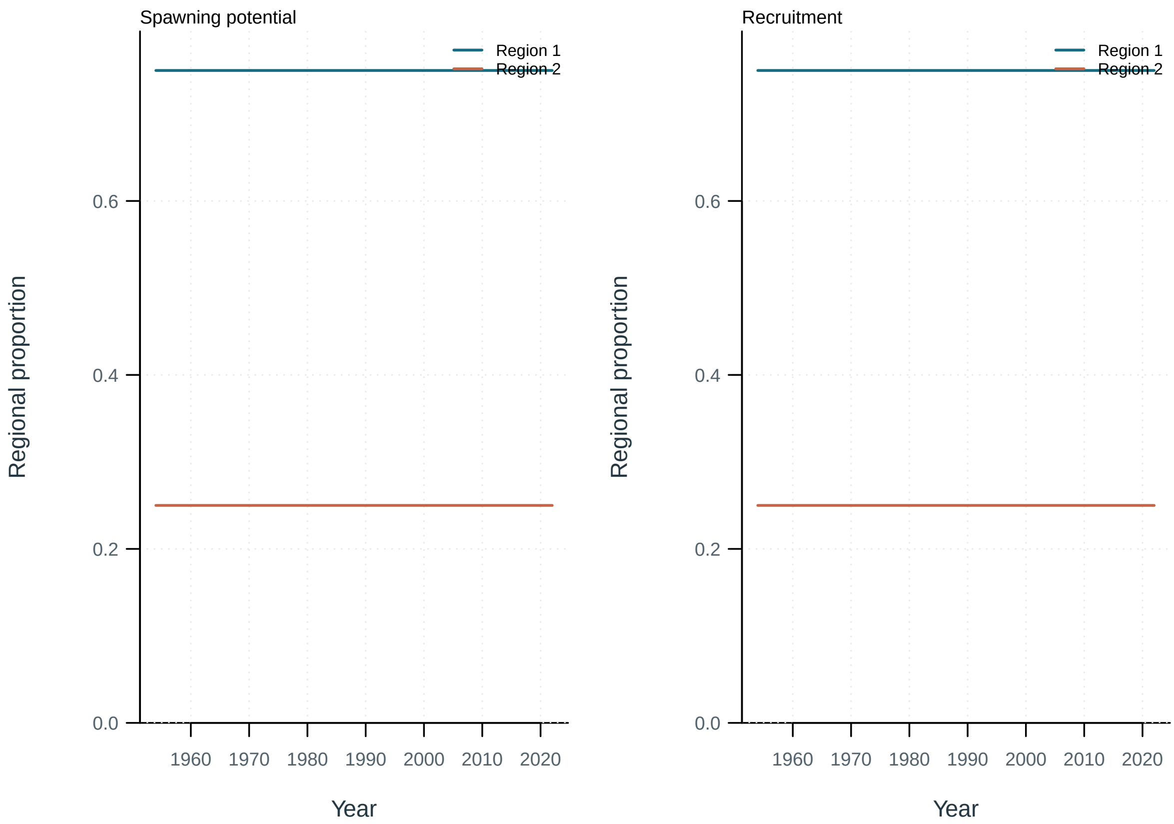

Figure 42: Regional proportions of spawning potential and recruitment.

Figure 43: Total biomass under estimated fishing and the dynamic no-fishing counterfactual.

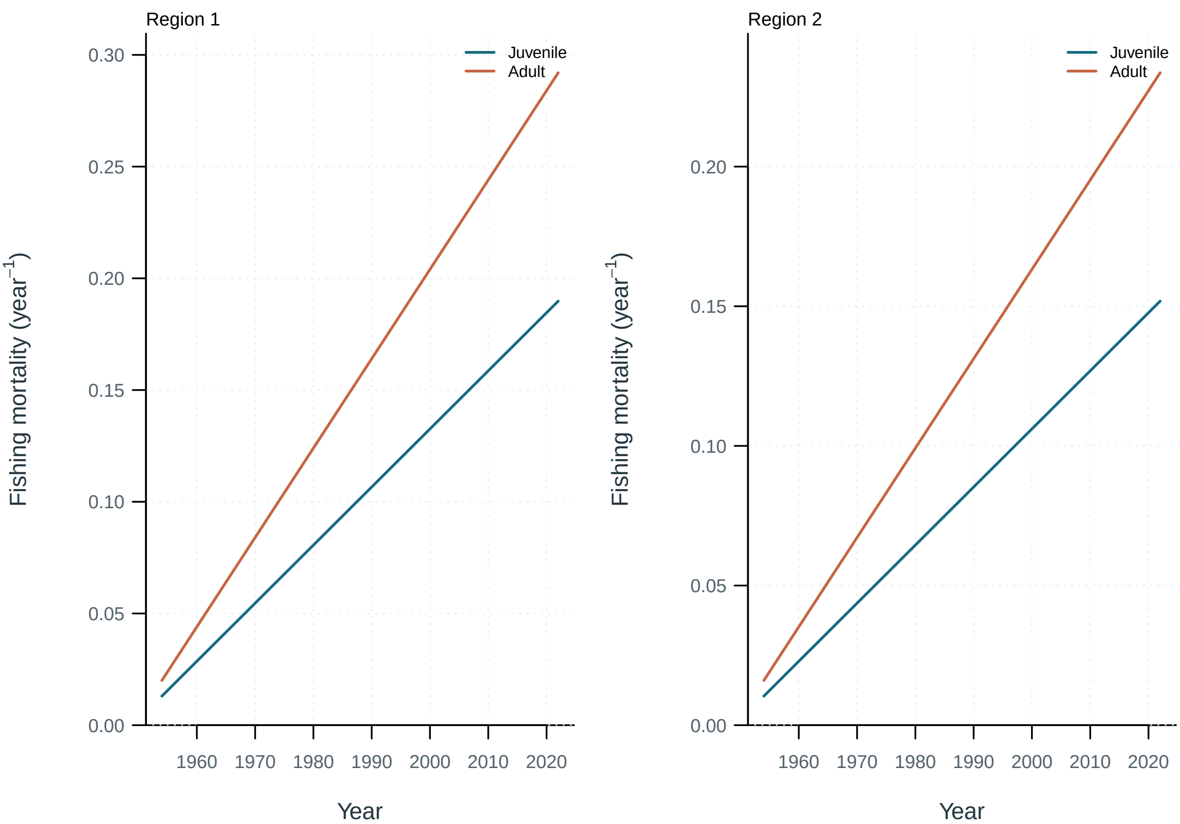

Figure 44: Annual juvenile and adult fishing mortality by region.

Figure 45: Spawning-depletion sensitivity to steepness.

Figure 46: Spawning-depletion sensitivity to natural mortality.

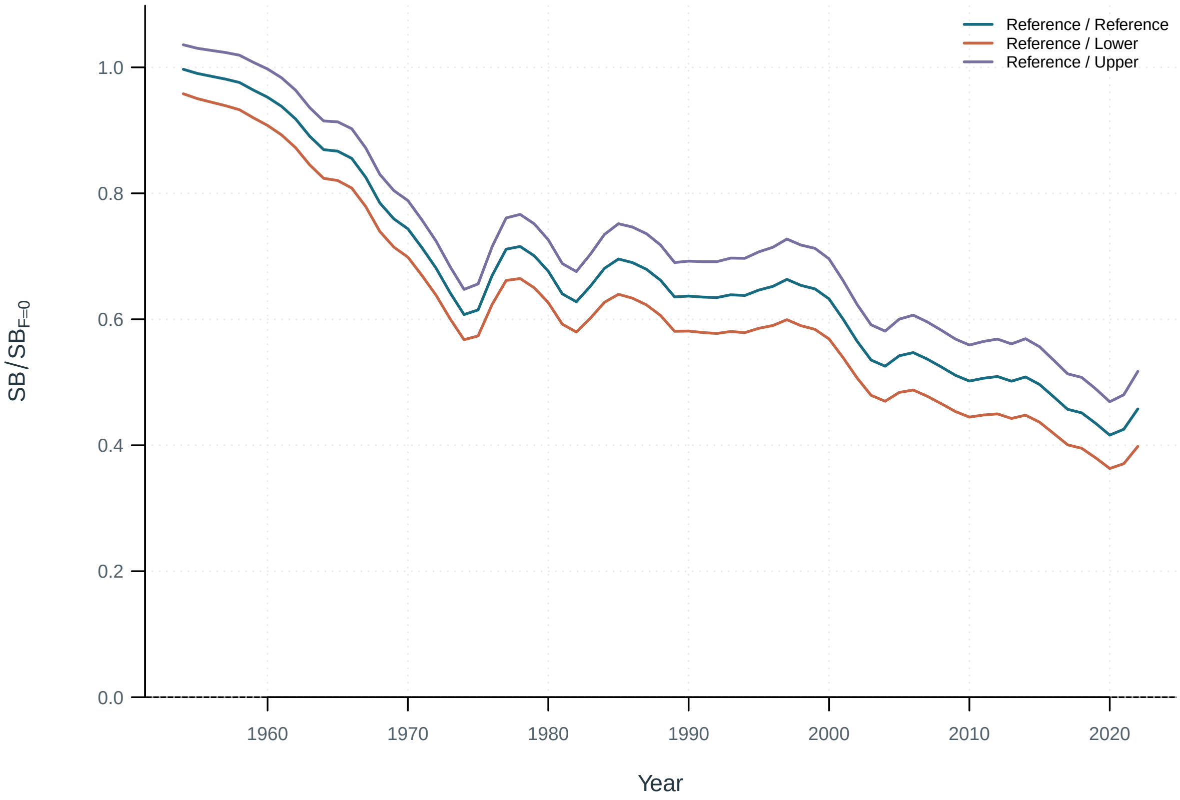

Figure 47: Spawning-depletion sensitivity to CAAL.

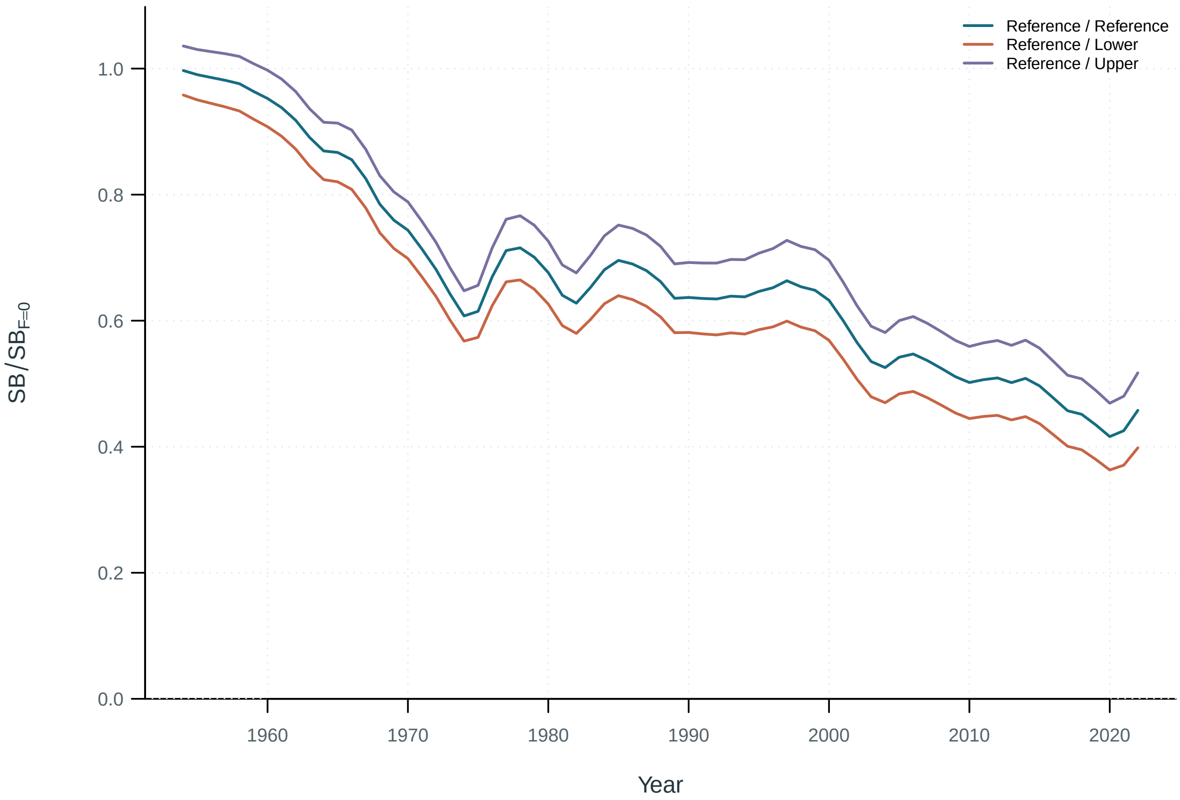

Figure 48: Spawning-depletion sensitivity to effort creep.

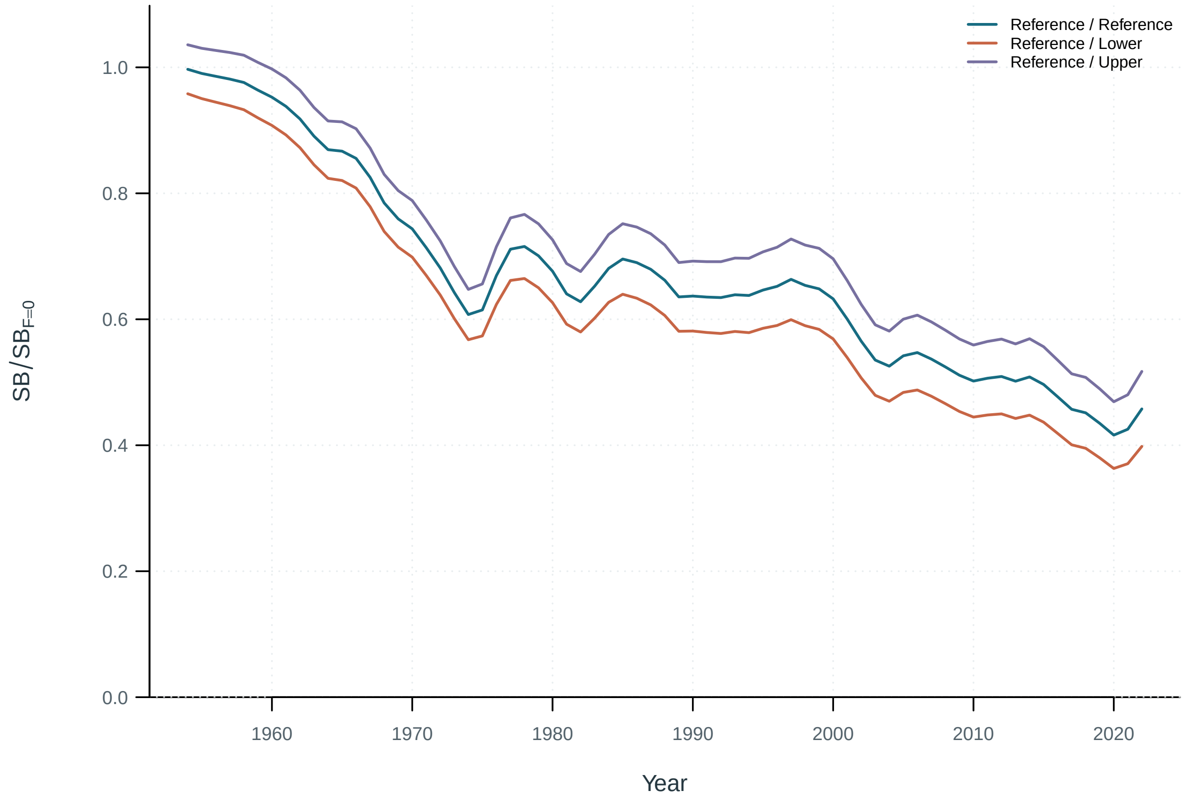

Figure 49: Spawning-depletion sensitivity to regional scaling.

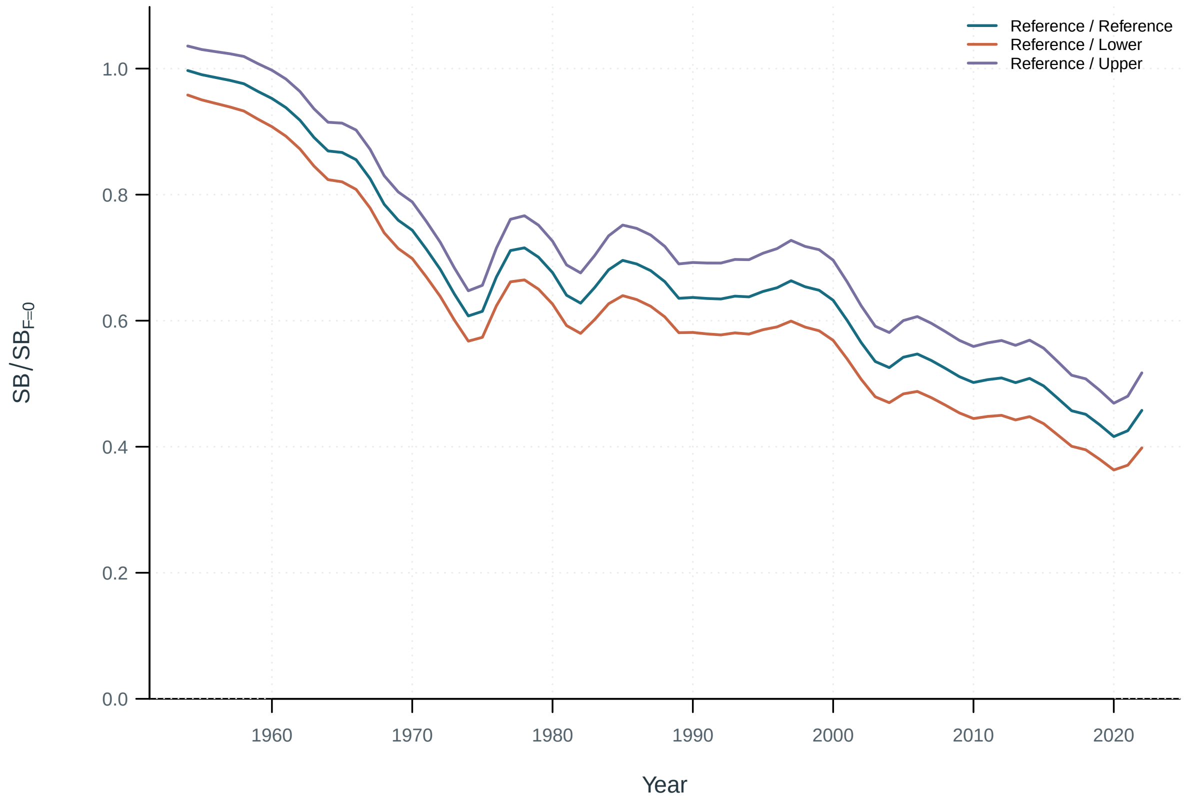

Figure 50: Spawning-depletion sensitivity to tag mixing periods Kstatistic cutoff.

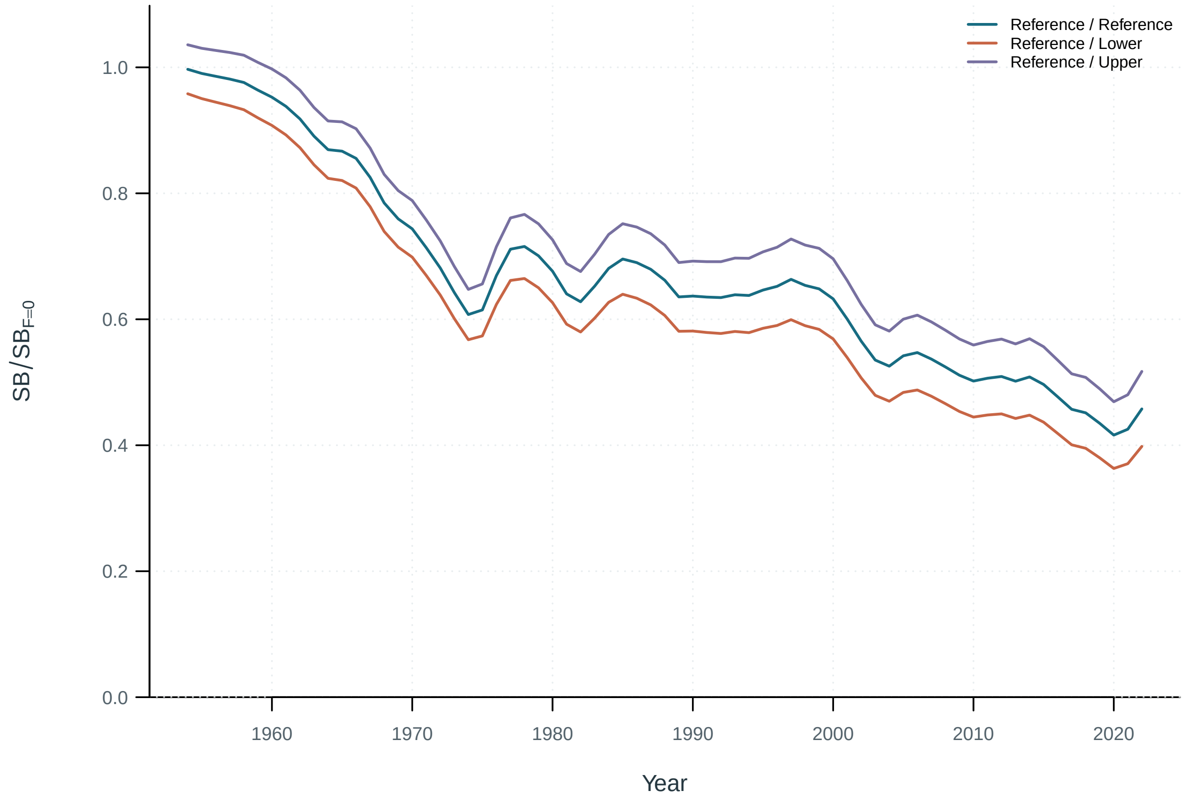

Figure 51: Spawning-depletion sensitivity to premixing tag reporting rate.

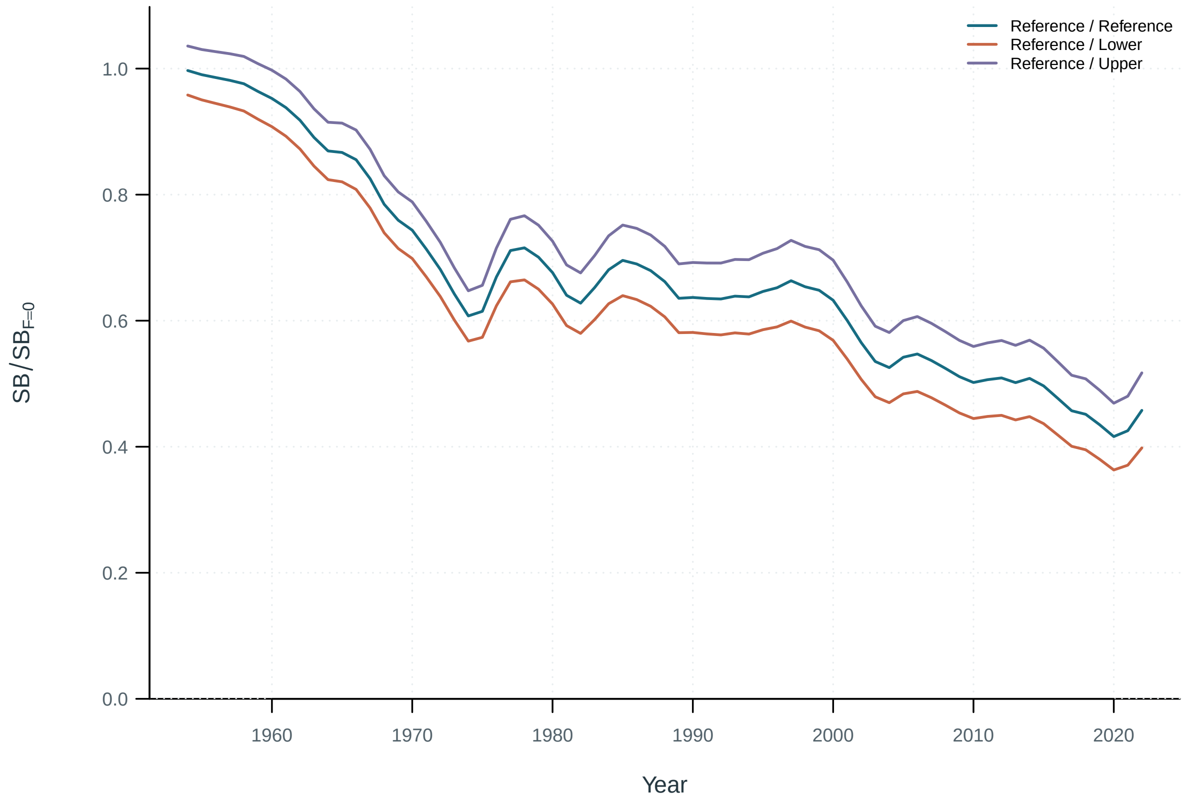

Figure 52: Spawning-depletion sensitivity to tag overdispersion.

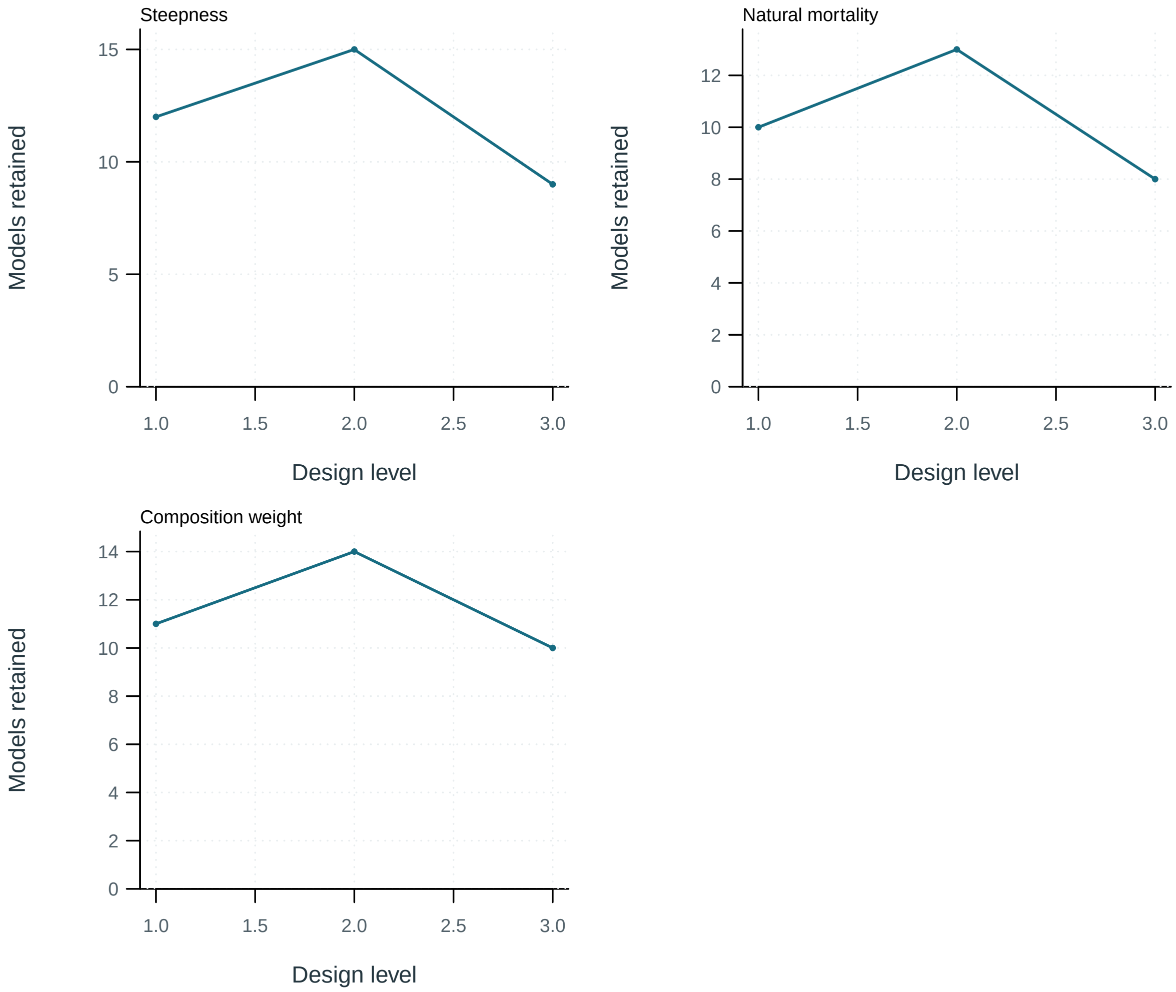

Figure 53: Retained model counts by design axis and level.

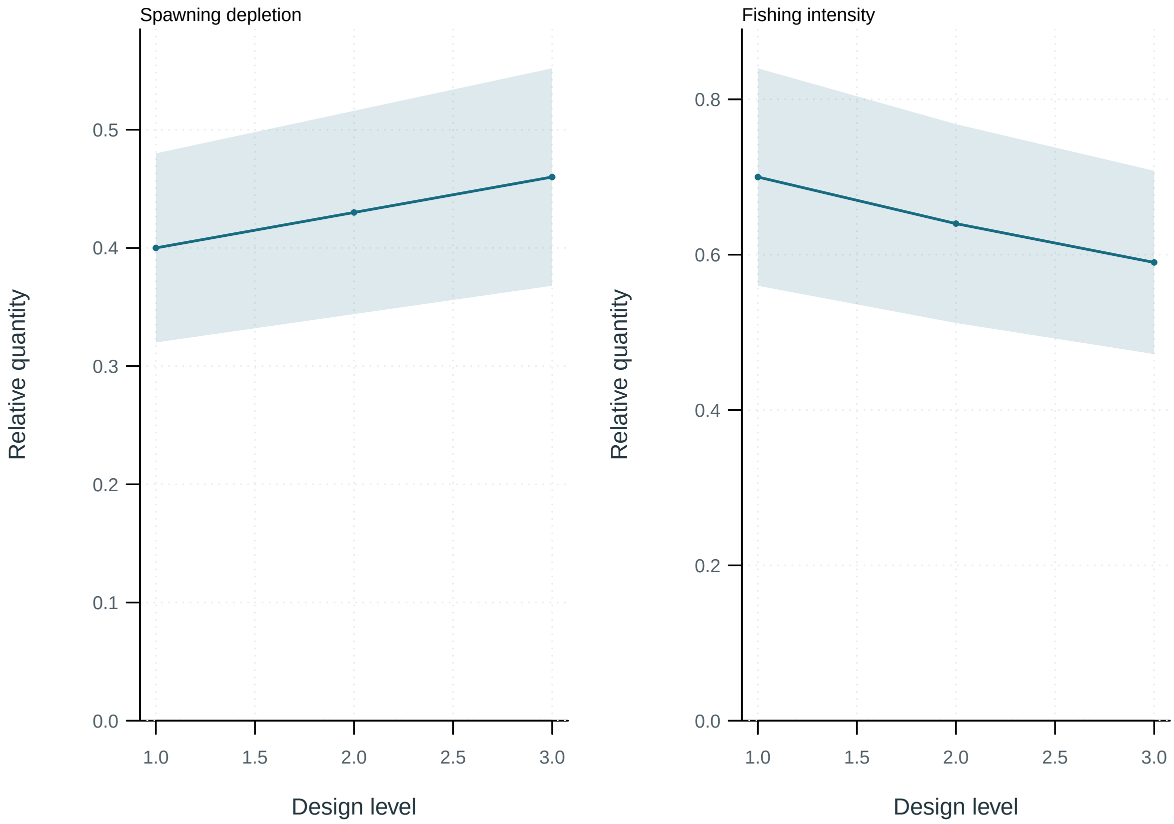

Figure 54: Management quantities across selected design levels; bands show the supplied model range.

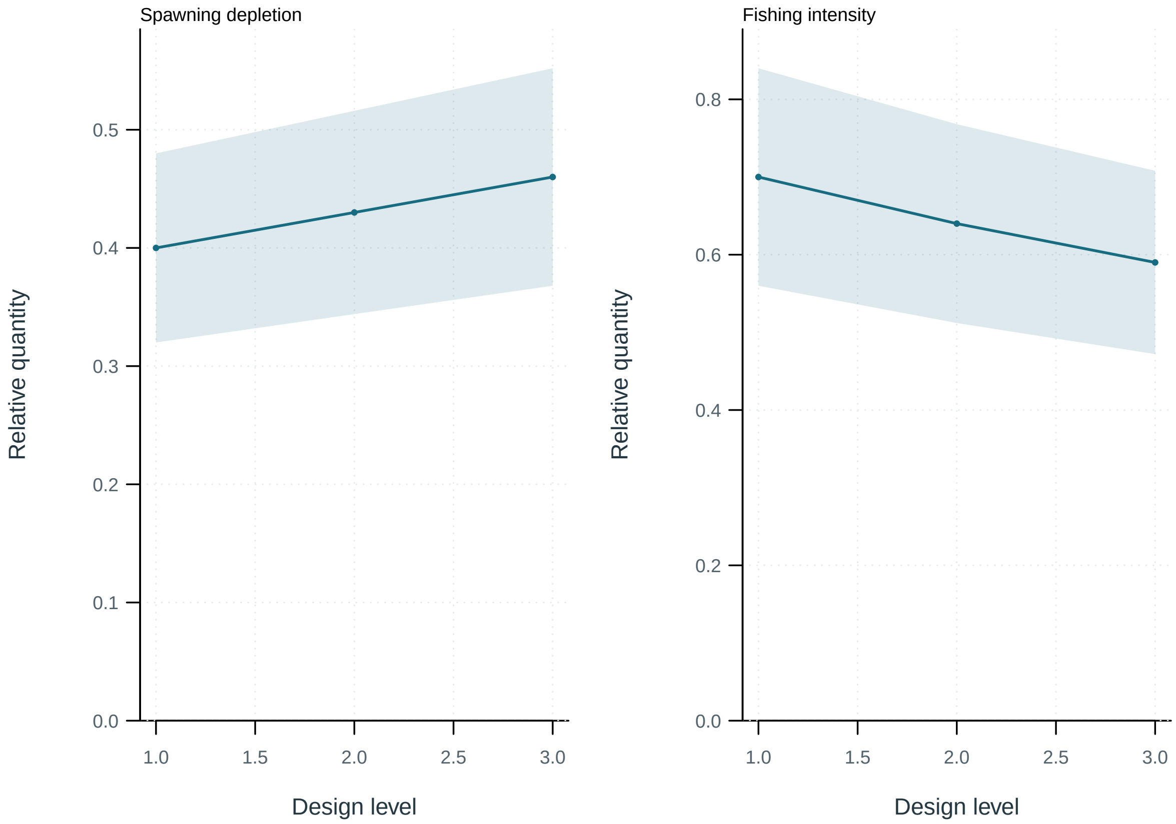

Figure 55: Management quantities across selected design levels; bands show the supplied model range.

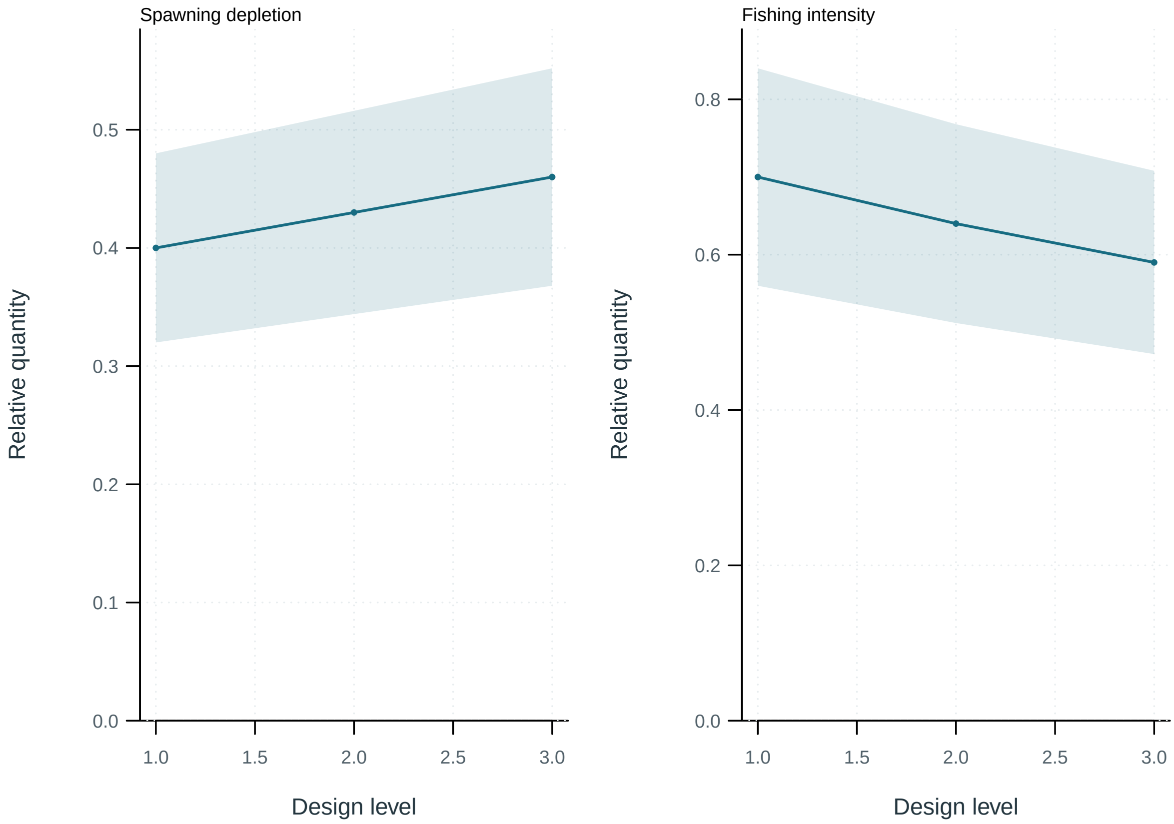

Figure 56: Management quantities across selected design levels; bands show the supplied model range.

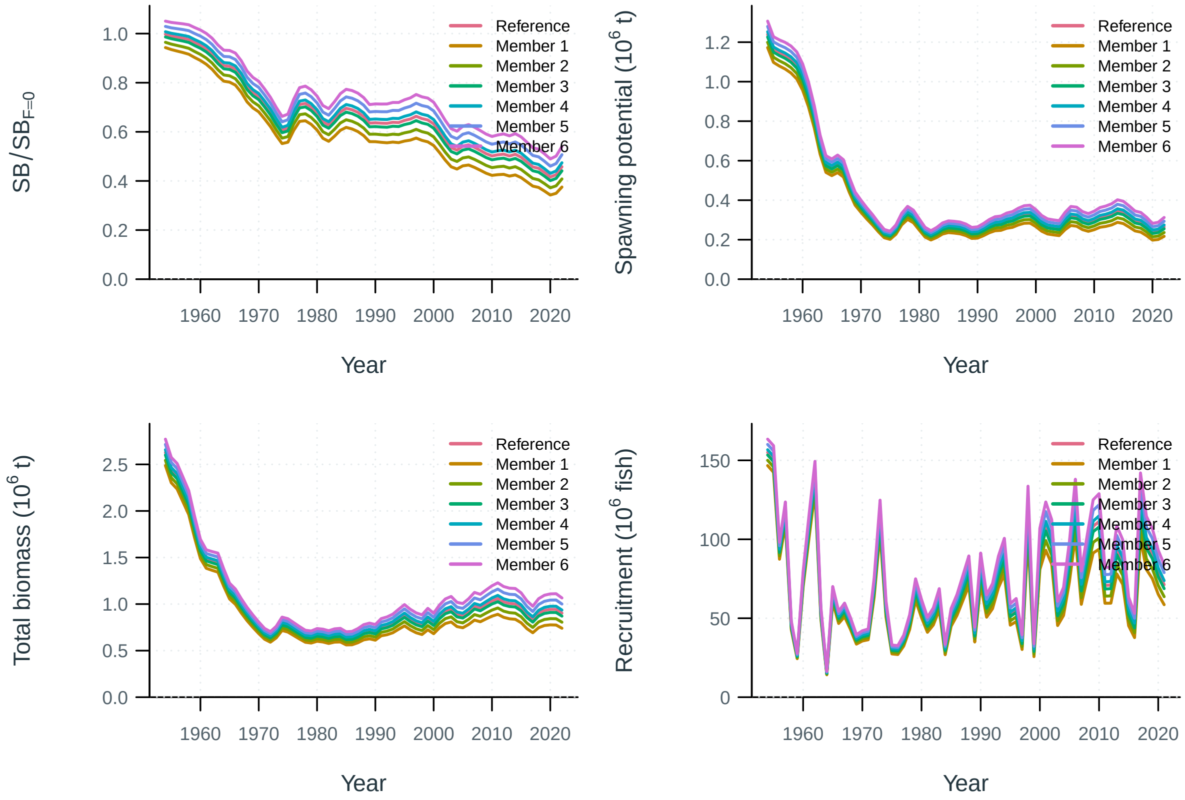

Figure 57: Ensemble model trajectories.

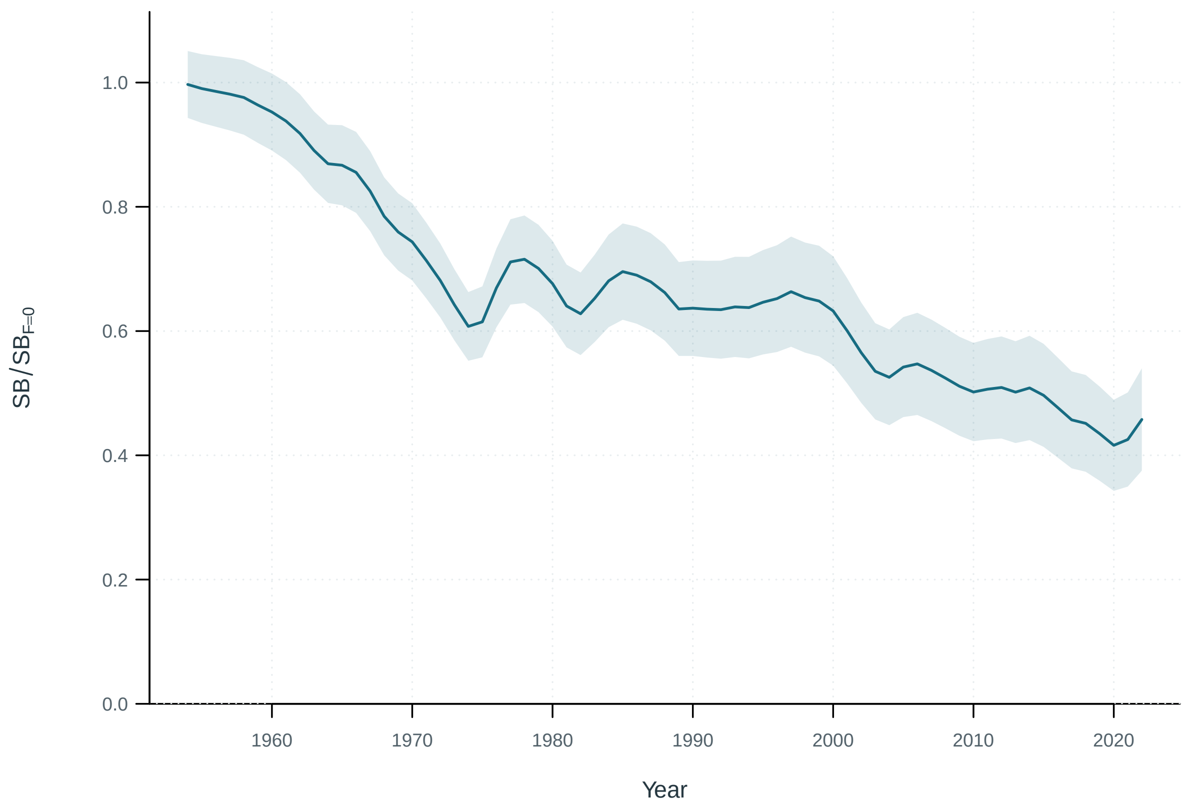

Figure 58: Median spawning depletion and range across the displayed models.

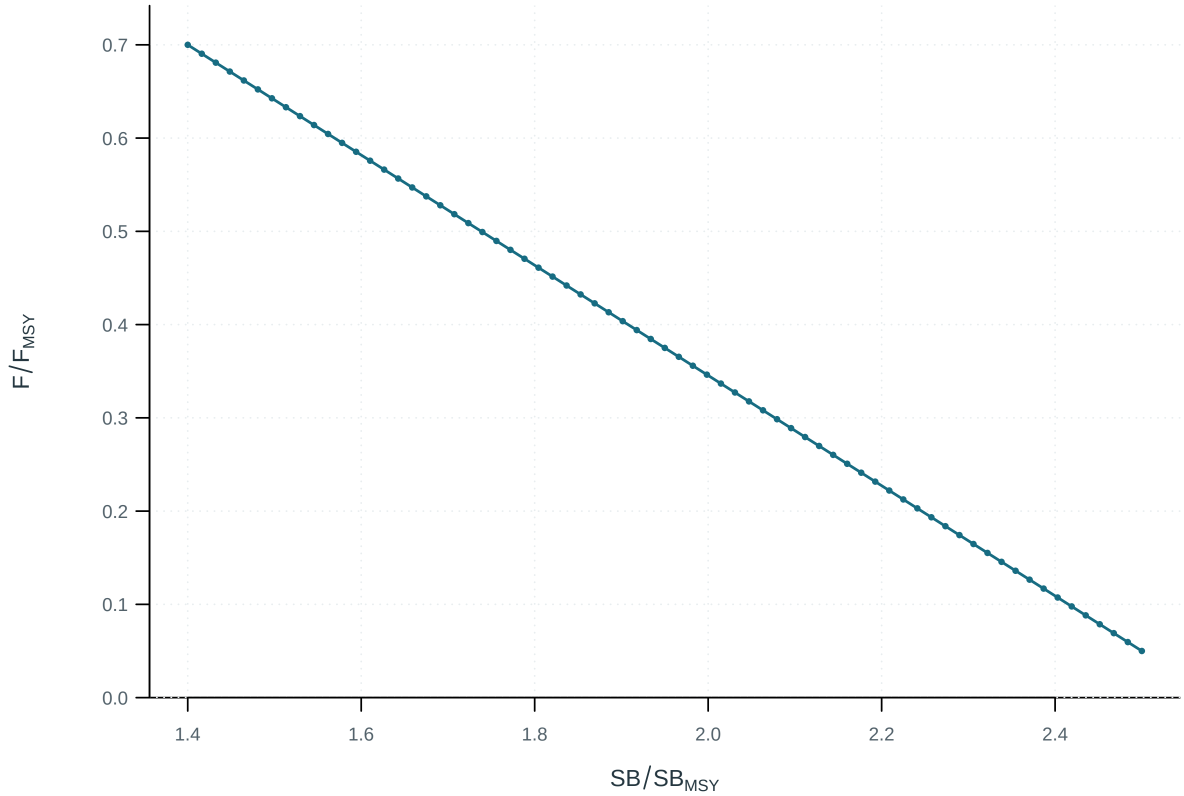

Figure 59: Stock-status trajectory relative to the supplied biomass and fishing-mortality reference points.

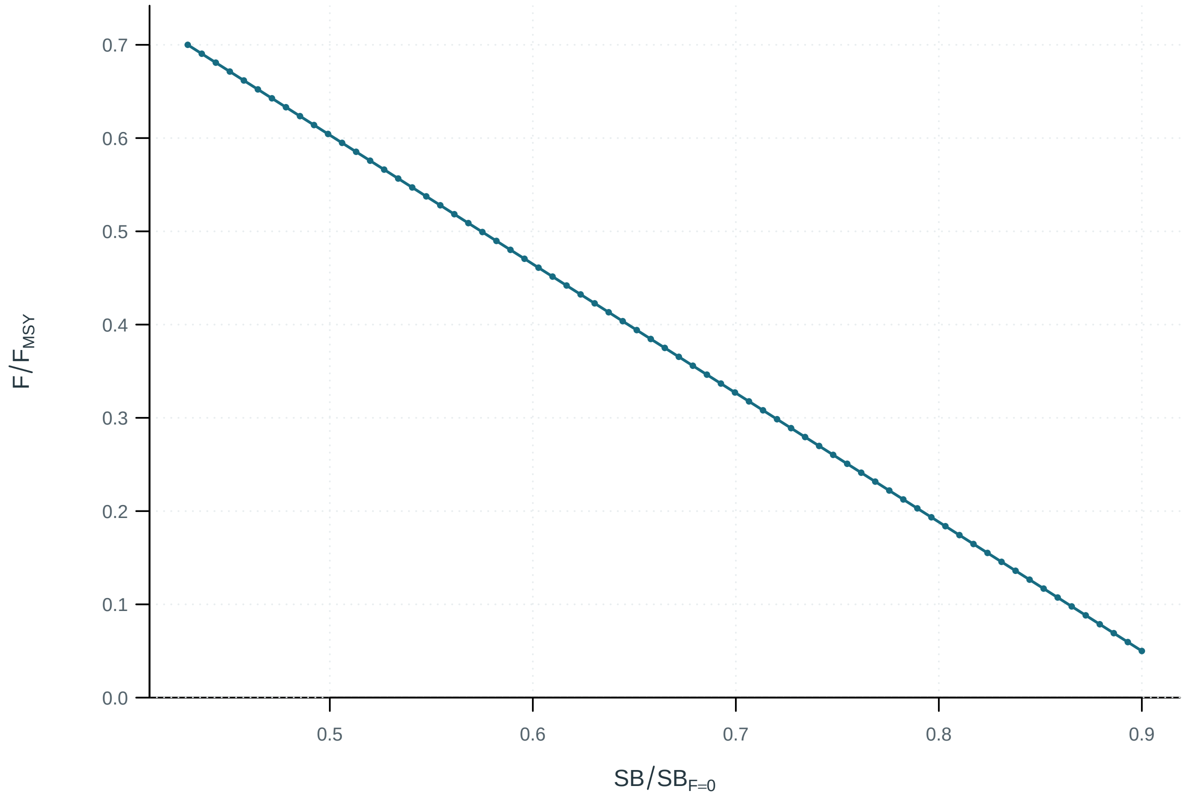

Figure 60: Fishing intensity and spawning depletion through the displayed stock-status trajectory.

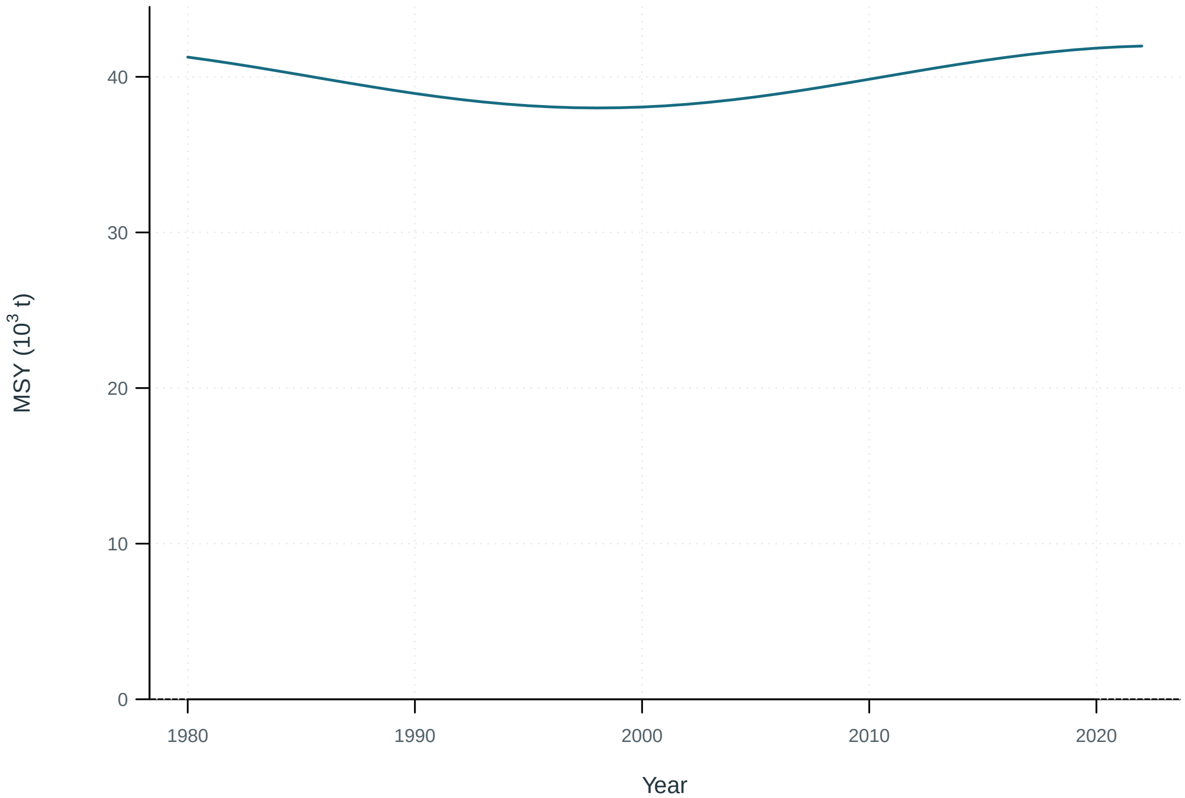

Figure 61: Time-varying maximum sustainable yield.

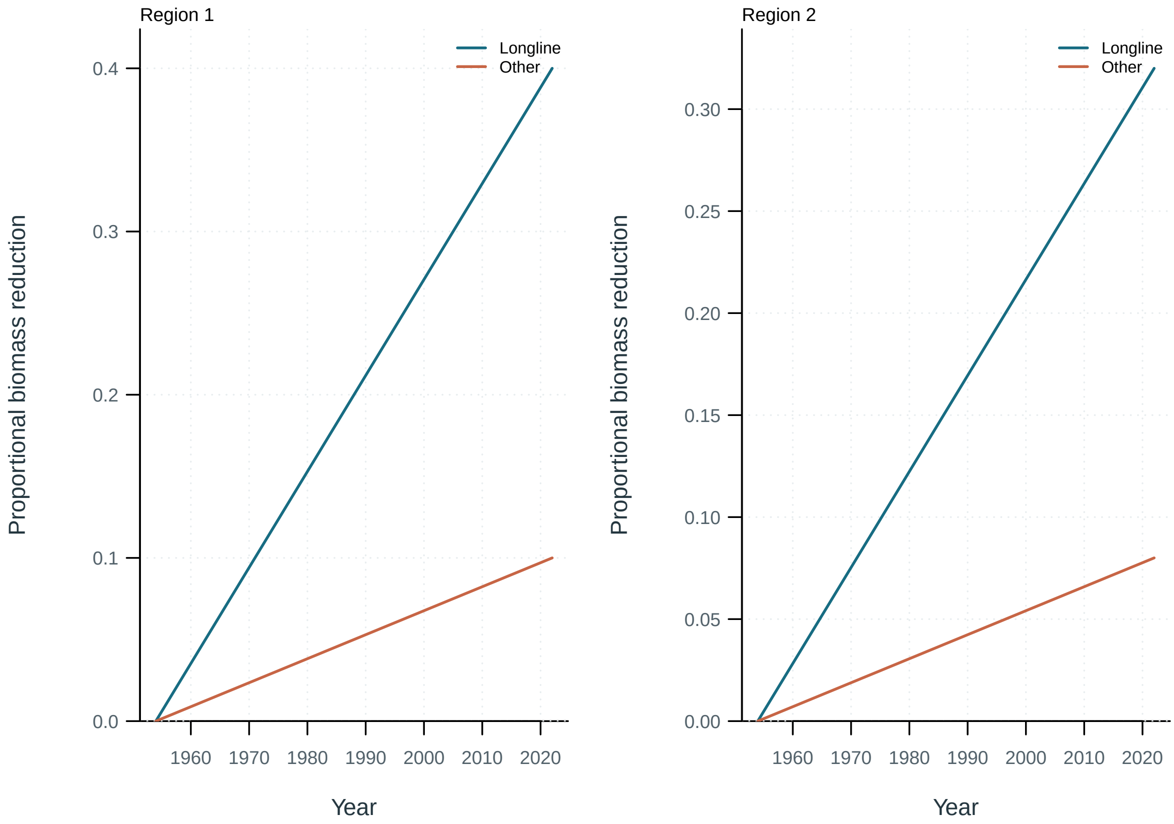

Figure 62: Reduction in spawning potential attributed to each fishing group, by region.

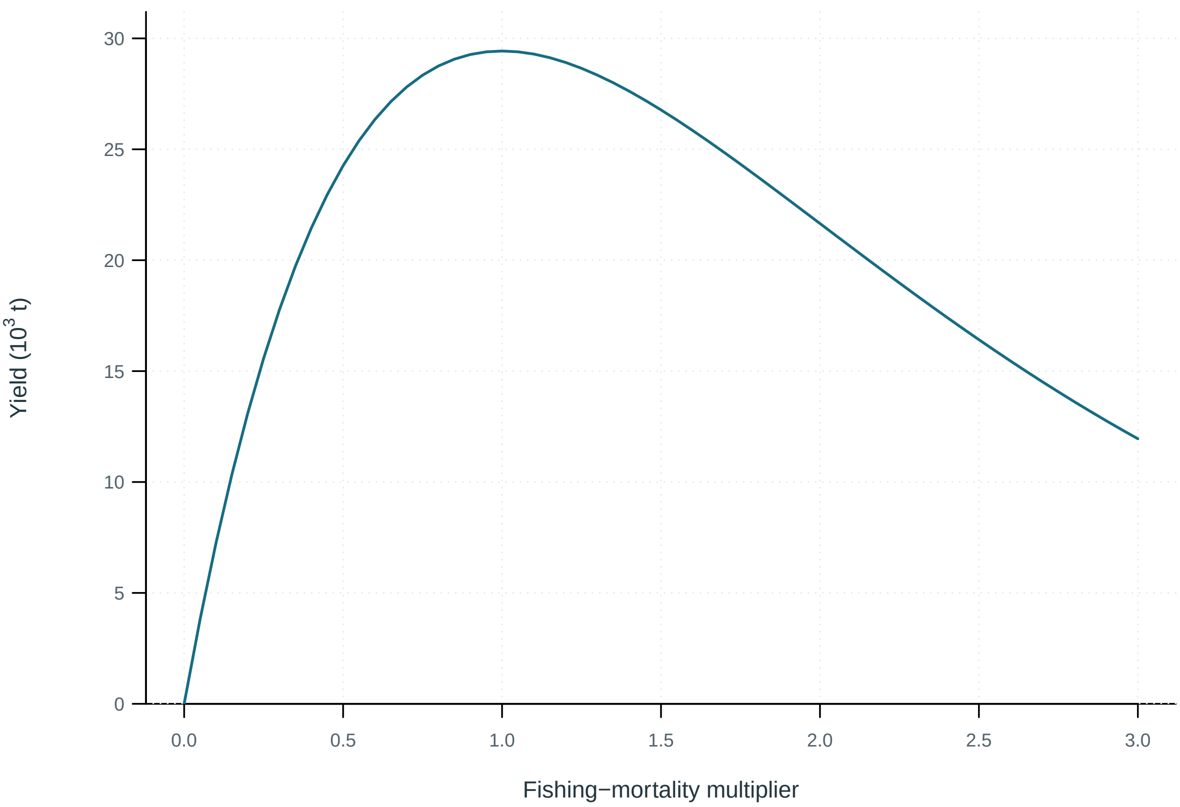

Figure 63: Equilibrium yield across fishing-mortality multipliers.

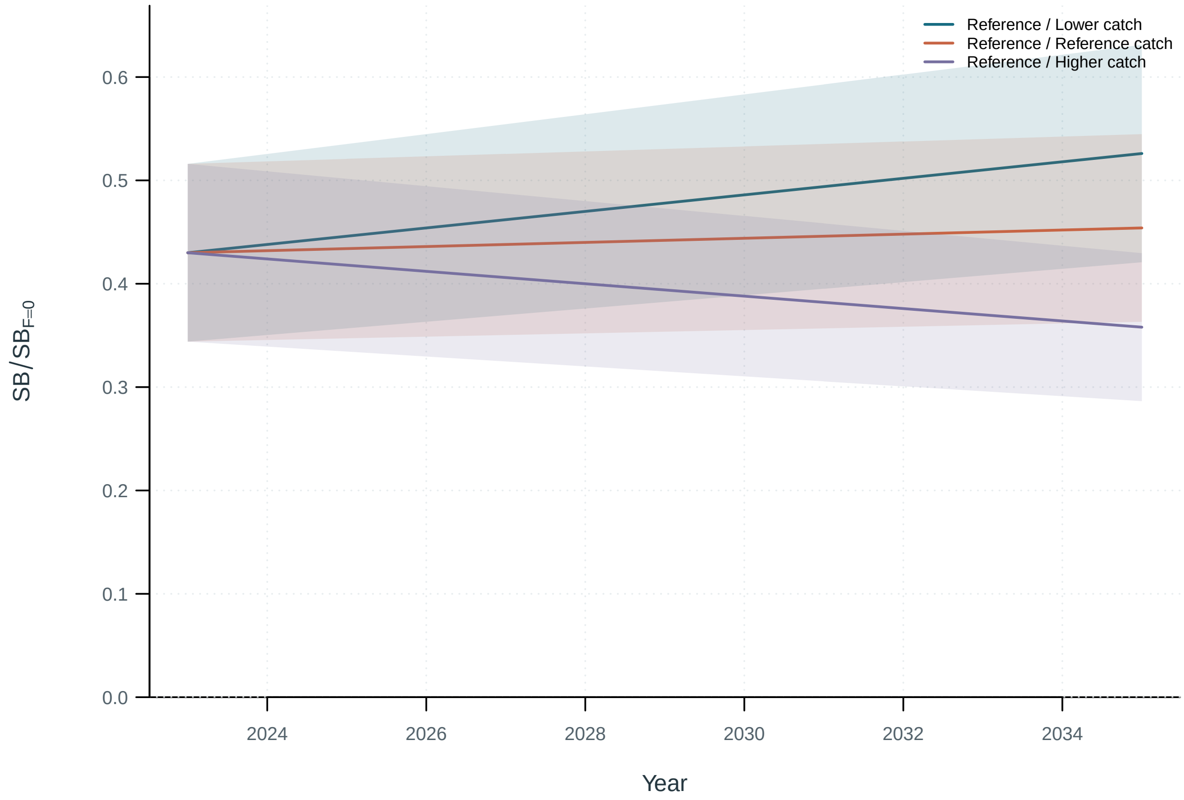

Figure 64: Projected spawning depletion by catch scenario, with the supplied central interval.

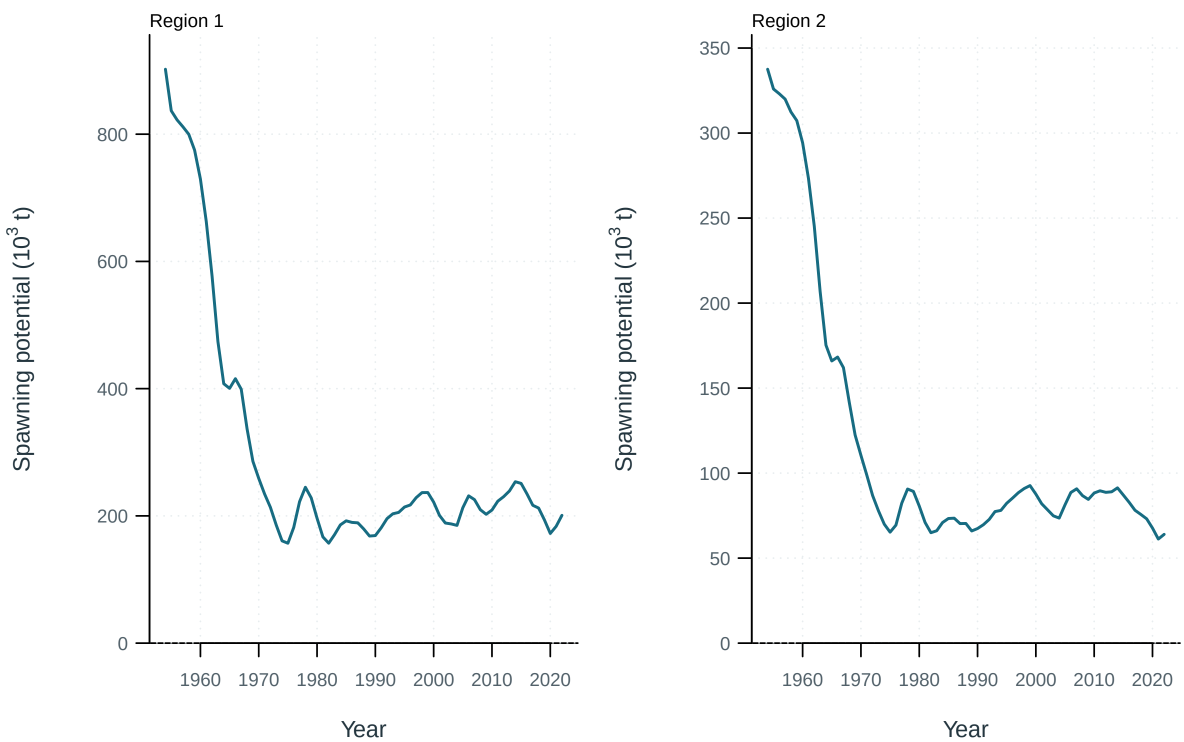

Figure 65: Spawning potential by model region.

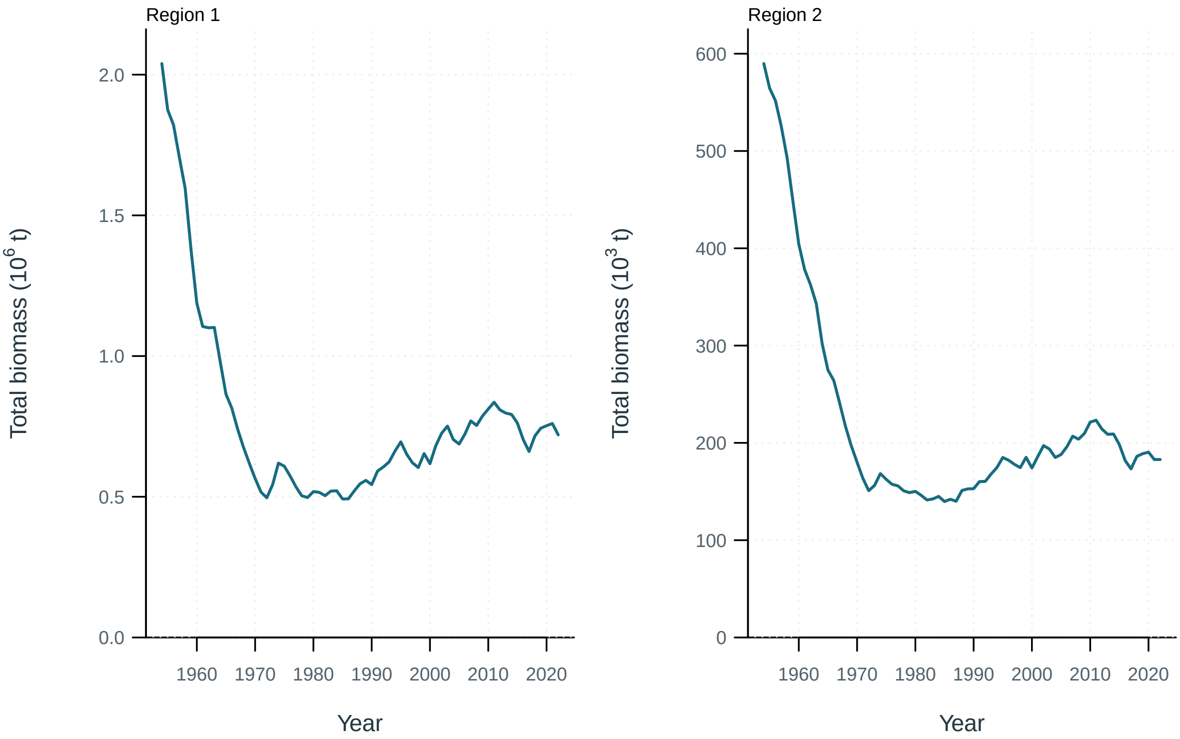

Figure 66: Total biomass by model region.

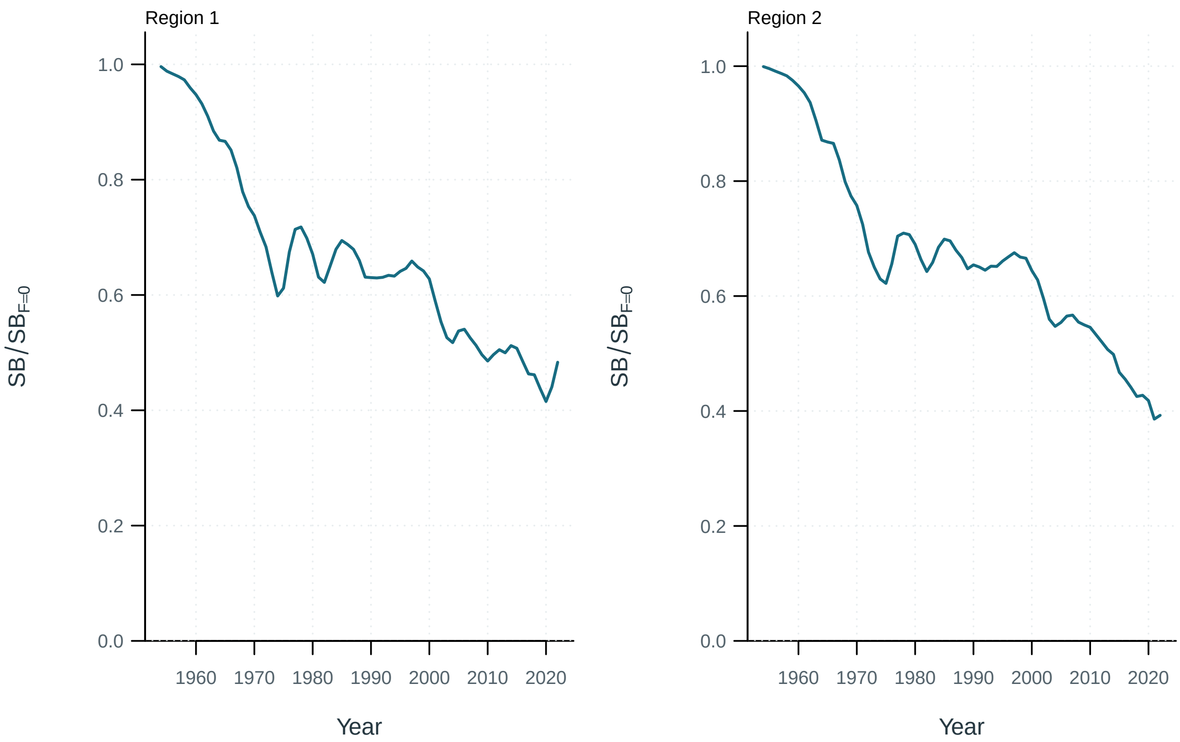

Figure 67: Dynamic spawning depletion by model region.

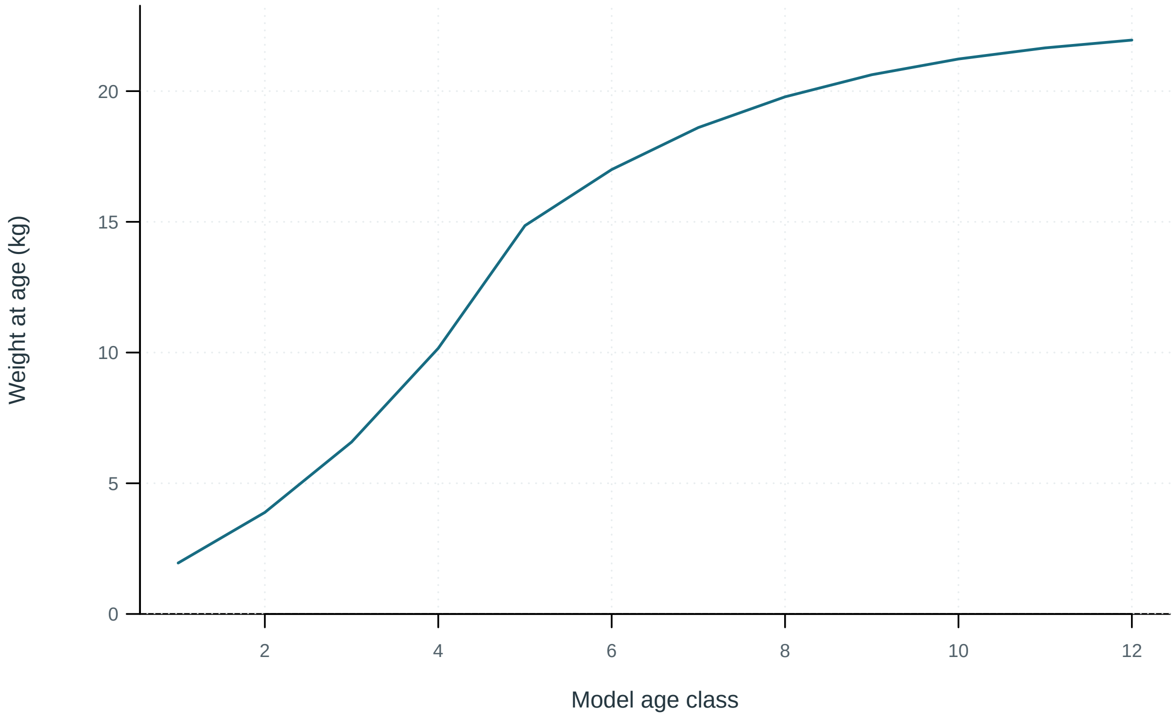

Figure 68: Mean weight at the start of each model age class.

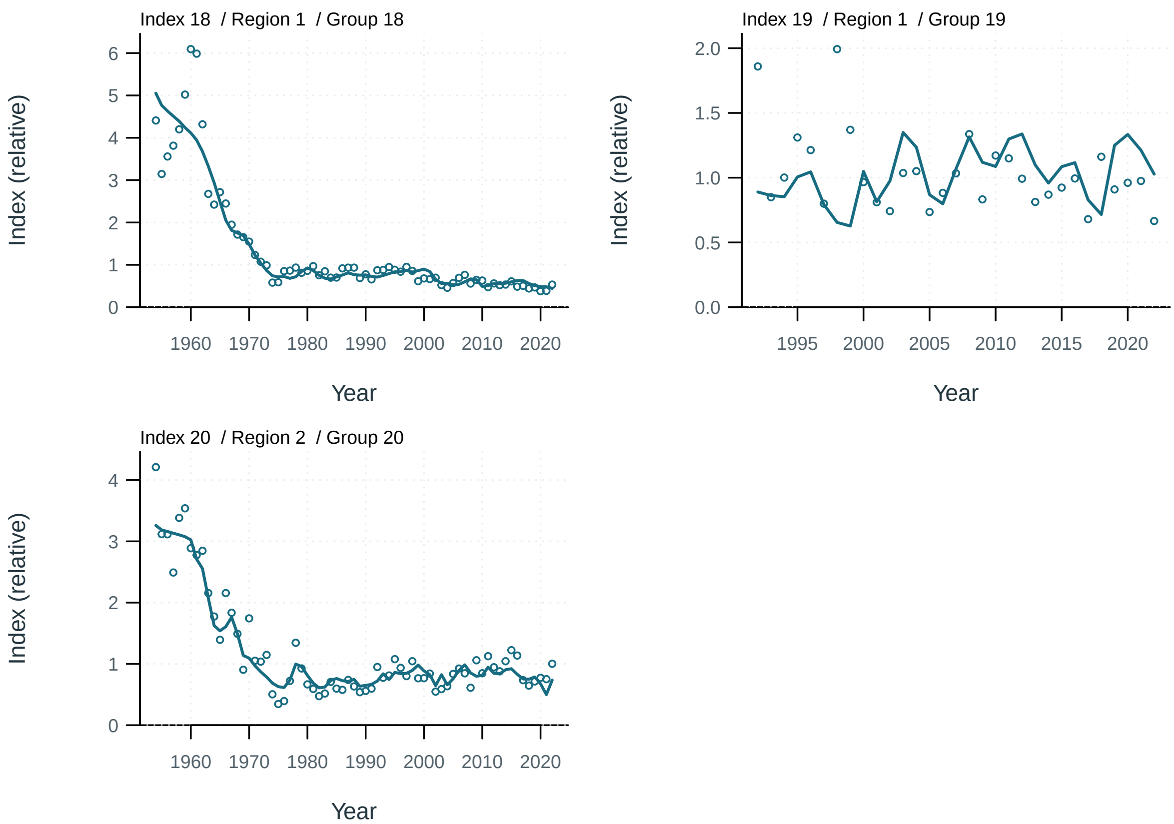

Figure 69: Index observations (points) and predictions (lines) by series.

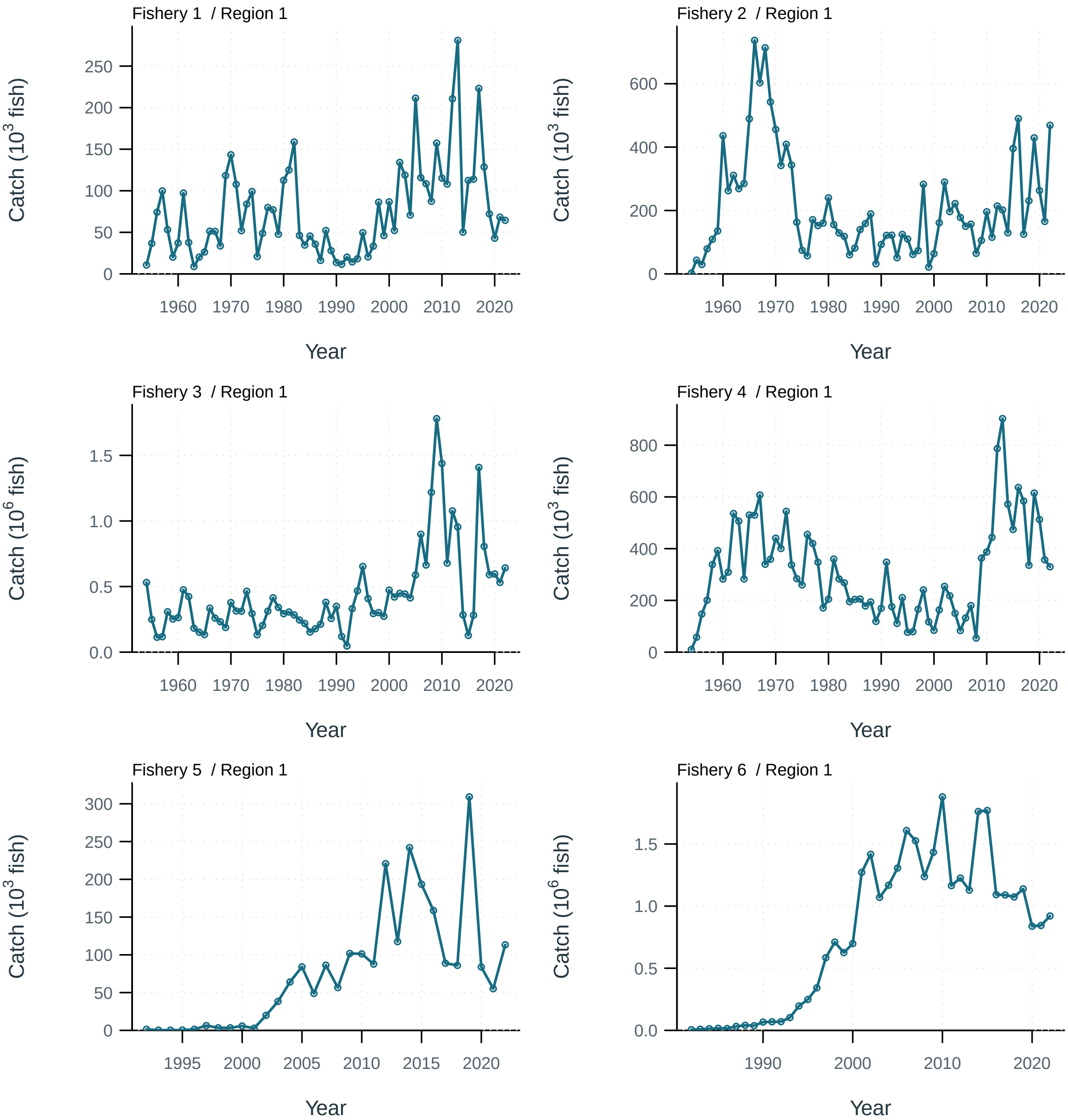

Figure 70: Catch observations (points) and predictions (lines) by series.

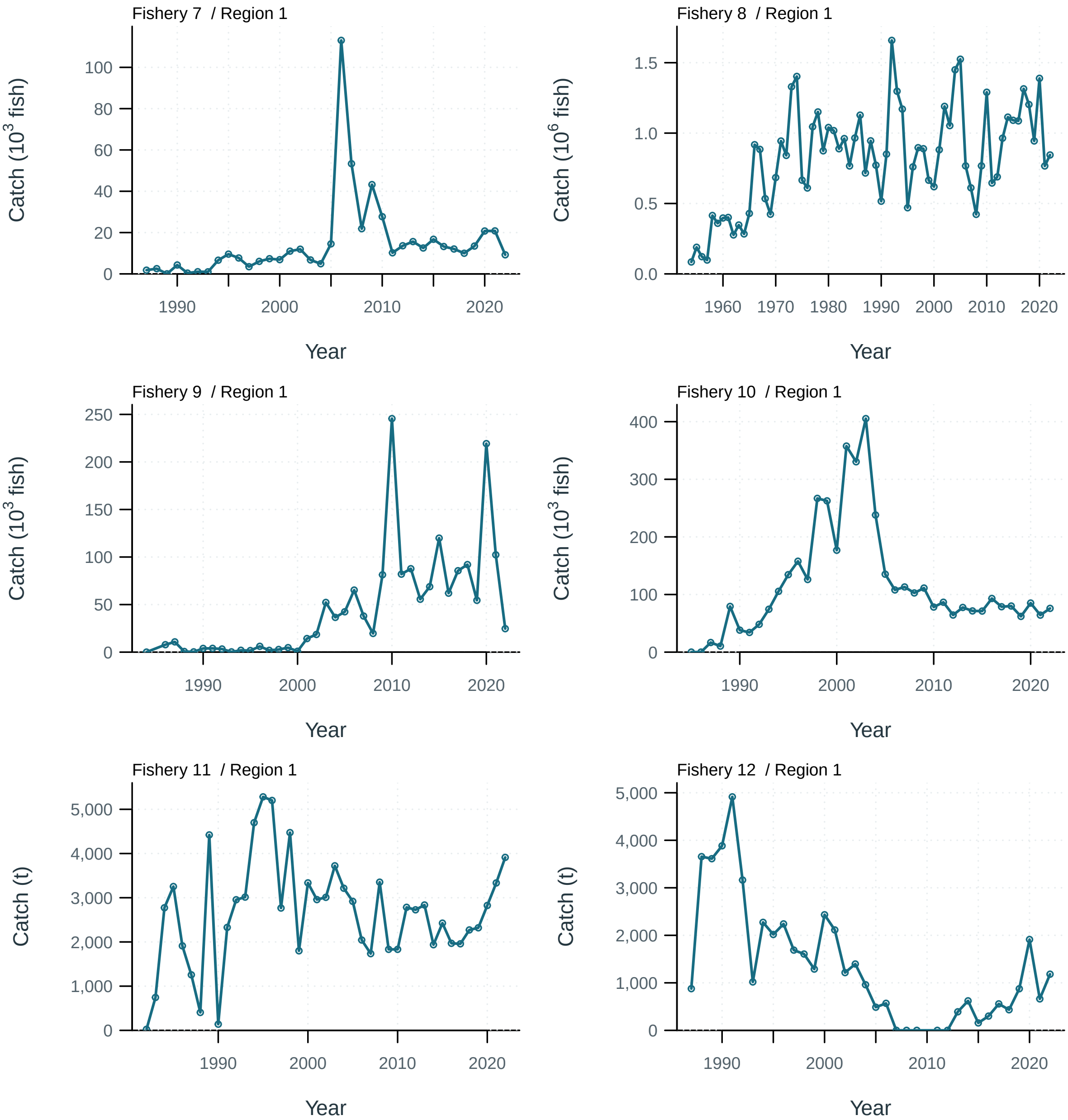

Figure 71: Catch observations (points) and predictions (lines) by series.

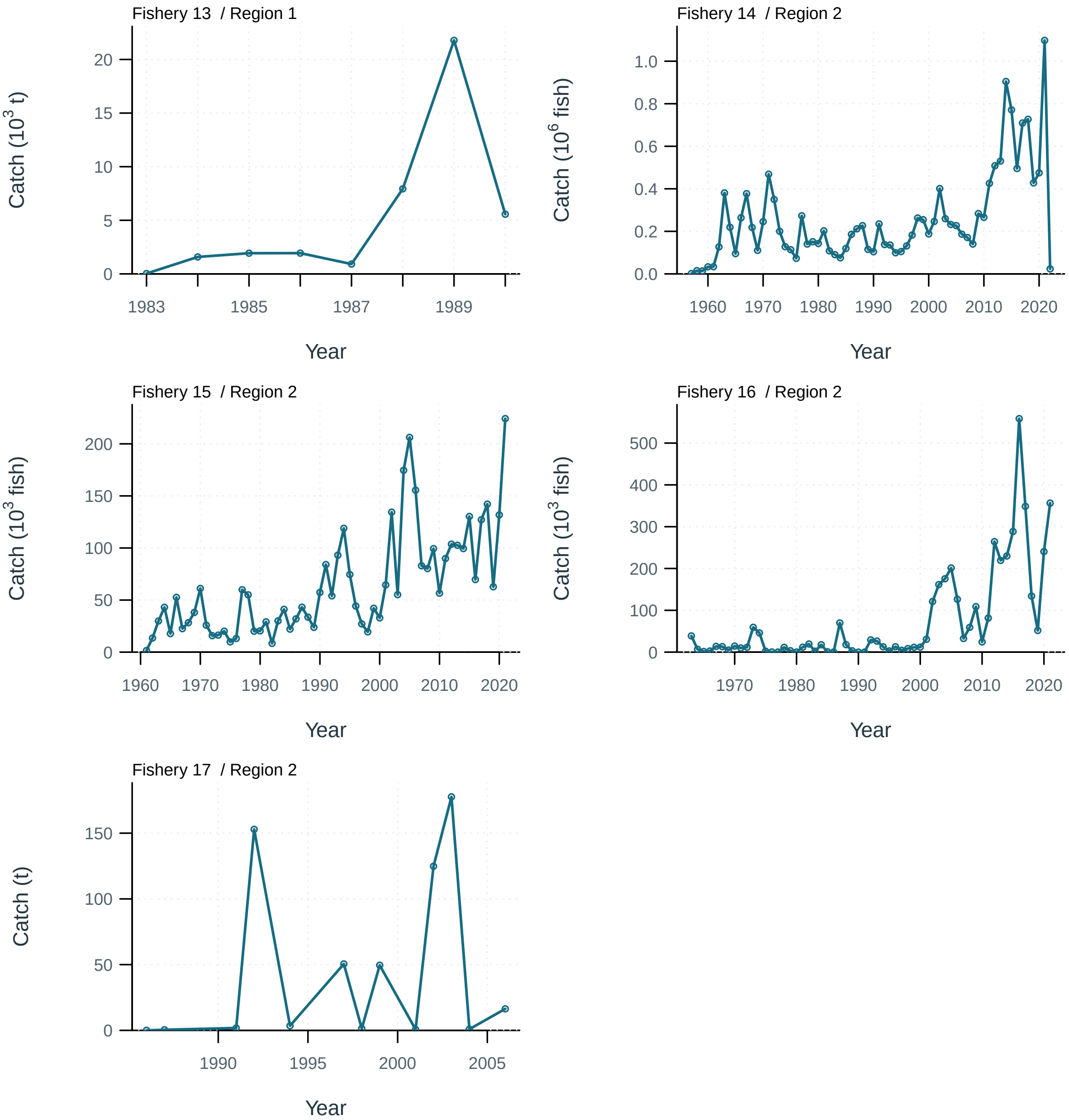

Figure 72: Catch observations (points) and predictions (lines) by series.

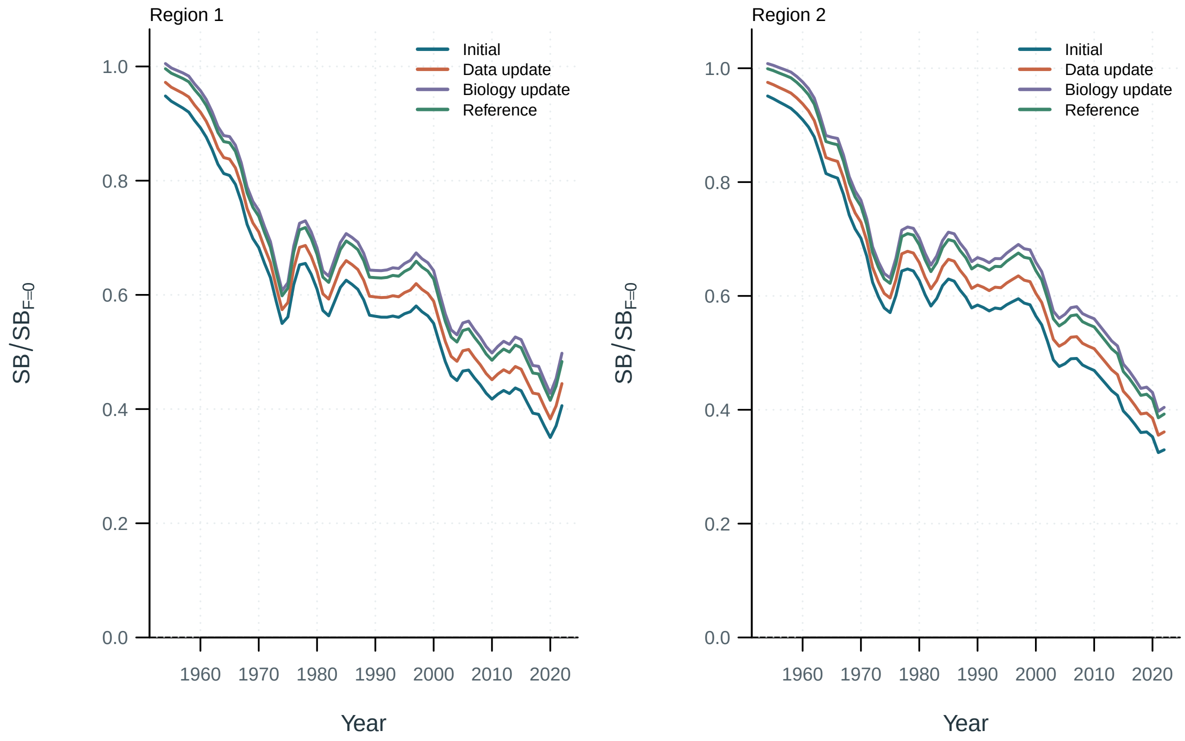

Figure 73: Spawning depletion by model and region, relative to each model's dynamic unfished reference.

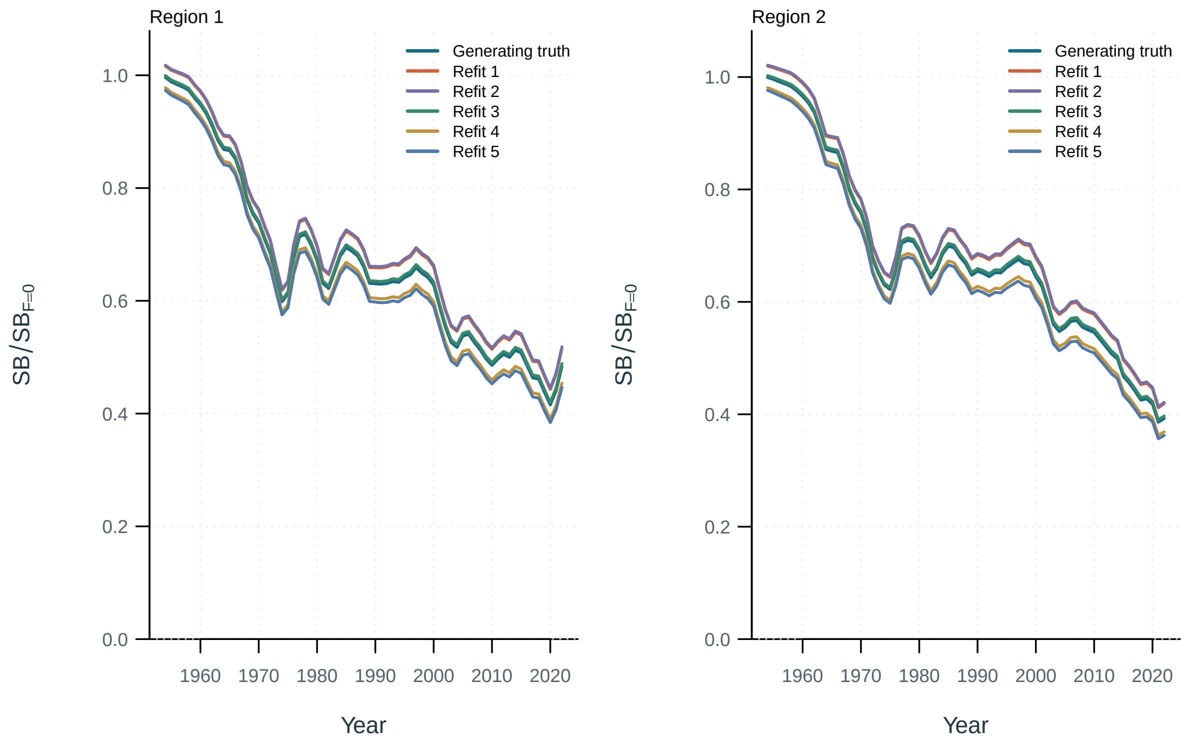

Figure 74: Spawning depletion by model and region, relative to each model's dynamic unfished reference.

Figure 75: Spawning depletion by model and region, relative to each model's dynamic unfished reference.

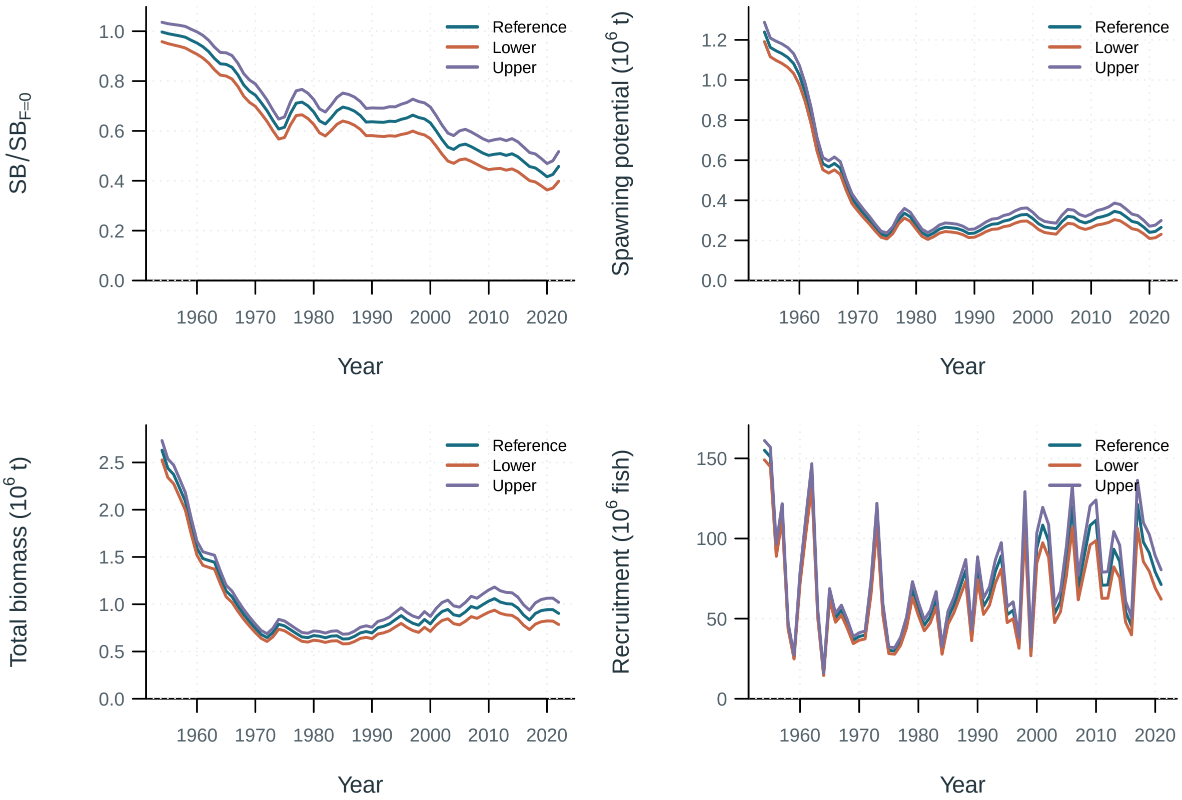

Figure 76: Sensitivity model trajectories.

Figure 77: Spawning depletion by model and region, relative to each model's dynamic unfished reference.

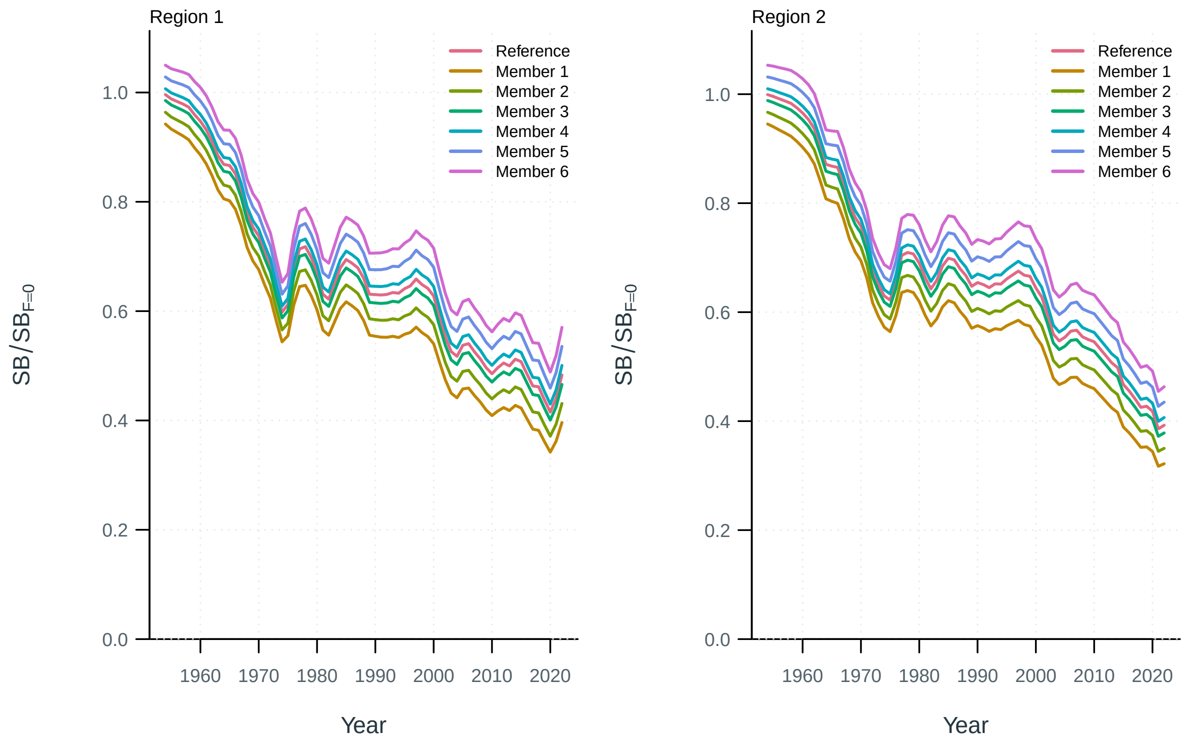

Figure 78: Spawning depletion by model and region, relative to each model's dynamic unfished reference.

11 Supplementary outputs

ALB · Diagnostic model interactive comparison

ALB · Diagnostic model figures and tables

ALB · Stepwise development interactive comparison

ALB · Stepwise development figures and tables

ALB · Starting-value trials interactive comparison

ALB · Starting-value trials figures and tables

ALB · Self-test recovery interactive comparison

ALB · Self-test recovery figures and tables

ALB · Retrospective analysis interactive comparison

ALB · Retrospective analysis figures and tables

ALB · Sensitivities interactive comparison

ALB · Sensitivities figures and tables

ALB · Uncertainty ensemble interactive comparison

ALB · Uncertainty ensemble figures and tables

ALB · Additional output formats interactive comparison

ALB · Additional output formats figures and tables

12 Appendix: supplementary methods

13 References